# When Silver Isn't Gold: Quantifying and Auditing Group-Based Disagreement in Human and Persona-Driven LLM Annotation

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LLM annotation is increasingly used as a scalable substitute for human annotations, including in polarized, subjective tasks where humans systematically disagree along group lines. Whether these systems reproduce, ignore, or distort human disagreement patterns is largely untested, in part because most existing methods for attributing Systematic Disagreement (SD) have not yet been tested in realistic, subjective annotation settings. This work represents the first empirical study which applies proposed, theoretical methods on large scale, realistic annotation settings. Observing the weaknesses of these methods, we develop a method that not only detects, but also quantifies attribution of Attributable Systematic Disagreement (ASD) to annotator SocioDemographic Dimensions (SDDs) and use it to audit nine open-source LLMs against five real-life datasets on subjective Natural Language Processing (NLP) tasks. Our results show that while persona-prompted LLMs can produce polarized annotations, the exhibited disagreement patterns are not aligned with those of human annotator SocioDemographic Dimensions (SDDs), even when tuning instruction prompts for this purpose. This limitation indicates that applications, studies and resources using persona prompting should include appropriate scoping and evaluation in order to avoid further model miscalibration. Our study also suggests that the structure of current subjective NLP datasets is insufficient, requiring more annotators per item, as we find that not all Systematic Disagreement (SD) can be attributed to any combination of known or latent annotator SocioDemographic Dimensions (SDDs).

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Unsupervised learning**; Language resources; • **Human-centered computing** → **Empirical studies in collaborative and social computing**; • **Social and professional topics** → *Race and ethnicity*; *Gender*; Sexual orientation; Age.

Additional Key Words and Phrases: Polarization, Clustering, Annotation, Sociodemographic Characteristics, Persona-Prompting, LLM Bias

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## 1 Introduction

The rise of internet spaces has necessitated research in subjective phenomena such as the detection of harmful content. Annotations are needed to support social science research, the training of automated classifiers and the proactive identification of vulnerable groups [70]. These annotations are frequently released as datasets, which due to their subjective nature typically contain items with very low Inter-Annotator Agreement (IAA) [70]. If ignored, this disagreement may lead to loss of valuable information [51] and provide us with the wrong signal, leading to significant misconfiguration in trained AI systems [68]. For example, toxicity models have disproportionately marked African American (AA) speech as "toxic" [59], or systematically ignored minority opinions [54] in the past.

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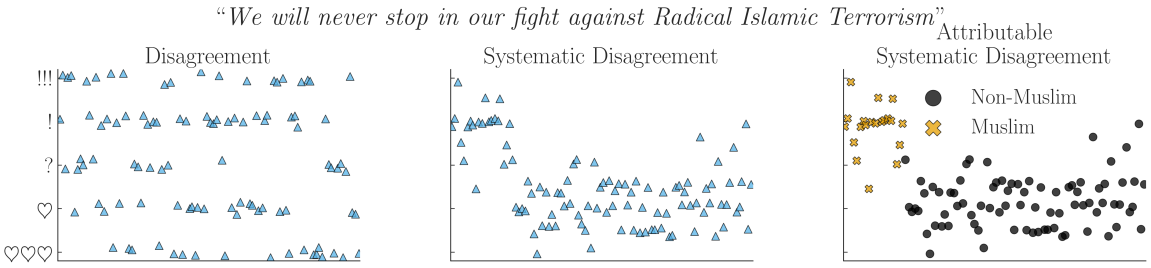



Fig. 1. Three kinds of disagreement in a hypothetical hate-speech annotations of a controversial comment. Each point is one annotator's rating, from ♡♡♡ (not hateful) to **!!!** (very hateful). *Left:* ratings spread uniformly across the scale, an extreme case of disagreement. *Middle:* SD, where a subset of annotators consistently rates the comment differently from the rest, although the cause is not visible from the ratings alone. *Right:* the same ratings as the middle panel, grouped by religion. Religion, one of the observable SDDs, acts as a "polarizer" that explains the observed differences in opinion, marking the disagreement as ASD.

Current literature lacks precise terminology to describe opinion differences between groups. We thus use the following definitions in this study: (1) **Disagreement** represents any variance among annotators' ratings for the same item, even if minute. (2) **Systematic Disagreement (SD)** is non-random and recurring disagreement. (3) **Attributable Systematic Disagreement (ASD)** is SD that can be attributed to differences in group identity (e.g., background, cultural context, ideology). (4) **Polarization** is the mechanism with which we can distinguish SD from ASD. In physics, a *polarizer* is an optical filter with which a set of tangled rays can be untangled to "pure" signals. In our study, the metaphorical polarizer is each of the SocioDemographic Dimensions (SDDs) of an annotator group. If SD (our "beam" in this case) is reduced when considering the groups emerging from an SDD (e.g., left-wing/right-wing when considering political orientation), it becomes ASD. Fig. 1 demonstrates this distinction.

Researchers traditionally attempted to distinguish *Disagreement* from SD, but few studies moved towards discovering ASD. In both cases, IAA metrics are typically used both to distinguish between general disagreement and SD and to attribute it to specific annotator groups. Traditional metrics however hold significant limitations, especially when few annotations per item are available and when the studied SDDs contain imbalanced groups. Both of these issues are typically present in subjective tasks featuring minorities, which is the exact case where our measures should be reliable.

Furthermore, as human annotation is generally expensive, LLM-as-judge systems are increasingly deployed as scalable substitutes for human annotators [32, 57] to acquire a cheaper, "silver" supervision signal in place of human-annotated "gold" labels.[1] It is therefore imperative that ASD by human annotators be preserved by LLMs whose annotations by default gravitate toward the statistical average [8]. Persona-prompting is sometimes used to bypass this limitation, but suffers from its own challenges such as stereotyping and inconsistent annotations [40, 42]. We therefore ask:

RQ 1: *How Do Current Theoretical Polarization Attribution Approaches Behave in Realistic Annotation Scenarios*? Our work is the first empirical study applying measures of polarization detection and attribution in real-life subjective datasets, finding that they tend to severely underperform. One of the reasons for this collapse is likely the assumption that annotations can be aggregated across items (Finding 1).

RQ 2: *How Can SD be Quantitatively Attributed to Annotator SDDs*? Since we can not reliably used established approaches, we introduce a new measure which *quantitatively* attributes SD to specific SDDs, while being conscious of the limitations present in realistic annotation scenarios (Fig. 2).

[1]What constitutes "gold" or "silver" labels is not standardized and the terms frequently change definitions between publications. Some advocate against using the term "gold labels" for subjective, human annotations [5, 20].

RQ 3: *Do LLM Annotators Exhibit ASD, and if so, Which Patterns are Shared Between Them and Humans*? Unlike prior work which framed the question of LLM annotation reliability in terms of human-LLM agreement [15, 23, 43] or task-specific performance [28], we compare human and LLM polarization patterns. We find that human ASD is explained by gender and ethnicity across five real-life subjective datasets (Finding 2). On the other hand, while LLMs using persona-prompting *can* produce polarized annotations, they fail to reproduce any ASD patterns regardless of instruction prompt, confirming literature suggesting that persona prompting does not consistently influence LLM outputs in the examined tasks (Finding 3).

Overall, our work indicates that *persona-prompting can not be used by itself to ensure fair and equitable judgments* across SDD groups by LLMs. Without careful scoping and task-specific evaluation, persona prompting can lead to LLMs not taking into account minority-group sensitivities, which could compromise proposed applications with autonomous agents in sensitive environments (e.g., content moderation, facilitation) and contaminate future LLM training, amplifying existing model biases, and lead to misleading findings in social science research. Our work also argues that even with recent methodological improvements (e.g., phasing out majority voting), *the field of subjective Natural Language Processing (NLP) has not adequately addressed the challenge of analyzing multiple perspectives* (§6). Most current datasets do not hold an adequate amount of annotator SDDs, nor enough annotators per item, an issue which has not yet been discussed in literature. Indeed, we discover "Unnatributtable Systematic Disagreement (USD)" which can not be observed even when exhaustively considering all possible annotator dimensions in real-life datasets and is caused by low annotation counts per item (Finding 4).

We also release a PyPI library implementing our measure, our LLM annotated datasets totaling more than 3,000,000 annotations, as well as the code and reproducibility details to aid future research in this topic.[2]

**Content Warning: Controversial and potentially harmful speech is used for demonstration purposes.**

## 2 Background and Related Work

### 2.1 Systematic Disagreement Detection

Prior work on analyzing annotations in subjective tasks has shown that SD may be caused by different points-of-view between groups [24, 46, 68]. While current research mostly rests on using IAA metrics between groups of a single annotator SDD [30, 34, 38, 46, 67, 69], these metrics suffer from severe limitations, which have motivated the development of new approaches.

*Limitations of Inter-Annotator Agreement.* IAA metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish SD disagreement from general disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup IAA without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds, producing results that are difficult to compare across studies [9, 10, 18, 69], being sensitive to class imbalance [18], and relying on questionable assumptions about chance agreement [10]. While recent work has addressed some of these limitations with methods that improve applicability and robustness [30, 69, 75], they are much more complex and less intuitive, indicating that IAA may not be the correct tool for analysis of minority viewpoints.

[2]Library https://anonymous.4open.science/r/apunim-5F70, HTML documentation https://anonymous.4open.science/r/apunim-page, replication code https://anonymous.4open.science/r/aposteriori-unimodality-E8C1. See Ethical Considerations for export measures relating to gated-release datasets.

*Alternatives.* A different approach [2] utilizes $\chi^2$ to estimate in-group vs out-group variance. Others frame SD detection as a cluster identification problem, applying either parametric (Gaussian Mixture Model (GMM)) [10], or non-parametric [45, 51] approaches to the annotation histograms. For our work, we use a non-parametric disagreement measure, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established measures [51], as we also verify in Fig. 2. nDFU (Eq. 1) takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). Intuitively, nDFU measures how prominent the maximum secondary peak is compared to the main peak.

$$\text{nDFU}(X) = \frac{\max_{i \neq m} \left( f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}, m = \arg\max_i f_i \tag{1}$$

where, $X$ denotes the set of annotations, $f_i$ the relative frequency of the $i$-th annotation (e.g., the frequency of 2 in a scale of 1–5), and $m$ the histogram peak.

*What About Attribution?* IAA analysis may be useful for exploration and identification of some controversial items, but fails to explain the reason *why* they are controversial [1, 71]. A purpose-designed method for the identification of ASD is the A Posteriori Unimodality (AU) method [51], which builds upon nDFU. It states that if a single item's annotations are multimodal (contains SD), but unimodal for all annotator subgroups in a specific SDD (contains disagreement but not SD), then polarization is explained by those groups (ASD). Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single SDD, which we prove to be impossible in many realistic cases (§6).

### 2.2 Using LLMs for Subjective Annotation

*Persona Prompting.* To address subjectivity in annotation, several NLP approaches incorporate annotator metadata into LLM annotation [14, 19, 27, 58]. A significant limitation is the requirement for annotator metadata, which can be side-stepped through embedding clustering [72], at greater computational cost and harder evaluation. Our approach instead directly clusters the annotations, which we believe provides a comparatively simpler and more interpretable mechanism.

*Persona Engineering.* At the time of writing, there is no established taxonomy of persona prompting techniques. Generally, personas may be authored by hand [44, 79], generated using observed annotator attributes [3], composed of answers to extensive questionnaires [50], or recently, generated entirely using LLMs [63]. The prompt itself may be structured [49, 73, 78] or described through natural language [39, 63, 79]. The effects of these methods on LLMs have not yet been studied and are largely outside the scope of this work. Since our goal is to explore disagreement patterns across LLMs and humans (RQ 3), we select attributes from the human annotators of each dataset and format them using JSON instead of natural language, in order to facilitate comparisons across experimental conditions.

*Persona Reliability.* Prior work suggests that LLMs are not necessarily reliable annotators when prompted with personas, as they tend to reproduce stereotypes [40, 42] and perform unreliably for minority groups [32, 42]. Additionally, some evidence suggests that task-specific improvements may be not be replicable outside of the datasets on which models are trained [37]. Our work ultimately supports the view that persona prompting is not reliable by itself in all tasks and must be separately evaluated when used.

Table 1. Datasets used in this study. *#Ann* refers to the number of annotators per item, while the **#SDDs** refers to the number of SDDs (e.g., age, gender, education). *Guid.* describers whether annotation guidelines are provided in the publication introducing the dataset (including documentation and supporting material) and the last two columns denote whether the datasets were usable for analysis of the original (human) annotations and LLM annotation respectively.

| Dataset | #Items | #Ann | #SDDs | Text Source | Task | Guid. | Human | LLM |
|---|---|---|---|---|---|---|---|---|
| Kumar [35] | 1000 | 5 | 10 | Twitter, Reddit, 4chan | Toxicity | Yes | No | Yes |
| Sap [60] | 585 | 5–6 | 3 | Twitter | Hate Speech | No | Yes | Yes |
| DICES-350 [4] | 350 | 123 | 4 | LLM responses | Hate Speech | No | Yes | No |
| DICES-990 [4] | 990 | 70–76 | 4 | LLM responses | Hate Speech | No | Yes | No |
| POPQUORN [52] | 1500 | 7–9 | 5 | Reddit | Hate Speech | Yes | Yes | Yes |

## 3 Datasets

Finding appropriate subjective datasets to test LLMs on is difficult for many reasons: (1) very few include annotator metadata [72], (2) most include only three annotators per item (e.g., [13, 36, 47, 65, 66]), (3) many datasets are sampled from the same source, negatively impacting the breadth of possible generalizable findings [70], and (4) annotation guidelines are frequently not included [70], complicating replication. Additionally, meta-studies on subjective datasets [53, 70] do not give special importance to the sparsity of annotations per item, which is an important consideration.[3]

We use five datasets from current NLP research that contain SDD information at the level of individual annotators:

- The *Kumar* dataset [35] which contains annotations for Toxicity detection.
- The *Sap* "Breadth of Posts" dataset with the "*toyou*" labels [60] for Hate Speech (HS) detection.
- The *DICES-350* (*Q3_bias_targeting_inherited_attributes* label) and *DICES-990* (*Q3_bias_incites_hatred* label) datasets [4], which pertain to LLM response safety. Our choice of labels allows us to use them as HS datasets, capturing conceptually similar SDDs despite the lack of shared labels between them. There is little-to-not information around these labels, since neither a description nor annotation guidelines are provided.
- The POPQUORN [52] dataset using the *"offensiveness"* label for HS detection.[4]

A brief overview of the dataset characteristics is provided in Table 1. The Kumar dataset is notable for its large scale, featuring over 100,000 annotated items and 10 distinct SDDs.[5] In contrast, the DICES datasets have fewer annotated items but involve more than an order of magnitude more annotators per item than the others. An exploratory analysis of these datasets is available in App. B. Note that not all datasets can be used for all stages of our study; the DICES datasets do not include enough reproducibility information to re-annotate using LLMs, while the Kumar dataset does not include enough human annotations for any methodology to find ASD (Table 2).

## 4 Empirically Testing Proposed Methods

### 4.1 On Real Data

We first test prior methods of attributing ASD to our real-life, human-annotated datasets. Through this comparison we aim to both explore the datasets themselves by finding SDDs on which multiple methods attribute ASD and as a sanity-check on the methods themselves.

[3] §6 shows that the real bottleneck in many NLP datasets is not the amount of recorded SDDs nor the amount of items, but rather the amount of information (number of annotators) per item.
[4] While "politeness" labels are included, they may not directly translate to toxicity; comments can simultaneously be toxic and phrased politely. Indeed, toxicity is not mentioned in the methodology of the original POPQUORN dataset and is considered a separate NLP task in other work [41, 80].
[5] In order to allow for a full comparison between human and LLM annotations, we use a sample of 1,000 items from the Kumar dataset (§5.3).

Table 2. Statistical significance of each SDD feature per method. ✓: significant at $p < 0.05$ after Bonferroni correction across the features of the dataset; ×: not significant.

| | DICES-350 | | | | DICES-990 | | | | Sap | | | Kumar | | | | | | | | | | | | | POPQUORN | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SDD | Age | Education | Gender | Race | Age | Education | Gender | Race | Age | Ethnicity | Gender | Age | Education | Ethnicity | Gender | Has Been Targeted | Is Parent | Is Transgender | Political Affiliation | Religion Important | Seen Toxicity | Sexual Orientation | Technology Impact | Toxicity Problem | Age | Education | Gender | Occupation | Race |
| apunim (ours) | × | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × | × | × | × | × | × | × | × | ✓ | × | × | ✓ |
| Krippendorff $\Delta\alpha$ | × | × | ✓ | ✓ | × | ✓ | ✓ | × | × | × | ✓ | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × |
| AU | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × | × |
| $\chi^2$ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Methods Considered.* We compare the effectiveness of our method (introduced in §5), the AU framework proposed by Pavlopoulos and Likas [51], the $\chi^2$ method presented in Akhtar et al. [2], GMM clustering used in Checco et al. [10], and a baseline using IAA ($\Delta\alpha$ Krippendorff) on simulated distributions. In this case, the GMM method is not run, as it requires the exact same annotators to annotate the items in each dataset, which is not the case for any of our datasets.[6]

*Results.* Table 2 shows whether each of the methods attributes ASD and on which SDDs. We observe that: (1) Two of the proposed methodologies do not yield any meaningful results. The AU method fails to find any ASD attribution pattern in *any* of the SDDs in any of the datasets, while the $\chi^2$ method reports that ASD can be attributed *all-but-one* SDD. (2) In contrast, both IAA and our own method report some shared findings across the DICES-350, DICES-990 and Sap datasets, while neither reports any attributable SDDs in Kumar. In fact, we observe that attributable SDDs (namely *Education, Race, Gender*) are commonly shared between datasets. (3) Unlike IAA, own method finds that the POPQUORN dataset does have SDDs on which ASD can be attributed. The fact that *three of the four currently proposed ASD attribution metrics are not usable* in any of our datasets is troubling and requires investigation.

### 4.2 Diagnosing Proposed Methods With Synthetic Data

*Generating ASD.* In both real-life human and LLM annotation we lack a ground truth for ASD. This limitation can be circumvented by constructing annotation distributions by hand, since we can know which SDD explains ASD and can test whether it is correctly attributed by each method. Additionally, by constructing these distributions we can test these methods in specific annotation scenarios reflected by our choice of parameters. We define an annotation $y_{ij}$ by annotator $i$ for item $j$ as $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$, where:

- $b_j$ represents a baseline rating around which all annotations for an item $j$ are clustered (e.g., $b_j = 5$ for an obviously toxic comment on a toxicity scale of 1–5).
- $g_i \sim \text{Uniform}(0, \frac{1}{2})$ denotes group identity. We use sampling to represent each annotator internalizing group identity differently.

[6]Regardless of empirical results, this limitation makes the GMM method extremely restrictive in realistic annotation settings.

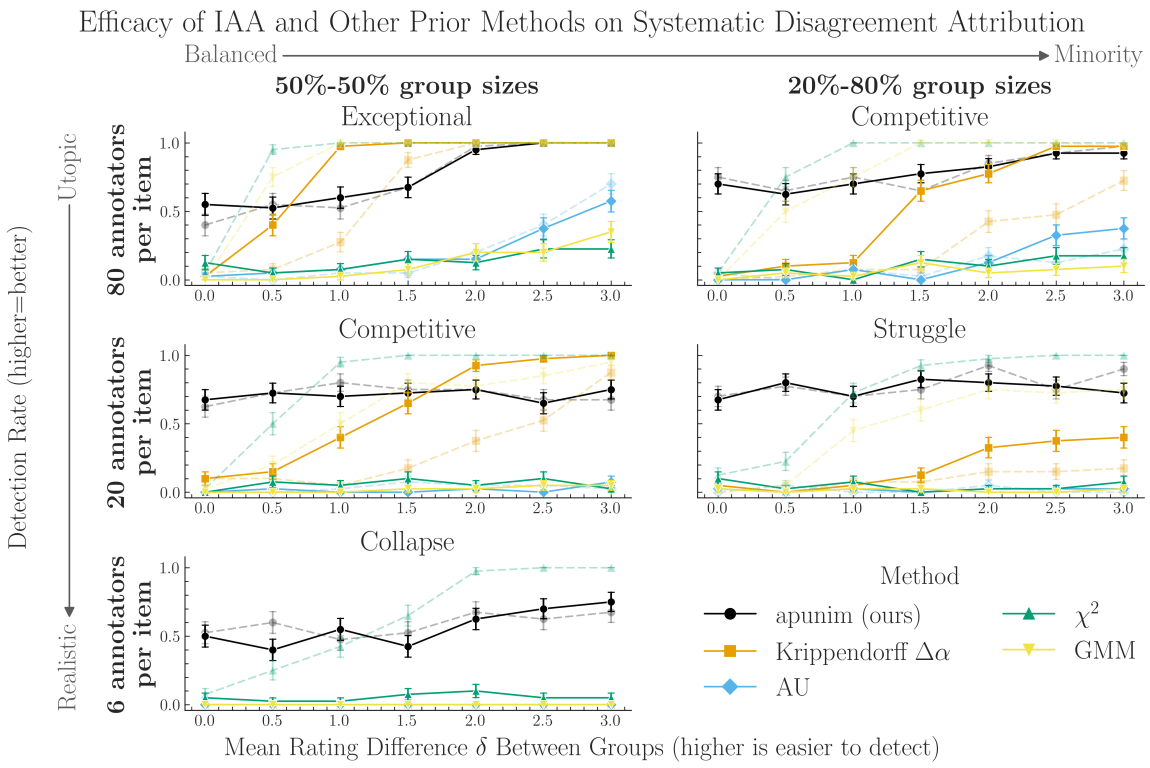



Fig. 2. ASD attribution quality on synthetic distributions. The x-axis represents the gap in opinion between groups ($\delta$ parameter used in $p_j$). Higher is easier to detect. The y-axis shows the accuracy of each method when attempting to recover the (known) SDD causing ASD. We find that both IAA and other proposed methods struggle when presented with realistic annotation scenarios, while our method remains consistent. Dashed lines show the results of a simplified experimental setup, where each group always prefers higher ratings than the other in all items and bars indicate standard error.

- $p_j$ represents real-life group polarization. We use two different formulations: (1) **Unidirectional**: $p_j \sim \text{Uniform}(0, \delta)$. The group always favors higher ratings across all items, though its magnitude varies per item. (2) **Bidirectional**: $p_j \sim \text{Uniform}(-\delta, +\delta)$. Annotators of the same group use systematically higher ratings in some items and lower in others. See Fig. 3 for an example.
- $\epsilon_{ij} \sim N(0, 1.8)$ represents general disagreement.

We use bootstrapping to determine whether a score is statistically significant and counts as a detection. For each experiment, we create 200 items across 40 independent runs, used to calculate standard error. Our hypothetical SDD features two groups, although ablations for $k = \{3, 5, 10\}$ groups per SDD are shown in Fig 10 in App. E.

*Sensitivity to per-Item Annotation Counts.* Fig. 2 shows the performance of each method on our synthetic annotations. IAA (orange squares) performs exceptionally in the case of eighty annotators per item even when differences in opinion (y-axis) are small (top-left). IAA however requires a larger difference in opinion to detect ASD when twenty annotators are available (middle-left) and collapses when only six annotators are available per item (bottom left). IAA is also sensitive to the size of the compared groups; detection performance degrades significantly when we compare a minority group (16 vs. 64 annotators; plots on the right). On the other hand, the AU method (blue diamonds) only detects

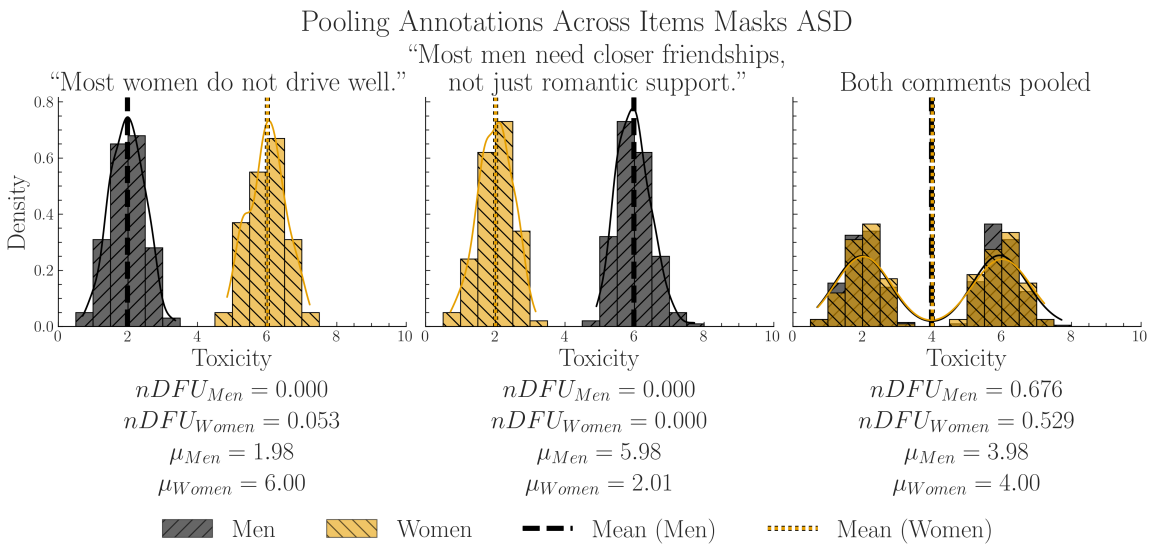



Fig. 3. Synthetic experiment showing two comments with annotators disagreeing about which one is toxic. Instead of comparing two unimodal distributions (comments viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions during aggregation, obscuring the source of the polarization (explained by gender, but not shown). This is true both for means-effects models, and nDFU comparisons between groups.

polarization when eighty annotators per item are available and differences in opinion are quantitatively large (top-left), but is resistant to minority group scenarios (top-right). This result is expected; AU utilizes clustering and is therefore largely invariant to the sizes of the opinion clusters. However, its absolute definition of ASD attribution (§2) renders it sensitive to noisy opinion histograms, which are common when few annotators per item exist.

*Inter-Item Annotation Aggregation.* The $\chi^2$ and GMM methods (green upright and yellow reverse triangles respectively) generally fail at attributing ASD to any group in the bidirectional polarization scenario (solid lines). However, when defining polarization unidirectionally (dashed, faded lines), they perform exceptionally. This observation suggests that these metrics assume that across items, ratings of group A have a different average than group B. This assumption is valid for the unidirectional polarization scenario, but not the bidirectional one.[7] Specifically, $\chi^2$ pools all annotations into a single contingency table, while GMM clusters annotators by their mean rating across items. Fig. 3 shows a synthetic scenario where both approaches would fail, as bidirectional polarization effects cancel out in the aggregate, making the two groups appear statistically indistinguishable.

*Results.* Overall, our results suggest that *current metrics for ASD attribution are sensitive to one or more conditions that may occur in realistic annotation setups* for subjective NLP datasets. IAA is sensitive to minority groups, which are the subject of many content moderation tasks, but are not evenly represented in most datasets (see Tables 9,10,11 in App. B) and on low annotator counts per item (also common, see Table 1). The latter is also required by AU.

Finding 1. *Current methodologies revealing ASD generally fail in realistic annotation conditions. One of the reasons is that disagreements may conflict across items, which may compromise cross-item annotation aggregation.*

[7]We control for our assumption of a non-fixed $g_i$ group polarization effect via the ablation shown in Fig. 11 in App. E.

## 5 Comparing Human and LLM Polarization

### 5.1 Quantifying Attribution to Annotator Dimensions

We estimate SD using nDFU (see Eq. 2). Specifically, we measure the SD of each group $g$ for an item $c$ as $P_g(c) = nDFU(A_g(c))$, where $A_g(c)$ is the set of annotations for item $c$ created by annotators belonging to a group $g$. We compare this quantity to the "apriori" polarization present in the item $c$, which is defined as the *expected* (average) unimodality of theoretical groups $\mathrm{P}_{apr}(c) = \frac{1}{k}\sum_1^k \mathrm{nDFU}(A_r(c))$, where $A_r$ is a set selected from a random partition of the original annotation set and $k$ is a hyperparameter (higher $k$ yields more accurate estimation, at a greater computational cost). Intuitively, a high $\mathrm{P}_{apr}(c)$ score means that much of the SD is likely not driven by any single group.

Since the $\mathrm{P}_{apr}(c)$ and $P_g(c)$ quantities represent very noisy estimates due to the lack of annotators per item, a direct comparison would not be reliable. We therefore exploit the Law of Large Numbers to compute the statistic on the dataset level and average both quantities for all items in the dataset $D$. Normalizing this difference yields the measure described in Eq. 1, which we call *apunim*:

$$apunim(D, g) = \frac{\mathbb{E}_{c\in D}\left[\mathrm{P}_{apr}(c)\right] - \mathbb{E}_{c\in D}\left[P_g(c)\right]}{1 - \mathbb{E}_{c\in D}\left[\mathrm{P}_{apr}(c)\right]} \tag{2}$$

The *sign* of apunim carries information agreement-based attribution does not:

- $apunim(g) \approx 0$: The group $g$ does not meaningfully disagree with the rest of the annotators.
- $apunim(g) > 0$: The group $g$ has fundamental disagreements with the rest of the annotators.
- $apunim(g) < 0$: The annotators of group $g$ are polarized between themselves.

Apunim thus measures the degree to which ASD can be attributed to an annotator group in a set of items. As demonstrated in Fig. 2, our measure is robust to few annotators per item and imbalanced (minority) group sizes. Furthermore, unlike some prior methods [2, 10], apunim allows us to attribute ASD to a specific group, not just a SDD (e.g., only "men" instead of "gender" as a whole), allowing fine-grained analysis. We also include a perturbation-based p-value estimation which we describe in App. C, which also contains a discussion of theoretical design decisions, limitations and computational details.

### 5.2 How Are Humans Polarized?

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDDs can explain the ASD shown in the five datasets (§3). Statistically significant results are presented in Table 3.[8][9] Note that none of the SDDs in the Kumar dataset cam show statistically significant values, but this is expected, since our methodology requires at least six annotators per item to function. See App. C for reproducibility details on choice of parameters and execution times.

We find that *ASD is consistently attributed to* ***ethnicity*** in DICES-990, DICES-350 and POPQUORN (Table 3), confirming prior work suggesting that it is a systematic source of annotator disagreement in HS tasks [24]. Specifically, in DICES-990, all four ethnic groups are significant, with the strongest value for AA annotators (0.0995), followed by White (−0.0649), Latino (−0.0631) and Other (0.0340). In DICES-350, AA (−0.0366) and Asian (0.0445) annotators are significant, and in POPQUORN, AA annotators (−0.1572). Additionally, in DICES-990, Sap and POPQUORN, ASD is

[8]Full results, including non-significant values are available in Tables 24,25,26,27 and 28 in App. G.
[9]Fig. 13 in App. E shows that our results are *robust to small annotator counts*, corroborating the synthetic experiments in Fig. 2.

Table 3. Statistically significant ($p < 0.05$) apunim results across the Sap and DICES datasets. Gray rows show `Ethnicity` and `Gender` groups. "Support" refers to number of annotations. Consult App. G for the full results.

| Group | Apunim | Support |
|---|---|---|
| **DICES-990** | | |
| Ethnicity=AfricanAm | 0.0995 | 5148 |
| Ethnicity=Latino | -0.0631 | 5877 |
| Ethnicity=Other | 0.0340 | 21669 |
| Ethnicity=White | -0.0649 | 10166 |
| Gender=Man | 0.0359 | 28817 |
| Gender=Woman | -0.0271 | 29990 |
| Age=GenZ | 0.0469 | 12214 |
| Age=Millennial | -0.0068 | 34433 |
| Age=GenX+ | -0.0282 | 12699 |
| Education=College | 0.0063 | 52727 |
| Education=High school | -0.0500 | 6619 |
| **DICES-350** | | |
| Ethnicity=AfricanAm | -0.0366 | 7627 |
| Ethnicity=Asian | 0.0445 | 6838 |
| Education=Other | -0.0230 | 1841 |
| **Sap** | | |
| Gender=Man | -0.1508 | 1254 |
| Gender=Woman | 0.2613 | 693 |
| Age=GenX+ | -0.1670 | 223 |
| **POPQUORN** | | |
| Gender=Man | 0.0081 | 2720 |
| Race=Black | -0.1572 | 222 |
| Education=College | 0.0496 | 2388 |
| Education=Graduate | -0.0808 | 468 |

Table 4. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under the default prompt, per dataset and pooled over all datasets (All). $n$ is the number of comments. Excluding the following models: llama8b, olmo7b, qwen7b.

| | Sap et al. 2022 | | Kumar et al. 2021 | | POPQUORN | | All | |
|---|---|---|---|---|---|---|---|---|
| | n | Mean (SE) | n | Mean (SE) | n | Mean (SE) | n | Mean (SE) |
| Model | | | | | | | | |
| Human | 585 | 0.38 (0.01) | 1000 | 0.42 (0.01) | 1500 | 0.31 (0.01) | 3085 | 0.36 (0.00) |
| olmo32b | 585 | 0.24 (0.01) | 1000 | 0.51 (0.01) | 1500 | 0.46 (0.01) | 3085 | 0.43 (0.01) |
| qwen32b | 585 | 0.18 (0.01) | 1000 | 0.19 (0.01) | 1500 | 0.25 (0.01) | 3085 | 0.22 (0.01) |
| mistral7b | 585 | 0.21 (0.01) | 1000 | 0.37 (0.01) | 1500 | 0.40 (0.01) | 3085 | 0.35 (0.01) |
| mistral24b | 585 | 0.25 (0.01) | 1000 | 0.25 (0.01) | 1500 | 0.33 (0.01) | 3085 | 0.29 (0.00) |
| gemma7b | 585 | 0.19 (0.01) | 1000 | 0.22 (0.01) | 1500 | 0.15 (0.01) | 3085 | 0.18 (0.00) |
| gemma31b | 585 | 0.04 (0.01) | 1000 | 0.09 (0.01) | — | — | 1585 | 0.07 (0.00) |

attributed to **gender**, which is consistent with previous findings [7, 74]. The effect is largest in Sap (Woman 0.2613, Man −0.1508), smaller in DICES-990 (Man 0.0359, Woman −0.0271), and smallest in POPQUORN (Man 0.0081).[10]

Finding 2. *We confirm prior literature by observing that ASD is consistently attributed to gender (DICES-990, Sap, POQUORN) and ethnicity (DICES-990, DICES-350, POPQUORN) in subjective human-annotated tasks.*

### 5.3 How Are LLMs Polarized?

In order to answer RQ. 3 we investigate the following questions: (1) Do LLMs exhibit any SD by default? (2) Does SD increase when using persona-prompting? (3) Can LLMs exhibit any ASD? (4) Does LLM ASD correspond to human ASD? These questions are linked sequentially: answering each one is a necessary step for answering the next.

[10]Note that AAs show differently signed apunim values in the two DICES datasets (0.0995 in DICES-990 vs. −0.0366 in DICES-350), which may be attributed to these datasets using different labels (see §3) and because different ethnic groups are considered in each dataset. Thus, while detection is constant across the two datasets, attribution is not. Additionally, in DICES-990 and POPQUORN men are a source of ASD (positive values: 0.0359 and 0.0081), while women disagree among themselves in DICES-990 (negative value: −0.0271), but the opposite is true for Sap (Man −0.1508, Woman 0.2613). Unfortunately we can not diagnose these discrepancies, since neither DICES nor Sap include annotation guidelines (Table 1).

Table 5. Cohen's d between the default and each adversarial instruction prompt, computed on the per-comment nDFU. Results are given per model, for each dataset separately and as the unweighted mean over datasets (All). Comments per model: Sap et al. 2022: 585; Kumar et al. 2021: 1000; POPQUORN: 1500 (gemma31b: 0). Excluding the following models: llama8b, olmo7b, qwen7b.

| | All | | | | Sap et al. 2022 | | | | Kumar et al. 2021 | | | | POPQUORN | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct |
| olmo32b | −0.04 | 0.02 | 0.07 | −0.10 | 0.09 | 0.06 | 0.26 | 0.06 | 0.14 | −0.13 | −0.29 | −0.31 | −0.35 | 0.14 | 0.25 | −0.05 |
| qwen32b | 0.19 | 0.16 | 0.18 | 0.13 | 0.32 | 0.13 | −0.04 | 0.09 | 0.17 | 0.21 | 0.51 | 0.31 | 0.08 | 0.13 | 0.07 | −0.01 |
| mistral7b | 0.13 | 0.03 | 0.10 | −0.20 | 0.21 | 0.10 | 0.08 | −0.21 | 0.07 | −0.01 | −0.10 | −0.05 | 0.11 | 0.01 | 0.31 | −0.35 |
| mistral24b | 0.59 | 0.25 | −0.32 | −0.03 | 0.64 | 0.04 | −0.88 | 0.17 | 0.98 | 0.97 | 0.57 | −0.25 | 0.17 | −0.26 | −0.66 | −0.00 |
| gemma7b | 0.16 | 0.27 | 0.38 | 0.11 | 0.18 | 0.48 | 0.70 | −0.08 | 0.10 | −0.08 | −0.40 | −0.07 | 0.20 | 0.41 | 0.85 | 0.49 |
| gemma31b | 0.36 | 0.21 | 0.02 | 0.07 | 0.36 | 0.21 | 0.02 | 0.07 | – | – | – | – | – | – | – | – |

Table 6. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under the default prompt, per dataset; $n$ is the number of dimensions. A dimension counts as significant if at least one of its groups is. †: ratio over fewer dimensions, as the model has results for only some of them. Excluding the following models: llama8b, olmo7b, qwen7b.

| Dataset | $n$ | Human | olmo32b | qwen32b | mistral7b | mistral24b | gemma7b | gemma31b |
|---|---|---|---|---|---|---|---|---|
| Sap | 3 | 66% | 33% | 66% | 66% | 100% | 33% | 66% |
| Kumar | 13 | 0% | 46% | 53% | 46% | 46% | 38% | 30% |
| POP | 5 | 60% | 100% | 80% | 100% | 100% | 60% | — |

Table 7. Statistically significant ($p < 0.05$) SDDs of each LLM compared with the human annotations, per dataset. A dimension counts as significant if at least one of its groups is. Each cell shows one symbol per prompt, in the order default, stereotype, persona, single, direct: ✓ both significant, with the same sign for every group that is significant in both; ✓ neither significant; ± both significant, but the sign of at least one group's effect differs; × significant for only one of the two; blank: no LLM result. Excluding the following models: llama8b, olmo7b, qwen7b.

| | | olmo32b | | | | | qwen32b | | | | | mistral7b | | | | | mistral24b | | | | | gemma7b | | | | | gemma31b | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ds | SDD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Sap | Age | ✓ | ✓ | ✓ | × | ✓ | ± | ± | ± | ± | ± | ± | ✓ | ± | ✓ | × | ± | ✓ | ± | ✓ | ± | × | × | ± | ± | × | ✓ | ✓ | ✓ | ✓ | ✓ |
| | Ethnicity | ✓ | ✓ | × | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × | × | × | ✓ | × | × | × | × | × | × | × | × | × | × | × | × | × |
| | Gender | × | × | × | ± | × | × | ✓ | × | × | × | × | ✓ | ✓ | × | ✓ | ± | ✓ | ✓ | ✓ | × | × | × | ± | ✓ | ± | × | × | × | × | × | × |
| POP | Age | × | × | × | × | × | × | × | × | × | × | × | | | ✓ | | × | × | ✓ | ✓ | × | × | × | × | × | × | | × | × | × | × |
| | Education | ✓ | ± | ± | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | | × | | ± | ✓ | × | ± | ± | ✓ | ± | ± | ± | ± | | ✓ | ✓ | × | × |
| | Gender | × | × | × | ✓ | × | ✓ | × | × | × | × | × | ✓ | | ✓ | | × | × | ✓ | ✓ | × | ✓ | × | ✓ | × | × | | × | × | × | × |
| | Occupation | ± | ✓ | ± | × | ± | ± | ± | ± | ± | ✓ | ✓ | × | | | | ± | ✓ | × | ✓ | ± | × | ± | ✓ | ✓ | ✓ | | ✓ | ✓ | × | ✓ |
| | Race | ✓ | ± | ✓ | × | ± | ✓ | ✓ | ± | ± | ± | ✓ | × | | | | ✓ | ± | × | × | ± | ✓ | ✓ | ✓ | ± | ✓ | | ± | ± | ± | ± |

*Experimental Procedure.* We ensure that our LLM annotation study is reproducible by selecting nine exclusively open-source, locally hosted models and fixing all relevant seeds (three not shown in main results because of prompt sensitivity issues[11]) using twenty annotators for the Kumar, Sap and POPQUORN datasets supplied with personas sampled from our human dataset.[12] We also do not annotate any of the DICES datasets since the LLM annotations are inconsistent (< 0.3 Cohen's $\kappa$ for all models but one when paraphrasing instructions; see Table 17 in App. F), an issue

[11]An ablation study where we rephrase the main annotation prompt three revealed that three models were especially sensitive to such changes (Table 17 in App. F), which led us to exclude those models from the main analysis. See Tables 31,32 for the full results in App. G. Additionally, a second ablation shows that we achieve 100% consistency across all models given the same prompt and input (which is generally not a given [8]).
[12]A compromise between practical computation costs and enough annotation density to facilitate exploration.

we can not fix as their original annotation guidelines are not public (Table 1). See App. F for the ablation experimental procedure, full results, model details, dataset selection, exploratory analysis, and comparison to human annotations.

We also test four adversarial prompts that stress the importance of the LLM's persona in different ways (App. H). *Persona-focused* asks the model to rate as its persona would perceive the comment. *Direct* makes the persona the only criterion. *Single* has the model pick the persona's most influential trait and rate based on that trait alone. *Stereotype* tells the model to weigh the persona's sensitivities over those of an average person. These prompts do not represent annotation strategies; they only attempt to steer the models towards persona-focused answers.

*Do LLMs Exhibit SD?* **Yes**. Table 4 shows that the nDFU values exhibited by LLMs when under standard persona prompting can even surpass those of humans. However, manipulating the amount of ASD through instruction prompting is not reliable. While an ANOVA[13] test shows that the choice of instruction prompt does indeed affect nDFU scores across most models, measuring Cohen's $d$ (Table 5) shows that the effect is generally small ($|d| \leq 0.3$) and inconsistent, with the sign and magnitude of the effect depending on the model and dataset.[14]

*Can LLMs exhibit any ASD?* **Yes**. Table 6 shows the percentage of SDDs where at least one of the constituent groups show statistically significant apunim results. We observe that LLM annotations frequently produce ASD.[15]

*Is LLM ASD Aligned With Humans?* **No**. Table 7 describes whether an SDD produces statistically significant results using apunim and compares it with the human apunim results (Table 3). We observe that alignment on ASD between humans and LLMs is rarely achieved. Specifically, we observe: (1) Low Precision: LLMs tend to produce polarized annotations for groups whose human annotators are not. (2) Low Recall: LLMs under-detect where humans find real effects, especially in Sap and POP Gender. (3) Where ASD matches, the direction does not (POP; Race, Education). (4) Where replication is achieved, it is generally on null results (e.g., *Mistral-7B*). We also find no connection between model size and family on the alignment of ASD.

Finding 3. *While persona-prompted LLMs can produce polarized annotations, they can not replicate human polarization patterns, even when we attempt to align them using prompting.*

### 5.4 What Does This Mean for LLM Annotation?

Our results may indicate that LLMs do not intrinsically capture the differences in opinion between human groups, or miscalculate the degree that each SDD characteristic influences human opinion. Our study thus joins a growing number of work questioning the capabilities of persona-prompting [40, 42] and the limitations of LLMs in creating data for subjective social science tasks [32, 57, 76]. While persona-prompting meaningfully increases the variance (here, disagreement) of LLM annotations, as argued in prior work [22], the direction of this variance is inconsistent.

This limitation would indicate the need for careful evaluation and calibration of LLM outputs on sensitive applications such as content moderation [12] and facilitation [33], which so far assume that LLMs can be used as autonomous agents. It also may indicate that research focusing on improving annotation diversity through persona prompting [21, 22] may lead to meaningless or miscalibrated data when applied outside of specific annotation scenarios with limited scope. Finally, there is a general concern that LLMs training on biased LLM-generated data may amplify already-existing

[13]We use a parametric test (ANOVA) in our analysis due to the abundance of observations ($N \gg 100$) as well as a multiple hypothesis p-value correction (Holm-Bonferroni [29]).
[14]See Table 30 in App. G for the actual mean nDFU values by prompt.
[15]Table 33 includes all LLMs and Table 34 distinguishes between different adversarial prompts in App. G.

biases [56, 62], which is especially important with the possible proliferation of persona-prompted mass-annotated synthetic datasets amplifying miscalibrated points of view across model iterations.

## 6 The Ideal Subjective Dataset

### 6.1 Categorizing Attributes of Annotation Datasets

As discussed in §3, there is a severe lack of datasets containing annotator metadata for subjective tasks. Given that the creation of these datasets is difficult, expensive and time-consuming, it is important to acknowledge and prioritize certain properties that are necessary for analyzing subjectivity across annotator groups. Since there is to our knowledge no taxonomy on the shape of annotated datasets, we distinguish three main properties:

- Data Coverage (DC): the number of *items* in a dataset. Due to the focus on training and benchmarking datasets in NLP, this property is usually the one with the most focus and importance [57].
- Group Coverage (GC): the number of *annotators* and their distribution across SDDs. A large annotator pool is required to get sufficient coverage on the examined groups and allows both higher fidelity of each SDD and more such SDDs to be examined. To our knowledge, there are no studies explicitly discussing this property.
- Annotation Density (AD): the number of *annotators per item*. This property is often neglected in practice, with many datasets [13, 36, 47, 65, 66] assuming "gold labels" can be inferred by three annotators. Similarly to GC, we can not find any discussion of this property, with some studies recommending more than three annotators but still defaulting to majority-voting [6].

While the field of subjective NLP tasks has thankfully moved away from single-annotated items and majority voting, the lack of discussion on GC and AD suggests that the shift has been incremental rather than substantive. A clear signal can not be always obtained when only a few cursory SDDs are reported instead of a comprehensive list of both innate characteristics and personal opinions. [16] Likewise, six annotators per item may not be adequate, even when a comprehensive list of SDDs is known, as we demonstrate in the next section.

### 6.2 The Case for Annotation Density

*Unnatributtable Systematic Disagreement (USD).* Some theoretical models such as AU make the assumption that all ASD is attributable to some annotator SDD, which has not been empirically verified in real-world annotation settings. We define this phenomenon "Unnatributtable Systematic Disagreement (USD)" for a single item $c$ as the minimum nDFU exhibited by any arbitrary dimension:[17] $\text{Pol}_{inh}(c) = \min_{S \in G_c} \{\text{nDFU}(S)\}$, where $G_c$ is the collection of all subsets $S$ of annotations for an item $c$ such that $|S| \geq 3$ and $S_j$ is a random sample drawn from $G_c$ for $j = 1, \ldots, k$. Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups, which is computationally intractable for large annotator counts. We can therefore use Monte Carlo sampling by drawing $k$ random partitions with random sizes: $\text{Pol}_{inh}^{MC}(c) = \min_{j=1}^{k} \{\text{nDFU}(S_j)\}$, which serves as an upper bound (as it considers a strict subset of $\text{Pol}_{inh}(c)$).

*Does It Exist?* We compute the USD for each item across the five datasets. For Sap and Kumar, we use the exhaustive formulation, whereas for the two DICES datasets we rely on the MC approximation with $k = 1000$. We also that USD typically exists only on smaller annotation counts, as evidenced in Fig 4; we observe USD for the Sap/Kumar datasets

[16] The Kumar dataset is an excellent example of a comprehensive subjective dataset (§3.

[17] This dimension need not be known in the dataset, as we consider all possible dimensions in our sample, both known and latent. This formulation also covers all possible combinations of such groups.

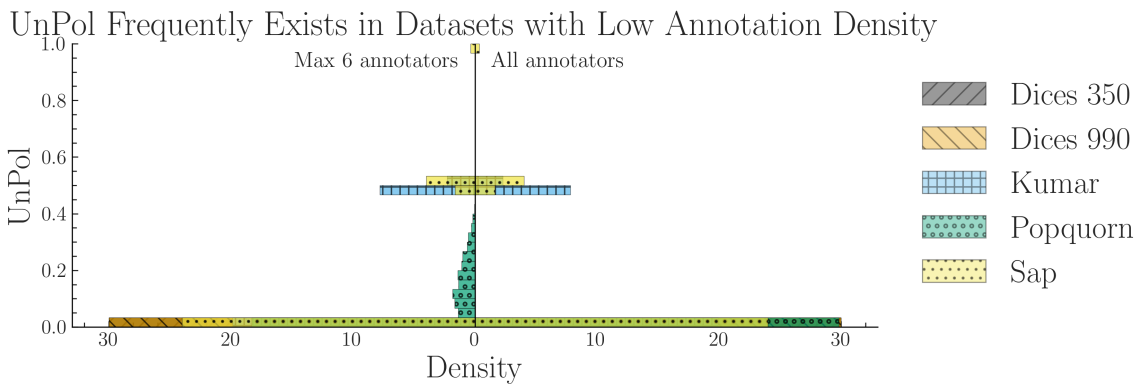



Fig. 4. Unnatributtable Systematic Disagreement (USD) across our datasets (§3). Contrary to assumptions made in prior work, Unnatributtable Systematic Disagreement (USD) is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts (see Table 1). The left side of the figure shows the same datasets, where annotations are subsampled 100 times down to six annotators and their Unnatributtable Systematic Disagreement (USD) is averaged.

(which are only annotated by six and five annotators respectively, see Table 1) and on POPQUORN only when it is subsampled down to six annotators per item. This pattern is also validated on both synthetic and LLM annotated data, as shown by Fig. 8 in App. D.[18][19]

Finding 4. *Items may exhibit a degree of USD that cannot be attributed to any known or latent annotator grouping.*

### 6.3 How Should We Design Subjective Datasets?

While DC may be convenient for training classifiers, relying on it without rigorous evaluation of SD can lead to misconfigured or biased models [59], evaluation which is impossible without the necessary DC). However, it comes at the expense of the other two (arguably more important) qualities. Generally, the *easiest-to-use resources for training LLMs and classifiers may be the riskiest*, as meaningful, SD cannot be reliably quantified (see the Kumar dataset in this study). We therefore recommend prioritizing dataset creation by first ensuring sufficient AD, then achieving necessary GC across underrepresented groups, and finally allowing budgetary constraints to determine DC.

While this tension between the need for massive data volumes in AI training and the requirement of responsible social science to use deeply annotated, subjective datasets is unlikely to be resolved, a practical step is providing full annotation guidelines in both cases. If the annotation target and definitions are unknown, comparing results across different datasets becomes extremely difficult. As shown in §5.3, without clear guidelines, differences in polarization patterns remain obscured when analyzing data from multiple studies.

## Conclusion

In this study, we showed that current disagreement attribution measures tend to underperform in realistic annotation settings through a large scale empirical study. We designed a new measure which quantifies attribution of Systematic Disagreement (SD) to annotator SDDs, taking into account the limitations of real-life annotation datasets. Using this

[18] See App. D for hypotheses on what creates USD. Also, examples of items with high USD can be found in Table 12 in App. D.

[19] It is unclear why the DICES datasets do not produce any USD even when repeatedly subsampled. We hypothesize than either $k$ is not large enough (which would make the computation prohibitively expensive), or the annotations themselves may not be as polarized. The latter hypothesis is plausible, as both DICES datasets do feature less polarized comments (see Fig. 5 in App. B) and also because their text targets (LLM responses) may not be as ambiguous as human comments.

measure, we identified shared polarization patterns among gender and ethnicity groups across five subjective NLP datasets. Additionally, we found that LLM annotators can create polarized annotations, but do not reproduce human polarization patterns, even when using prompt engineering. Our research is limited due to the information present in the datasets; namely, we discovered that due to low annotation counts per item, some polarization cannot be attributed to any combination of theoretical dimensions ("Unnatributtable Systematic Disagreement (USD)"). Overall, our work warns against using LLM persona prompting without rigorous, task-specific evaluation, and motivates the development of "wider" datasets with fewer but better annotated items.

## AI Use Statement

In this work we used proprietary models such as Anthropic's Claude Sonnet 5 and OpenAI's GPT-5 for partially implementing the synthetic experiments in §4 and generating parts of the library, experiment and graph-creation code. The unit-tests used in our library were similarly generated by these models, although new tests were added as edge-cases and bugs were discovered. All such tasks were executed with direct author supervision, and all relevant code files (library, experiments, graphs) have been edited manually at multiple points by the authors to ensure reproducibility, accuracy and software quality. We also used Scholar Labs to find citations for targeted claims (e.g., for known issues with persona prompting), although all such citations were manually filtered by us. Finally, we used a locally hosted Gemma-4 model (specifically the `gemma-4-E4B-it-Q4_K_M.gguf` variant) for rephrasing this manuscript in order to improve readability. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

## Ethics Statement

*Dual Use Concerns.* The primary objective of this study is to identify systemic sources of ASD in subjective NLP datasets to facilitate the construction of more fair and reliable AI systems. However, given that our methodology allows for the quantitative attribution of specific annotation patterns to SDDs, we acknowledge three primary risks inherent in this research. (1) *Discriminatory Profiling*: If the measure is applied outside of a dataset quality analysis, its ability to correlate specific opinion patterns with annotator groups could be misused for surveillance or unjust profiling. (2) *Automated Bias Amplification*: The diagnostic patterns of disagreement we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. (3) The knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict. We stress that our measure functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system. In any case, despite the risks, we believe that the benefits of this research outweigh the risks outlined above.

*Data Sourcing and Privacy.* All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors, nor have we provided any to proprietary LLMs. Publicly accessible datasets will be provided as is, while gated-release datasets will be provided with their text contents hashed. Thus, our annotations can still be matched against the actual dataset, if it is acquired

separately by the original authors. Finally, we note that our data is limited to English posts from a U.S. cultural context and our results may not be representative for other cultures.

*Examples Used.* Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors. We have partially censored explicit slurs in Table 12 in order to both mitigate uncomfortable or triggering content, while also showing enough content for the examples to be effective.

*Prompts Used.* This study uses adversarial prompts which explicitly instruct the model to change its outputs according to their persona prompts. The purpose of these prompts is to test the LLMs under an extreme case of prompt engineering and are not to be used for generalized usage. The use of these prompts may degrade model performance, increase the risk of stereotyping and produce misleading results.

## A Acronyms

### Acronyms

**AA** African American
**AD** Annotation Density
**ASD** Attributable Systematic Disagrement
**AU** A Posteriori Unimodality
**DC** Data Coverage
**GC** Group Coverage
**GMM** Gaussian Mixture Model
**GPLv3** GNU General Public Licence v3
**HS** Hate Speech
**IAA** Inter-Annotator Agreement
**nDFU** normalized Distance From Unimodality
**NLP** Natural Language Processing
**SD** Systematic Disagrement
**SDD** SocioDemographic Dimension
**USD** Unnatributtable Systematic Disagreement

## B Exploratory Analysis

The number of annotators per item for all datasets presented in §3 can be found in Fig. 6, and descriptive statistics can be found in Table 8. The DICES-990 dataset has exactly 123 annotations per item, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few items that vastly exceed the promised 5 number of annotators. We do not include these items, since they are very likely a data error. Tables 9,10,11 show the annotation counts for each SDD group in each dataset.

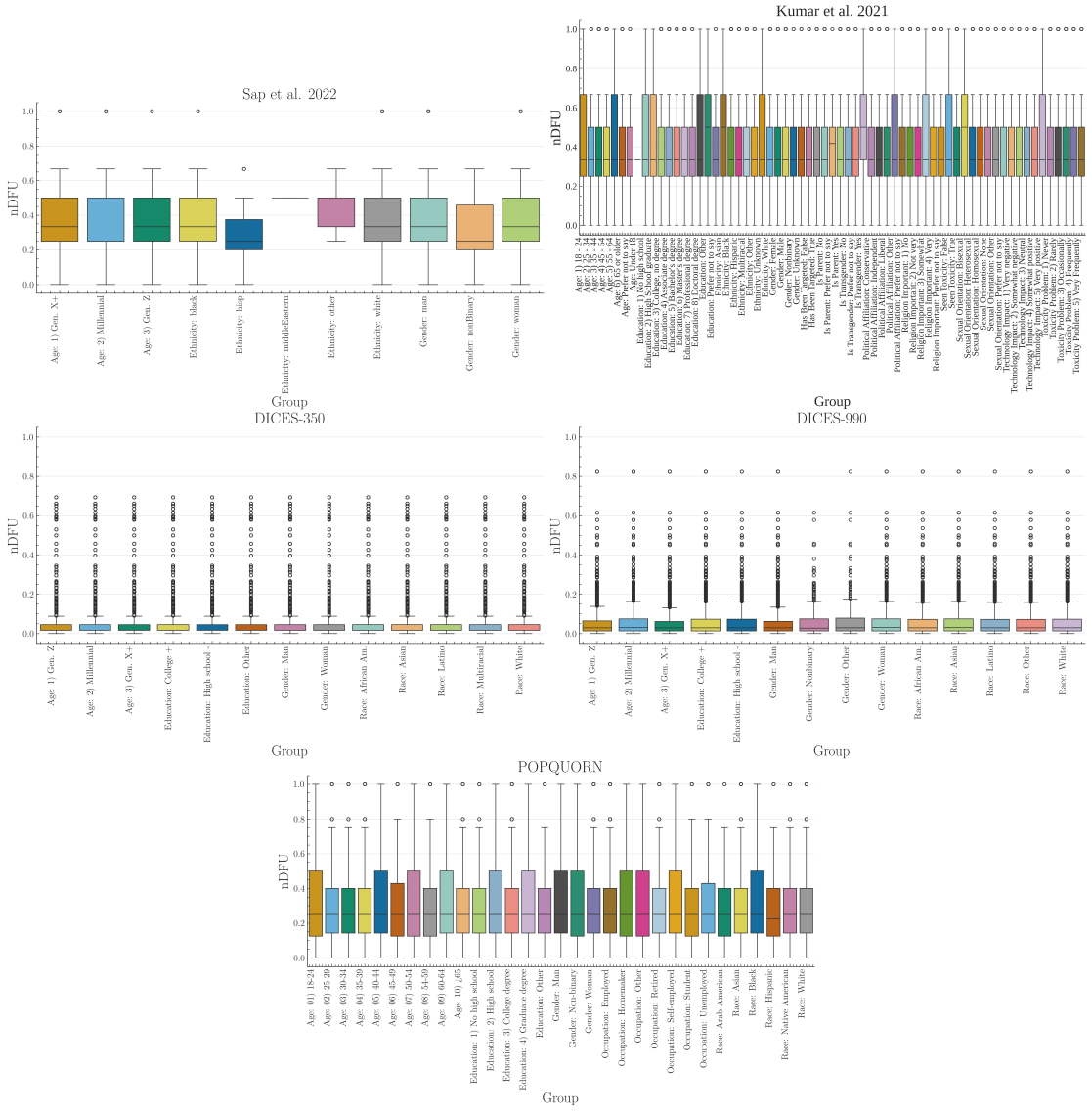



Fig. 5. Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined SDD.

## C Apunim Details

### C.1 Design Decisions

*Defining Apriori Polarization.* In §5, we explained how our methodology needs to compare the polarization between the set of all annotations and those made by a specific group $g$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also

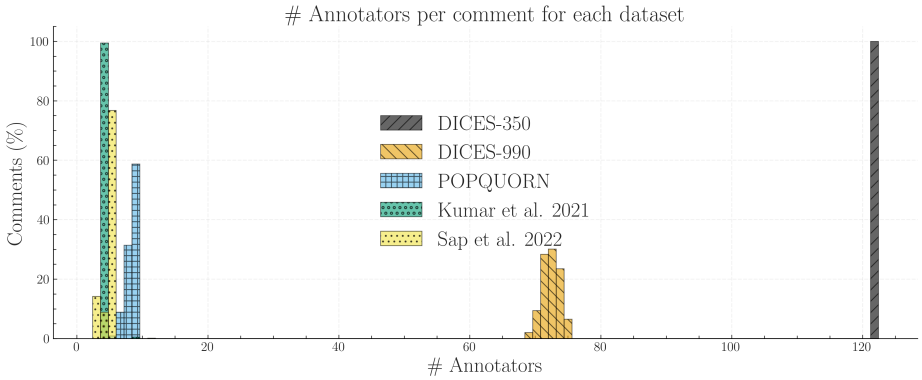



Fig. 6. We calculate the ratio of items for each dataset (y-axis) that had $x$ number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few items in the Kumar dataset exceeding this threshold (see Table 8), which is likely caused by a data error.

Table 8. Descriptive statistics for the number of annotations per comment, grouped by dataset.

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| DICES-350 | 350 | 123 | 0 | 123 | 123 | 123 | 123 | 123 |
| DICES-990 | 990 | 72.8313 | 1.1652 | 69 | 72 | 73 | 74 | 76 |
| POPQUORN | 1500 | 8.6573 | 0.9087 | 5 | 8 | 9 | 9 | 10 |
| Kumar et al. 2021 | 105910 | 5.0261 | 0.3604 | 5 | 5 | 5 | 5 | 10 |
| Sap et al. 2022 | 585 | 5.6359 | 0.7713 | 3 | 6 | 6 | 6 | 12 |

Table 9. Subgroup Counts for DICES-990 (left) and DICES-350 (right). "Support" refers to annotation count.

| Category | Subgroup | Support |
|---|---|---|
| Gender | Woman | 36050 |
| | Man | 35392 |
| | Nonbinary | 331 |
| | Other | 330 |
| Race | Other | 26394 |
| | Asian | 19885 |
| | White | 12322 |
| | Latino | 7170 |
| | African Am. | 6332 |
| Age | Gen. Z | 14899 |
| | Millennial | 41542 |
| | Gen. X+ | 15662 |
| Education | College + | 64054 |
| | High school - | 8049 |

| Category | Subgroup | Support |
|---|---|---|
| Gender | Woman | 21700 |
| | Man | 21350 |
| Race | White | 10500 |
| | African Am. | 10150 |
| | Asian | 9100 |
| | Latino | 7700 |
| | Multiracial | 5600 |
| Age | 1) Gen. Z | 19600 |
| | 2) Millennial | 12600 |
| | 3) Gen. X+ | 10850 |
| Education | College + | 26250 |
| | High school - | 14350 |
| | Other | 2450 |

would not work; if we apply the subset-vs-complement method in the example of Fig. 7, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

Table 10. Subgroup Counts for the Sap (left) and POPQUORN (right) datasets. "Support" refers to annotation count.

| Category | Subgroup | Support |
|---|---|---|
| Age | Gen. Z | 252 |
| | Millennial | 2259 |
| | Gen. X+ | 786 |
| Race | White | 2168 |
| | Black | 1091 |
| | Hisp | 32 |
| | Other | 5 |
| | MiddleEastern | 1 |
| Gender | man | 1859 |
| | woman | 1412 |
| | nonBinary | 26 |

| Category | Subgroup | Count |
|---|---|---|
| Gender | Woman | 6470 |
| | Man | 6166 |
| | Non-binary | 350 |
| Race | White | 9754 |
| | Black | 1685 |
| | Asian | 949 |
| | Native American | 299 |
| | Hispanic | 250 |
| | Arab American | 49 |
| Age | 18-24 | 1545 |
| | 25-29 | 1236 |
| | 30-34 | 1494 |
| | 35-39 | 1547 |
| | 40-44 | 1142 |
| | 45-49 | 1091 |
| | 50-54 | 947 |
| | 54-59 | 1793 |
| | 60-64 | 747 |
| | >65 | 1444 |
| Education | No high school | 237 |
| | High school | 3934 |
| | College degree | 6036 |
| | Graduate degree | 2481 |
| | Other | 298 |
| Occupation | Student | 995 |
| | Homemaker | 594 |
| | Retired | 1446 |
| | Employed | 6325 |
| | Self-employed | 1692 |
| | Unemployed | 1684 |
| | Other | 250 |

*Filtering Ineligible items.* Because *apunim* is fundamentally designed to quantify polarization, we exclude items that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude items annotated exclusively by a single group (i.e., items where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per item. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

### C.2 Statistical Estimation

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized

Table 11. Subgroup Counts for the Kumar dataset. "Support" refers to annotation count.

| Category | Subgroup | Support |
|---|---|---|
| Gender | Female | 8011 |
| | Male | 6832 |
| | Unknown | 124 |
| | Nonbinary | 93 |
| Ethnicity | White | 10699 |
| | Black | 1940 |
| | Asian | 862 |
| | Multiracial | 664 |
| | Hispanic | 418 |
| | Other | 247 |
| | Unknown | 230 |
| Age | Under 18 | 3 |
| | 18 - 24 | 1711 |
| | 25 - 34 | 5969 |
| | 35 - 44 | 3785 |
| | 45 - 54 | 1977 |
| | 55 - 64 | 1101 |
| | 65 or older | 471 |
| | N/A | 43 |
| Education | No high school | 71 |
| | High School | 1353 |
| | College | 3090 |
| | Associate's | 1676 |
| | Bachelor's | 6071 |
| | Master's | 2261 |
| | Professional | 236 |
| | Doctoral | 198 |
| | N/A | 71 |
| | Other | 32 |
| Sexual Orient. | Heterosexual | 12391 |
| | Bisexual | 1665 |
| | Homosexual | 515 |
| | N/A | 302 |
| | Other | 131 |
| Is Trans | No | 14549 |
| | Yes | 363 |
| | N/A | 148 |
| Politics | Liberal | 5997 |
| | Independent | 4143 |
| | Conservative | 3981 |
| | N/A | 609 |
| | Other | 330 |
| Is Parent | Yes | 7889 |
| | No | 7015 |
| | N/A | 156 |
| Tech Impact | Very negative | 149 |
| | Somewhat negative | 1411 |
| | Neutral | 1905 |
| | Somewhat positive | 7690 |
| | Very positive | 3905 |
| Tox Problem | Never | 869 |
| | Rarely | 2485 |
| | Occasionally | 5307 |
| | Frequently | 4441 |
| | Very Frequently | 1958 |
| Rel Important | No | 4735 |
| | Not very | 1808 |
| | Somewhat | 3548 |
| | Very | 4741 |
| | N/A | 228 |
| Seen Toxicity | True | 11505 |
| | False | 3555 |
| Targeted | False | 10741 |
| | True | 4319 |

partitions and compare the observed apunim against this distribution using a one-sample Student-$t$ test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 &: \; \bar{apunim}_{\text{rand}}(d, g) = apunim(d, g) \\ H_a &: \; \bar{apunim}_{\text{rand}}(d, g) \neq apunim(d, g) \end{aligned} \tag{3}$$

where $\bar{apunim}_{\text{rand}}(d, g)$ denotes the average apunim values obtained from $t$ randomized annotator partitions.

We account for the presence of multiple subgroups in each SDD using a multiple-hypothesis correction. Since our test is parametric (App. **??**), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per SDD group to estimate nDFUs, (3) the apunim values of the random partitions are independent and

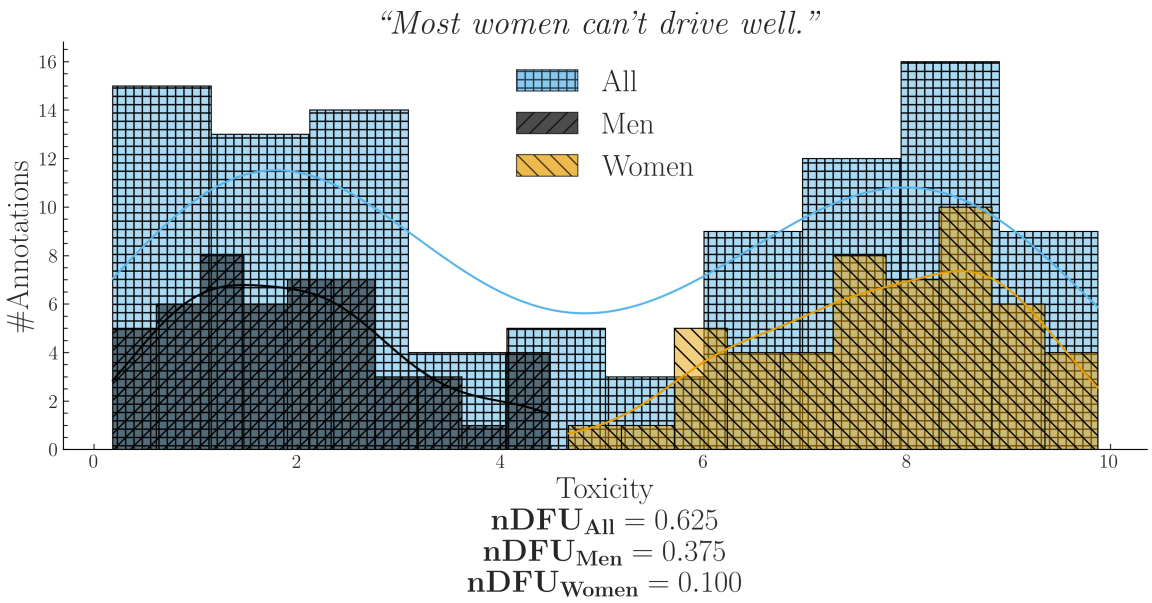



Fig. 7. Simulated experiment describing a polarizing comment where men and women agree between themselves, but disagree with the opposite gender.

identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific SDD. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holm method [29]. The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [11]).

### C.3 Software

*Installation and Usage.* We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (https://anonymous.4open.science/r/apunim-page).

We use version `1.0.6` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation and the p-value estimation (App. **??**), and set a confidence level of $\alpha = 0.05$. The relative library setup is

provided as a conda environment file to ensure reproducibility. The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (https://anonymous.4open.science/r/aposteriori-unimodality-E8C1). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

*Implementation Details.* By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple SDDs, each of which is computationally expensive. In order to lower the computational cost required, we re-use the random partitions, which are originally calculated to obtain the apriori polarization values, for the p-value calculations (App. **??**). This means that in our current implementation, the number of groups for estimating apriori polarization, and the number of apunim values used in the p-value calculation have to be the same.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of items, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in Fig. 13 required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each item, for each SDD. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each SDD subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package [61] for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered (this is not a requirement for loading the sample used in this study). The CPU-bound experiments require approximately 3 hours to run, assuming a computer capable of efficient multiprocessing.

Finally, we note that in Fig. 13, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of items in DICES-990 were annotated by more than 73 annotators—see App. B).

## D Exploring USD

*Why Does it Exist?* Unlike disagreement, USD does not provide any information, but rather represents noise or uncertainty translated in terms of polarization. In order to minimize it, we need to find its cause. Since the DICES datasets are the only ones without *any* items with USD leads us to two hypotheses:

Hypothesis 1 (Low annotator count). *A low number of annotators per item may provide our model with insufficient information.*

Hypothesis 2 (Inaccessible groups). *USD may arise from the presence of partially represented groups within the annotator pool which may go undetected if they are represented by only one or two annotators, as nDFU is undefined for these groups.*

Table 12. Sample of comments from the Sap dataset and their USD. Note that unambiguously toxic comments can be assigned with zero USD, while more nuanced cases tend to have higher values.

**Content Warning: Slurs, offensive, toxic and hateful speech.**

| Comment | USD |
|---|---|
| The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!. | 1 |
| There should be such thing as a b[redacted]ch resistance for us women | 1 |
| There's a big difference between being a b[redacted]ch and being a woman | 1 |
| i just watched a video of @joisnotonfire being an annoying h[redacted]e on skype i'm laughing | 1 |
| n[redacted]as love to commit crime | 1 |
| I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a n[redacted]a | 0.5 |
| I will become a cold ass b[redacted]ch because of my treatment | 0.5 |
| I would happily lay out a n[redacted]a | 0.5 |
| Shoot yourself in the head you hypersensitive fuckin b[redacted]ch. | 0.5 |
| not all women are b[redacted]ches, but all b[redacted]ches are women | 0.5 |
| "you ugly b[redacted]ch!!!"- that seems to be my real name these days" | 0 |
| "I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)" | 0 |
| "Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible" | 0 |
| someone plz get this n[redacted]er some help. He's playing up like mad | 0 |
| stop talking you fucking n[redacted]er ! | 0 |

Both of these explanations are plausible and reflect known limitations of most NLP datasets. Unfortunately, we can not confirm any of these hypotheses.

*How Does it Look Like?* Table 12 shows a random sample of items taken from the Sap dataset and their respective USD values (§4). Note that explicitly toxic or racist items tend to have USD values of zero, alongside clearly unproblematic items. Inversely, more ambiguous and less comprehensible items tend to have larger USD values. This pattern may suggest that USD may be tied to the overall ambiguity of the statements, especially in cases where items may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2). We include the USD values for each individual item for all four datasets in the supplementary material.

*Can We Produce it Synthetically?* Fig. 8 shows the USD of LLM annotations and of the synthetic simulations of §4, using the same layout as for the human datasets: all annotators (right) against at most six annotators per comment (left). Looking at the synthetic annotations, we note that USD is generally produced irrespective of number of groups per hypothetical SDD. With respect to LLM annotation, we observe that prompt-unstable models are the largest contributors of USD, indicating that the latter may be linked with general disagreement (variance) rather than SD.

In all cases, including human annotations (Fig. 4), low annotator counts per item seem to be the cause of SD. Indeed, we observe USD from most models when repeatedly subsampling six annotators (Fig. 9). In the case of the Sap and Kumar datasets, human annotators show significantly higher USD values, which is not the case for POPQUORN.

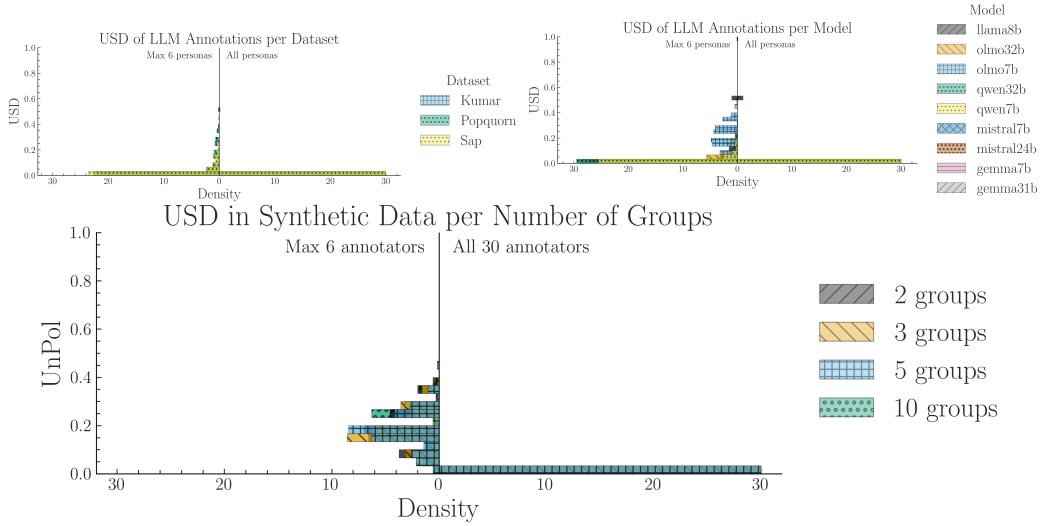



Fig. 8. USD for LLM annotations grouped by dataset (top-left) and model (top-right) as well as for the synthetic simulations used in §4 (bottom). LLM annotations include models that were excluded due to prompt sensitivity.

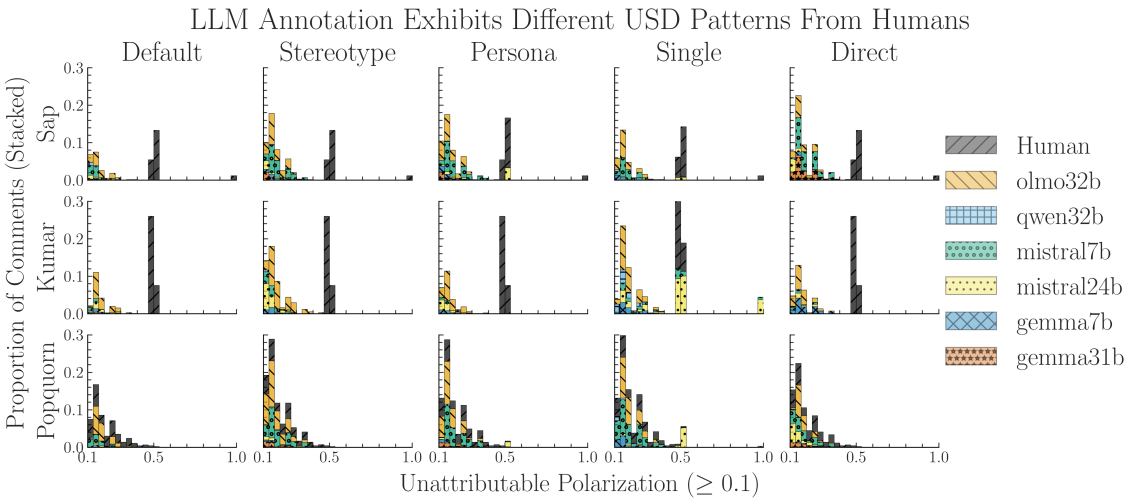



Fig. 9. Normalized histograms showing the USD exhibited by each model when truncating annotation to six annotations, compared to human annotations for each dataset. Some LLMs fail to show any USD, while the rest show far less than humans. Note that the graph does not include zero USD, and the y-axis is truncated to $[0, 0.3]$.

Instruction prompts can influence the extent of USD, but inconsistently. Providing adversarial instruction prompts seems to increase USD, although the effect of each prompt seems to be dataset-dependent.

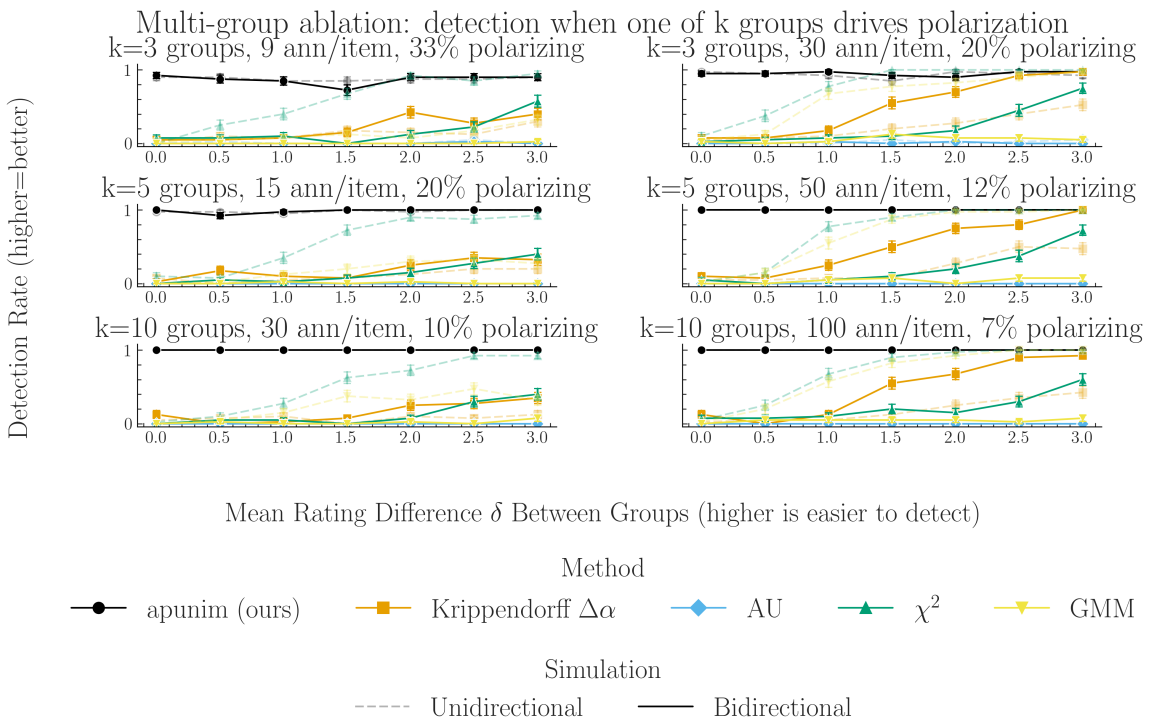



Fig. 10. Detection rate of each method on synthetic data with $k$ annotator groups per item, only one of which is polarizing. The setup is otherwise identical to Fig. 2. Panel titles give $k$, the number of annotators per item, and the share of annotators in the polarizing group. Solid lines show bidirectional polarization; dashed, faded lines show unidirectional polarization, where the effect always has the same direction. Bars show the standard error.

## E Ablations

### E.1 Synthetic Experiments

*Multigroup Synthetic Experiments.* We repeat the comparison of Fig. 2 with $k$ annotator groups per item, only one of which is polarizing. As Fig. 10 shows, apunim remains competitive with multiple groups. Most methods detect polarization more often when it is unidirectional, so the difficulty that bidirectional polarization poses for prior methods is likely not limited to two groups.

*Do Fixed Group Effects Affect our Experiments?* In our synthetic experiments, annotators within a group do not share the same opinion: each annotator's group effect is sampled, with $g_i \sim U(-1/2, 0)$ for one group and $g_i \sim U(0, 1/2)$ for the other. In the multigroup experiments, each annotator in the polarizing group has a strength $h_i \sim U(0, 1)$. To test whether our conclusions depend on this choice, we rerun both experiments with a fixed group effect, where every annotator in a group shifts by exactly the same amount ($g_i = \pm 1/2$ and $h_i = 1$). All other settings, including the conditions, $\delta$ values, and number of repetitions, are unchanged. Note that the fixed setting also doubles the expected difference between groups at a given $\delta$, so it is both more homogeneous and a stronger signal. Fig. 11 shows the results.

### E.2 Dataset Sampling

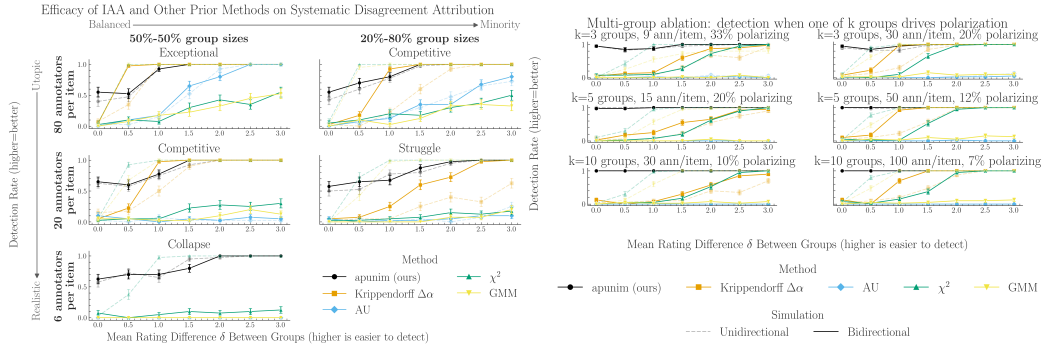



Fig. 11. The experiments of Fig. 2 (left) and Fig. 10 (right), rerun with a fixed group effect: every annotator in a group shifts by exactly the same amount ($g_i = \pm 1/2$, $h_i = 1$) instead of a sampled amount. All other settings are unchanged. Solid lines show bidirectional polarization; dashed, faded lines show unidirectional polarization. Bars show the standard error.

Table 13. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

| | | 1000 | 3000 | 10000 | 30000 |
|---|---|---|---|---|---|
| | **SDD** | | | | |
| Age | 1) 18 - 24 | 0.014 | -0.017 | -0.037 | -0.016 |
| | 2) 25 - 34 | -0.003 | -0.000 | 0.000 | 0.000 |
| | 3) 35 - 44 | 0.000 | -0.011 | -0.001 | -0.001 |
| | 4) 45 - 54 | 0.022 | 0.033 | 0.013 | 0.003 |
| | 5) 55 - 64 | 0.000 | 0.000 | -0.057 | 0.001 |
| | 6) 65 or older | — | — | — | 0.948 |
| Education | 2) High School graduate | 0.000 | 0.000 | -0.005 | 0.014 |
| | 3) College, no degree | 0.010 | 0.014 | 0.010 | 0.006 |
| | 4) Associate degree | 0.000 | 0.000 | 0.005 | 0.009 |
| | 5) Bachelor's degree | 0.001 | -0.000 | -0.001 | -0.000 |
| | 6) Master's degree | 0.000 | -0.006 | 0.004 | -0.000 |
| Ethnicity | Asian | -0.000 | 0.000 | 0.011 | 0.007 |
| | Black | 0.000 | -0.024 | -0.004 | -0.012 |
| | Hispanic | — | — | 0.000 | -0.099 |
| | Multiracial | 0.000 | 0.000 | 0.000 | 0.000 |
| | Other | — | — | — | -0.306 |
| | White | 0.000 | -0.000 | -0.000 | 0.000 |
| Gender | Female | 0.001 | 0.000 | 0.002 | 0.001 |
| | Male | 0.001 | -0.004 | -0.002 | -0.001 |
| Has Been Targeted | False | 0.001 | 0.000 | -0.000 | 0.000 |
| | True | -0.009 | -0.015 | -0.003 | -0.002 |

Continued on next page

Table 13. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

| | | 1000 | 3000 | 10000 | 30000 |
|---|---|---|---|---|---|
| | **SDD** | | | | |
| Is Parent | No | -0.002 | 0.002 | 0.001 | 0.001 |
| | Prefer not to say | — | — | 1.000 | 0.000 |
| | Yes | -0.000 | 0.000 | -0.000 | 0.000 |
| Is Transgender | No | 0.000 | 0.000 | 0.000 | 0.000 |
| Political Affiliation | Conservative | -0.013 | 0.003 | -0.003 | -0.002 |
| | Independent | 0.000 | -0.001 | 0.001 | 0.002 |
| | Liberal | -0.001 | 0.006 | 0.001 | 0.001 |
| | Other | — | 0.000 | 0.000 | 0.000 |
| | Prefer not to say | — | — | 0.000 | 0.022 |
| Religion Important | 1) No | 0.002 | 0.004 | 0.002 | 0.001 |
| | 2) Not very | 0.000 | -0.006 | 0.010 | 0.006 |
| | 3) Somewhat | 0.000 | 0.002 | 0.003 | -0.002 |
| | 4) Very | 0.000 | -0.003 | -0.003 | -0.002 |
| Seen Toxicity | False | -0.007 | 0.002 | -0.001 | -0.001 |
| | True | -0.001 | 0.000 | 0.001 | 0.000 |
| Sexual Orientation | Bisexual | -0.068 | -0.029 | -0.040 | -0.013 |
| | Heterosexual | -0.001 | -0.000 | -0.000 | 0.000 |
| | Homosexual | — | — | 0.000 | 0.053 |
| | Prefer not to say | — | — | — | 0.370 |
| Technology Impact | 2) Somewhat negative | 0.000 | 0.000 | 0.000 | 0.003 |
| | 3) Neutral | 0.124 | 0.037 | -0.014 | 0.000 |
| | 4) Somewhat positive | 0.004 | 0.001 | -0.000 | 0.001 |
| | 5) Very positive | -0.015 | -0.004 | -0.004 | -0.003 |
| Toxicity Problem | 1) Never | 0.000 | 0.000 | -0.037 | -0.081 |
| | 2) Rarely | 0.000 | 0.000 | 0.003 | -0.001 |
| | 3) Occasionally | 0.000 | -0.004 | -0.003 | -0.001 |
| | 4) Frequently | 0.000 | 0.000 | -0.001 | -0.000 |
| | 5) Very Frequently | 0.000 | -0.002 | 0.030 | 0.015 |

*How Does Dataset Size Influence Apunim?* Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 3,000, 10,000, and 30,000 items), shown in Table 13. Sample size does not seem to have a consistent effect on apunim values, other than large dataset sizes uncovering significant polarization effects in sparsely populated groups (e.g., 65+ yo. in *Age*, "Prefer Not to Say" in *Sexual Orientation*, "Never" , "Very Frequently" in *Toxicity Problem*). The apunim direction stays broadly the same. We also test whether the results in Table 3 are an artifact of the dataset split, because of the small sample size used in the study. We repeat the apunim experiment with ten different

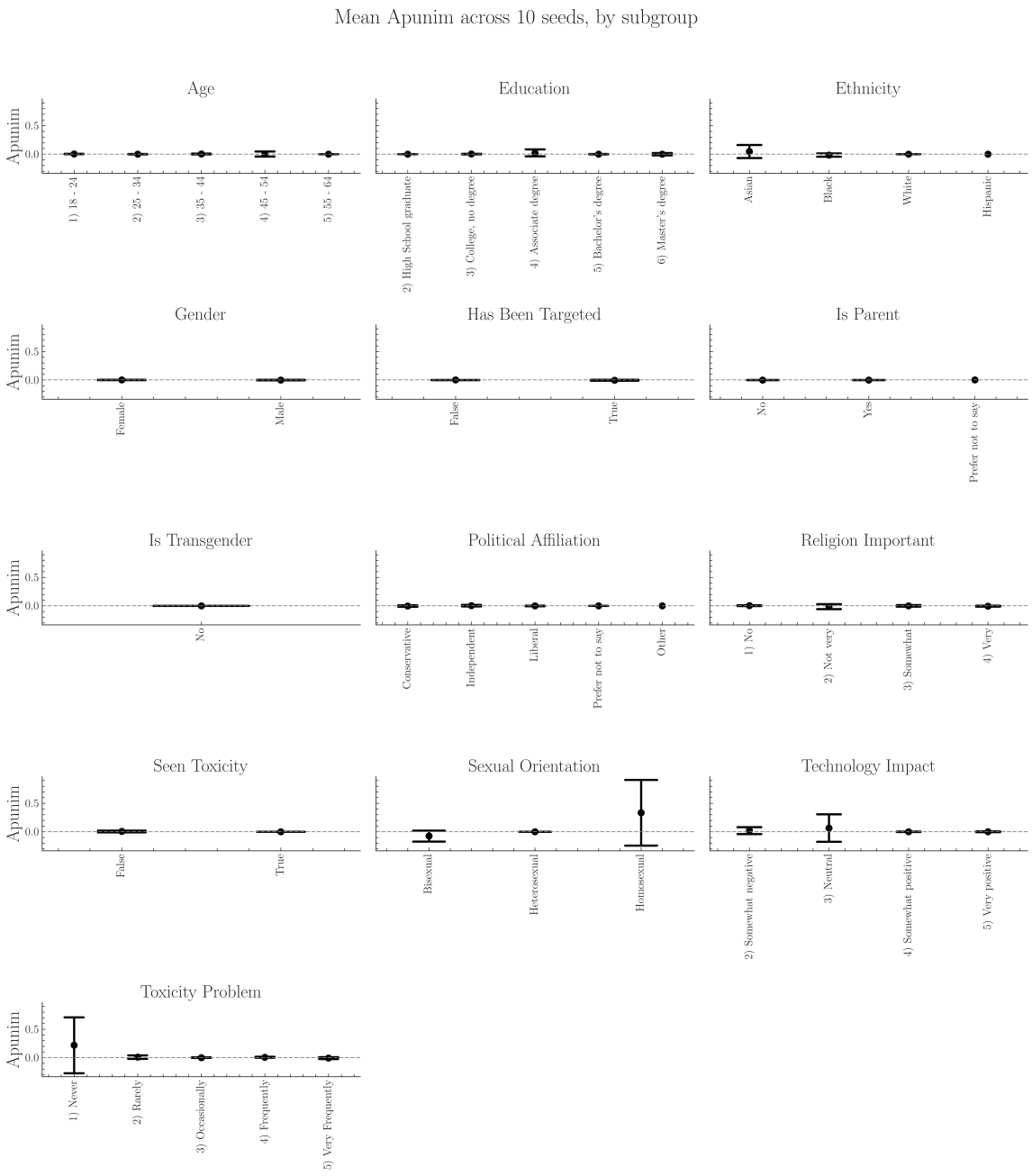



Fig. 12. Apunim values on the Kumar dataset ($n_{samples} = 1000$) across ten runs with different subsets. Black bars indicate two standard deviations. SDDs with many groups are more volatile, although all SDDs broadly keep the same apunim sign across subsets.

subsets of the same dataset and record their apunim values, shown in Fig. 12. We find that most apunim values are stable, save for "Never" in *Toxicity Problem*, "Heterosexual" in *Sexual Orientation* and "Neutral" in *Technology Impact*.

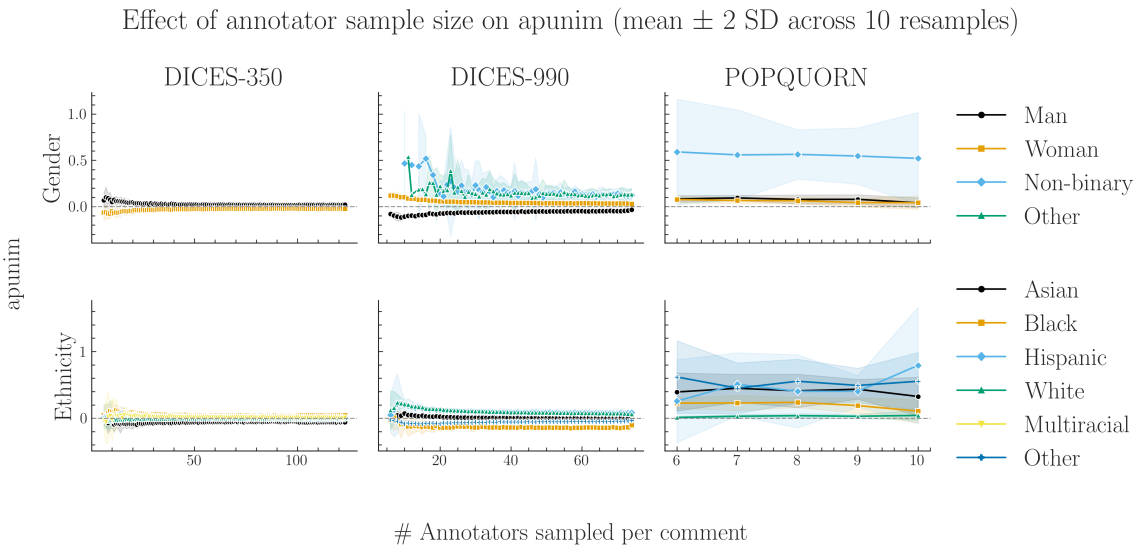



Fig. 13. Standard deviation of *apunim* across items, averaged across 10 runs, where each item is resampled 100 times. Shaded areas represent two standard deviations.

*How Does Sample Selection Influence Apunim?* Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero. In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 12, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few items may genuinely change the value of the apunim measure. These groups however are largely not detected by apunim as significant either way (see Table 3).

### E.3 How Many Annotators Does Apunim Need?

Fig. 13 sows that non-statistically significant results from Table 3 remain systematically close to zero, while statistically significant results can be distinguished from very small annotator counts (six annotators for well-balanced groups, $\approx$ 20 for poorly split groups such as non-binary annotators in DICES-990). Sampling annotators with replacement also uncovers a new group which systematically disagrees with the others; the Non-Binary annotators in the POPQUORN and DICES-990 datasets. This group was undetectable when partitioning annotator groups with the methodology described in Eq. 2 since only one item is annotated by three *Non-Binary* annotators in POPQUORN and none for DICES-990, but sampling with replacement uncovers a very strong attribution. It is unclear whether this is a methodological artifact, or a new finding unexplored in prior analysis concerning these datasets.

# F LLM Annotation Procedure

## F.1 Models Used

We use only open-source models, as proprietary models such as OpenAI's GPT-5 yield much less reproducible results [8]. Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings [57]. It is also worth noting that since the datasets used are not publicly licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation [16, 17]. Table 14 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the `unsloth huggingface` repository.

Table 14. LLMs used in this study.

| Family | Version | #Params | Year |
|---|---|---|---|
| OLMo [26] | 2 (0325) | 32B | 2025 |
| Qwen [77] | 2.5 | 32B | 2024 |
| OLMo [48] | 3 | 7B | 2025 |
| Qwen [77] | 2.5 | 7B | 2024 |
| LLaMa [25] | 3.1 | 8B | 2024 |
| Mistral [31] | Small 3 (2501) | 24B | 2025 |
| Mistral [31] | v0.3 | 7B | 2024 |
| Gemma [64] | 4 | E2B | 2026 |
| Gemma [64] | 4 | 31B | 2026 |

## F.2 Experimental Setup

*Default Prompt.* Since each of the datasets in §3 was annotated using different annotation guidelines and targets we use different instruction prompts for each dataset. The "default" annotation prompt is directly extracted from the studies in the case of Sap and Kumar, and inferred from the instruction guidelines used in the DICES datasets with minor modifications. We use standard persona prompting procedures sampled randomly from the available SDDs in each respective dataset. Twenty LLM annotators are selected for each item and model.

*Ablation Prompts.* The prompts used in the ablation study were provided by a GPT-5 and a local Gemma-4 (quantized) model by providing the original prompt and instructing it to produce four variants. The tested prompts use different methods to restrict the model's behavior. Besides the default prompt, we test four prompts that make the persona more central to the rating. All four ask the same question about the comment as the default prompt and append the persona description after the comment. They differ only in how they instruct the model to use the persona. *Persona-focused* asks the model to rate the comment as its persona would personally perceive it, based on the persona's characteristics, attitudes, experiences, and sensitivities. *Direct* states that the persona's characteristics are a mandatory prerequisite to consider and that the model must evaluate the comment exclusively as the persona would classify it. *Single* asks the model to first identify the single most influential trait of its persona, and then to derive the rating by matching the comment to that trait alone. *Stereotype* asks the model to be very sensitive to its persona's characteristics and to give

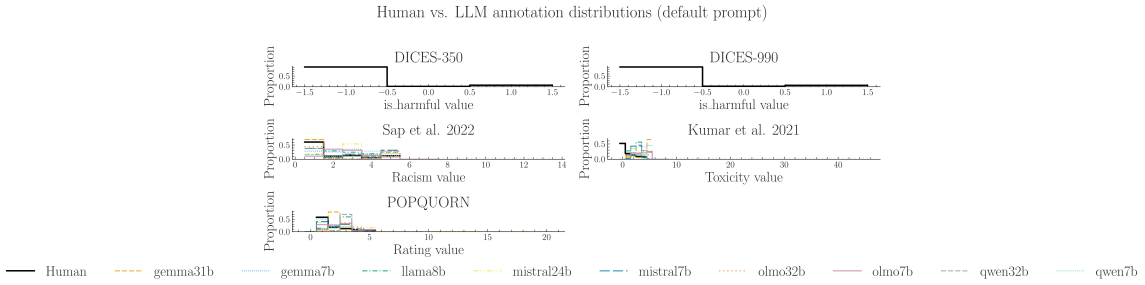



Fig. 14. Annotation frequencies for humans and each LLM in the primary annotation task using the default prompt.

substantial weight to the persona's sensitivities, values, experiences, and likely reactions, rather than to the judgment of a generic or average person. The full prompts are in App. H.

*Hardware and Execution Times.* We use a single Quadro RTX 6000 GPU for tasks requiring small (7-8B) models and two such GPUs for the rest. The annotation task, including ablations, requires approximately 300 hours of computation to complete. We use the `transformers` library to load and manage the LLMs. We release the relevant code in the supplementary material. Our codebase is platform-independent except for three high-level bash scripts and a full environment file is included for reproduction.

### F.3 Exploratory Analysis

*Frequencies.* Fig. 14 shows the different annotation frequencies of LLMs and human annotators in the same dataset split. Across all datasets, we observe a consistent mismatch between human and LLM annotation distributions. For the DICES-350 and DICES-990 harmfulness tasks, human annotations are heavily concentrated at the lowest harmfulness value, whereas most LLMs assign substantially more probability to moderate and high harmfulness values. A similar pattern appears in the Sap et al. [60] racism dataset: human judgments are concentrated at low racism scores, while LLMs produce broader distributions with considerably more mass at higher scores, particularly values 4–5. The [35] toxicity results show the same general trend, with humans assigning most probability to zero toxicity while LLMs distribute their predictions across higher toxicity levels, often peaking around values 2–4. Importantly, the magnitude of this discrepancy varies across models; some LLMs are closer to the human distribution, while others show substantially stronger shifts toward higher scores.

*Dropped Responses.* Table 15 shows a breakdown of failure modes across models and datasets. Most failed answers stem from the model refusing to provide any number in their response. Some attempt to initiate a Chain of Thought response, which is truncated due to us forcing a three-token limit on LLM answers. Others refuse to respond due to safety tuning. Finally, some (and especially the smaller Gemma model) seem to produce degenerate integer outputs. The full file of failed responses can be found in the Supplementary Material.

### F.4 Annotation Ablations

We examine the following conditions for selecting models and datasets for our LLM annotation experiment. All ablation experiments use a sampled 20% fraction of the full LLM annotation dataset used in the main text.

Table 15. Number of non-standard LLM annotations per model and dataset, summed over all instruction prompts. Unparsed: no integer found in the reply (dropped from the analysis); Out of range: an integer outside the dataset's annotation scale. Incidence: percentage of the model's annotations on that dataset.

| Model | Dataset | Status | Count | Incidence (%) |
|---|---|---|---|---|
| gemma31b | popquorn | Unparsed | 104 | 0.09 |
| gemma31b | sap | Unparsed | 8 | 0.01 |
| gemma7b | kumar | Unparsed | 12 | 0.01 |
| gemma7b | popquorn | Out of range | 1 | 0.00 |
| gemma7b | popquorn | Unparsed | 340 | 0.23 |
| gemma7b | sap | Out of range | 20 | 0.03 |
| gemma7b | sap | Unparsed | 2 | 0.00 |
| llama8b | kumar | Out of range | 153 | 0.15 |
| llama8b | kumar | Unparsed | 33352 | 33.35 |
| llama8b | popquorn | Out of range | 1 | 0.00 |
| llama8b | popquorn | Unparsed | 47661 | 31.77 |
| llama8b | sap | Unparsed | 20552 | 35.13 |
| mistral24b | kumar | Out of range | 1 | 0.00 |
| mistral24b | kumar | Unparsed | 14745 | 14.74 |
| mistral24b | popquorn | Unparsed | 28599 | 19.07 |
| mistral24b | sap | Unparsed | 16783 | 28.69 |
| mistral7b | kumar | Out of range | 5 | 0.01 |
| mistral7b | kumar | Unparsed | 12152 | 12.15 |
| mistral7b | popquorn | Out of range | 62 | 0.04 |
| mistral7b | popquorn | Unparsed | 6174 | 4.12 |
| mistral7b | sap | Unparsed | 2771 | 4.74 |
| olmo32b | kumar | Out of range | 25 | 0.03 |
| olmo32b | kumar | Unparsed | 270 | 0.27 |
| olmo32b | popquorn | Out of range | 10 | 0.01 |
| olmo32b | popquorn | Unparsed | 1504 | 1.00 |
| olmo32b | sap | Out of range | 2 | 0.00 |
| olmo32b | sap | Unparsed | 953 | 1.63 |
| olmo7b | kumar | Out of range | 96 | 0.10 |
| olmo7b | kumar | Unparsed | 613 | 0.61 |
| olmo7b | popquorn | Out of range | 110 | 0.07 |
| olmo7b | popquorn | Unparsed | 4849 | 3.23 |
| olmo7b | sap | Out of range | 37 | 0.06 |
| olmo7b | sap | Unparsed | 3867 | 6.61 |
| qwen32b | kumar | Unparsed | 1 | 0.00 |
| qwen7b | kumar | Out of range | 137 | 0.14 |
| qwen7b | kumar | Unparsed | 22 | 0.02 |
| qwen7b | popquorn | Unparsed | 42 | 0.03 |
| qwen7b | sap | Unparsed | 5 | 0.01 |

*The LLMs should be consistent given the same prompt.* While we have fixed all relevant seeds, it is possible that generation deviates across repeated model prompting given the same input [8]. We check this condition by providing the model with the main annotation prompt (see App. H) five times and find that generation is indeed deterministic under our setup for all models (Table 16).

*The LLMs should be prompt insensitive.* Small changes in the prompt have been shown to drastically affect LLM output in some tasks depending on the experimental setup [55]. In order to verify whether any models are unreliable under our setup, we create two alternative formulations of each prompt without a change in substance using a GPT-5 model

Table 16. Consistency (Krippendorff's $\alpha$, ordinal) of each model with itself across repeated runs of the same (default) prompt.

| Dataset | Model | Runs | Items | Krippendorff's alpha |
|---|---|---|---|---|
| DICES-350 | gemma31b | 5 | 210 | 1.0000 |
| DICES-350 | gemma7b | 5 | 210 | 1.0000 |
| DICES-350 | llama8b | 5 | 210 | 1.0000 |
| DICES-350 | mistral24b | 5 | 210 | 1.0000 |
| DICES-350 | mistral7b | 5 | 210 | 1.0000 |
| DICES-350 | olmo32b | 5 | 209 | 1.0000 |
| DICES-350 | olmo7b | 5 | 210 | 1.0000 |
| DICES-350 | qwen32b | 5 | 210 | 1.0000 |
| DICES-350 | qwen7b | 5 | 210 | 1.0000 |
| DICES-990 | gemma31b | 5 | 594 | — |
| DICES-990 | gemma7b | 5 | 594 | 1.0000 |
| DICES-990 | llama8b | 5 | 594 | 1.0000 |
| DICES-990 | mistral24b | 5 | 592 | 1.0000 |
| DICES-990 | mistral7b | 5 | 594 | — |
| DICES-990 | olmo32b | 5 | 594 | 1.0000 |
| DICES-990 | olmo7b | 5 | 594 | 1.0000 |
| DICES-990 | qwen32b | 5 | 594 | 1.0000 |
| DICES-990 | qwen7b | 5 | 594 | 1.0000 |
| Sap et al. 2022 | gemma31b | 5 | 348 | 1.0000 |
| Sap et al. 2022 | gemma7b | 5 | 348 | 1.0000 |
| Sap et al. 2022 | llama8b | 5 | 287 | 1.0000 |
| Sap et al. 2022 | mistral24b | 5 | 348 | 1.0000 |
| Sap et al. 2022 | mistral7b | 5 | 345 | 1.0000 |
| Sap et al. 2022 | olmo32b | 5 | 343 | 1.0000 |
| Sap et al. 2022 | olmo7b | 5 | 338 | 1.0000 |
| Sap et al. 2022 | qwen32b | 5 | 348 | 1.0000 |
| Sap et al. 2022 | qwen7b | 5 | 348 | 1.0000 |
| Kumar et al. 2021 | gemma31b | 5 | 600 | 1.0000 |
| Kumar et al. 2021 | gemma7b | 5 | 600 | 1.0000 |
| Kumar et al. 2021 | llama8b | 5 | 458 | 1.0000 |
| Kumar et al. 2021 | mistral24b | 5 | 600 | 1.0000 |
| Kumar et al. 2021 | mistral7b | 5 | 597 | 1.0000 |
| Kumar et al. 2021 | olmo32b | 5 | 599 | 1.0000 |
| Kumar et al. 2021 | olmo7b | 5 | 600 | 1.0000 |
| Kumar et al. 2021 | qwen32b | 5 | 600 | 1.0000 |
| Kumar et al. 2021 | qwen7b | 5 | 600 | 1.0000 |
| POPQUORN | gemma7b | 5 | 900 | 1.0000 |
| POPQUORN | llama8b | 5 | 771 | 1.0000 |
| POPQUORN | mistral24b | 5 | 900 | 1.0000 |
| POPQUORN | mistral7b | 5 | 899 | 1.0000 |
| POPQUORN | olmo32b | 5 | 893 | 1.0000 |
| POPQUORN | olmo7b | 5 | 880 | 1.0000 |
| POPQUORN | qwen32b | 5 | 900 | 1.0000 |
| POPQUORN | qwen7b | 5 | 900 | 1.0000 |

(free version), whose outputs we manually verified (see App. H). Table 17 shows that while most LLMs are relatively robust to changes in formulation, the smaller ones (OLMo-7B, Qwen-7B and LLaMa-8B) are not. They are thus excluded from the annotation study.

*The annotation guidelines should be unambiguous.* The studies that provided the datasets used in our work (see §3) do not include LLM annotation, nor equivalent LLM prompts, which meant we had to generate them from the human annotation guidelines ourselves. This was not a problem in the case of the Sap and Kumar datasets, since the annotation guidelines were short, simple and required minimal edits to be transformed into LLM annotation guidelines. However, the DICES datasets use poorly documented guidelines and the target label we used is an aggregation of multiple annotation tasks. For these reasons the prompt creation is non-trivial and prone to errors. Indeed, Table 18 shows that inter-model agreement is extremely low for the DICES datasets, even when we remove the sensitive models (Table 19). Given that the DICES datasets are (1) problematic w.r.t. inherent polarization (see App. D), (2) conditioned on several dozen annotations per item (3) difficult to find reliable annotation prompts for LLMs, we decided to exclude them from the annotation study.

### F.5 LLM Annotation Results

Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| Age | 1) 18 - 24 | llama8b | -0.049** | 0.013 | 0.060** | — | -0.054* |
| | | olmo32b | -0.068*** | 0.090 | — | — | -0.017 |
| | | olmo7b | 0.031** | 0.102*** | 0.052*** | -0.115 | 0.026* |
| | | qwen32b | 0.023 | 0.177 | — | -0.018 | -0.073 |
| | | qwen7b | 0.012 | -0.075*** | -0.000 | 0.042 | 0.077*** |
| | | mistral7b | 0.014 | 0.052*** | 0.104*** | 0.014 | 0.215*** |
| | | mistral24b | 0.048*** | 0.031* | 0.167*** | -0.107 | -0.004 |
| | | gemma7b | 0.051* | 0.045* | -0.019 | -0.006 | -0.031 |
| | | gemma31b | 0.035 | — | — | — | — |
| | 2) 25 - 34 | llama8b | 0.048** | 0.012 | 0.000 | — | -0.140*** |
| | | olmo32b | -0.010 | 0.068 | — | — | 0.051** |
| | | olmo7b | 0.042*** | 0.055*** | 0.031*** | 0.149 | -0.008 |
| | | qwen32b | 0.035* | 0.105 | — | -0.046* | -0.304 |
| | | qwen7b | 0.036** | 0.112*** | 0.082*** | 0.118*** | 0.074*** |
| | | mistral7b | 0.049*** | 0.036** | 0.100*** | 0.095 | 0.162*** |
| | | mistral24b | 0.059*** | 0.040** | 0.153** | 0.146 | -0.026 |
| | | gemma7b | 0.009 | 0.011 | -0.020 | 0.060* | -0.033 |
| | | gemma31b | 0.073* | — | — | — | — |
| | 3) 35 - 44 | llama8b | -0.046** | 0.015 | 0.022 | — | -0.035 |
| | | olmo32b | 0.029* | — | — | — | -0.024 |
| | | olmo7b | -0.018 | 0.057*** | -0.016 | 0.182 | 0.021 |
| | | qwen32b | 0.042** | 0.122 | -0.213 | 0.058* | 0.101 |
| | | qwen7b | 0.051*** | 0.008 | 0.103*** | -0.027 | -0.000 |
| | | mistral7b | 0.068*** | 0.024 | -0.007 | 0.153 | 0.029 |
| | | mistral24b | 0.007 | 0.030* | -0.041 | -0.056 | -0.044** |
| | | gemma7b | -0.029 | 0.008 | -0.016 | 0.000 | -0.004 |
| | | gemma31b | 0.031 | — | — | — | — |
| | 4) 45 - 54 | llama8b | 0.051** | 0.001 | 0.097*** | — | 0.043** |
| | | olmo32b | 0.002 | -0.254 | — | — | -0.012 |
| | | olmo7b | 0.057*** | -0.003 | 0.004 | -0.160 | -0.025* |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | qwen32b | 0.044** | 0.153 | — | 0.047* | -0.081 |
| | | qwen7b | 0.084*** | -0.040 | -0.059 | 0.007 | 0.070*** |
| | | mistral7b | -0.004 | 0.012 | -0.001 | -0.458 | -0.103*** |
| | | mistral24b | 0.040** | 0.006 | 0.177*** | -0.090 | -0.008 |
| | | gemma7b | 0.029 | -0.004 | -0.039 | 0.116*** | -0.030 |
| | | gemma31b | 0.037 | — | — | — | — |
| | 5) 55 - 64 | llama8b | 0.025 | -0.006 | 0.007 | — | 0.065*** |
| | | olmo32b | -0.026 | 0.089 | — | — | -0.016 |
| | | olmo7b | 0.049*** | -0.027* | -0.031*** | — | -0.015 |
| | | qwen32b | 0.089*** | 0.084 | 0.527 | 0.059* | 0.077 |
| | | qwen7b | 0.014 | 0.016 | 0.023 | -0.053** | -0.006 |
| | | mistral7b | 0.041** | 0.020 | 0.074*** | -0.213 | -0.135*** |
| | | mistral24b | 0.045** | 0.049*** | 0.094** | -0.111 | 0.046** |
| | | gemma7b | 0.024 | 0.024 | 0.052 | 0.032 | 0.001 |
| | | gemma31b | -0.030 | — | — | — | — |
| | 6) 65 or older | llama8b | 0.020 | -0.035*** | -0.058** | — | 0.073*** |
| | | olmo32b | -0.035** | 0.333 | — | — | 0.005 |
| | | olmo7b | 0.036*** | -0.034** | 0.002 | — | 0.020 |
| | | qwen32b | 0.053** | -0.210 | — | 0.061* | 0.146 |
| | | qwen7b | 0.049*** | 0.086*** | 0.034 | 0.009 | 0.079*** |
| | | mistral7b | 0.007 | -0.034** | -0.141*** | -0.114 | 0.034 |
| | | mistral24b | 0.053*** | -0.015 | -0.100*** | 0.055 | 0.192*** |
| | | gemma7b | -0.007 | 0.004 | 0.021 | -0.036 | -0.002 |
| | | gemma31b | -0.012 | — | — | — | — |
| | Prefer not to say | llama8b | -0.003 | 0.039*** | 0.003 | — | -0.079** |
| | | olmo32b | -0.043** | 0.099 | — | — | 0.011 |
| | | olmo7b | 0.029* | 0.139*** | 0.041*** | 0.120 | 0.017 |
| | | qwen32b | 0.032* | 0.102 | 0.097 | 0.017 | -0.174 |
| | | qwen7b | 0.010 | 0.002 | -0.017 | -0.012 | 0.001 |
| | | mistral7b | 0.047*** | -0.016 | -0.019 | 0.227 | 0.036 |
| | | mistral24b | 0.091*** | 0.031* | -0.046 | -0.024 | 0.011 |
| | | gemma7b | -0.007 | -0.000 | 0.077** | -0.034 | -0.021 |
| | | gemma31b | -0.027 | — | — | — | — |
| | Under 18 | llama8b | 0.013 | 0.017 | 0.009 | — | -0.105*** |
| | | olmo32b | 0.040** | -0.127 | — | — | -0.018 |
| | | olmo7b | 0.010 | -0.063*** | 0.036*** | — | -0.019 |
| | | qwen32b | 0.036* | -0.110 | -0.225 | -0.062* | 0.005 |
| | | qwen7b | 0.014 | -0.031 | -0.021 | 0.057** | -0.005 |
| | | mistral7b | -0.001 | 0.005 | -0.063* | -0.186 | 0.028 |
| | | mistral24b | 0.019 | -0.064*** | -0.149*** | 0.103 | 0.046** |
| | | gemma7b | 0.009 | 0.051** | -0.084*** | 0.017 | -0.012 |
| | | gemma31b | 0.018 | — | — | — | — |
| Education | 1) No high school | llama8b | 0.029 | -0.001 | 0.059* | — | 0.025 |
| | | olmo32b | -0.007 | — | — | — | 0.068*** |
| | | olmo7b | 0.004 | -0.124*** | 0.092*** | 0.113 | -0.014 |
| | | qwen32b | -0.009 | -0.170 | 0.399 | -0.042 | -0.106 |
| | | qwen7b | 0.028 | 0.088** | 0.003 | 0.057* | 0.014 |
| | | mistral7b | 0.004 | -0.005 | 0.021 | -0.083 | -0.078* |
| | | mistral24b | -0.012 | -0.028 | 0.086 | -0.159 | -0.025 |
| | | gemma7b | -0.027 | 0.002 | 0.159*** | -0.015 | 0.036 |
| | | gemma31b | 0.037 | — | — | — | — |
| | 2) High School graduate | llama8b | 0.027 | 0.062*** | -0.021 | — | 0.084* |
| | | olmo32b | -0.100*** | — | — | — | 0.022 |
| | | olmo7b | -0.038* | 0.011 | -0.083*** | -0.219 | 0.058*** |
| | | qwen32b | -0.016 | -0.006 | 0.307 | -0.076* | 0.212 |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | qwen7b | 0.028 | 0.082** | 0.052 | -0.002 | 0.038 |
| | | mistral7b | -0.032 | -0.005 | -0.073 | 0.062 | 0.101** |
| | | mistral24b | 0.012 | 0.040 | 0.166 | 0.502 | 0.050* |
| | | gemma7b | 0.031 | -0.032 | 0.080* | 0.032 | -0.012 |
| | | gemma31b | -0.054 | — | — | — | — |
| | 3) College, no degree | llama8b | -0.025 | -0.025* | -0.035 | — | -0.005 |
| | | olmo32b | 0.070*** | — | — | — | -0.052* |
| | | olmo7b | -0.020 | -0.006 | -0.035** | — | -0.000 |
| | | qwen32b | 0.017 | 0.045 | — | -0.044 | 0.137 |
| | | qwen7b | 0.022 | -0.068* | -0.014 | -0.031 | -0.033 |
| | | mistral7b | -0.020 | -0.032 | -0.025 | -0.064 | 0.009 |
| | | mistral24b | -0.033 | -0.029 | 0.061 | 0.044 | 0.024 |
| | | gemma7b | 0.025 | 0.030 | -0.009 | 0.080* | -0.008 |
| | | gemma31b | 0.049 | — | — | — | — |
| | 4) Associate degree | llama8b | 0.058* | 0.032* | -0.034 | — | -0.116** |
| | | olmo32b | 0.023 | 0.079 | — | — | -0.053** |
| | | olmo7b | 0.033 | -0.000 | 0.017 | — | 0.021 |
| | | qwen32b | 0.030 | 0.132 | — | 0.133*** | 0.156* |
| | | qwen7b | -0.016 | -0.018 | 0.017 | -0.047 | 0.045* |
| | | mistral7b | 0.030 | 0.041 | -0.024 | -0.103 | 0.025 |
| | | mistral24b | 0.053** | 0.003 | 0.135** | -0.109 | -0.026 |
| | | gemma7b | 0.012 | -0.003 | 0.062* | -0.033 | -0.051 |
| | | gemma31b | -0.060 | — | — | — | — |
| | 5) Bachelor's degree | llama8b | 0.053* | -0.036** | 0.049* | — | -0.051 |
| | | olmo32b | -0.034 | 0.141 | — | — | -0.108*** |
| | | olmo7b | -0.011 | 0.012 | 0.010 | 0.242 | -0.032* |
| | | qwen32b | 0.011 | -0.353 | -1.000 | 0.036 | -0.097 |
| | | qwen7b | -0.022 | 0.016 | 0.064* | 0.031 | 0.071*** |
| | | mistral7b | -0.062*** | -0.015 | 0.015 | 0.160 | 0.092** |
| | | mistral24b | 0.059** | 0.028 | -0.018 | 0.378** | 0.066** |
| | | gemma7b | -0.032 | 0.025 | -0.116** | -0.080* | -0.011 |
| | | gemma31b | 0.029 | — | — | — | — |
| | 6) Master's degree | llama8b | 0.003 | 0.028* | -0.004 | — | 0.022 |
| | | olmo32b | 0.006 | — | — | — | 0.097*** |
| | | olmo7b | -0.003 | -0.014 | 0.022 | -0.346 | 0.029* |
| | | qwen32b | -0.014 | 0.242 | — | 0.034 | 0.319 |
| | | qwen7b | 0.030 | 0.009 | 0.015 | -0.027 | -0.008 |
| | | mistral7b | 0.008 | 0.014 | -0.031 | — | -0.076* |
| | | mistral24b | 0.017 | 0.003 | -0.063 | 0.267 | 0.002 |
| | | gemma7b | 0.033 | 0.006 | -0.022 | -0.009 | -0.040 |
| | | gemma31b | 0.042 | — | — | — | — |
| | 7) Professional degree | llama8b | -0.058* | -0.002 | 0.088** | — | -0.015 |
| | | olmo32b | 0.007 | — | — | — | -0.009 |
| | | olmo7b | 0.033 | 0.008 | 0.008 | 0.136 | 0.000 |
| | | qwen32b | -0.004 | 0.158 | 0.268 | -0.093** | -0.387** |
| | | qwen7b | -0.014 | 0.030 | 0.050 | 0.055* | -0.006 |
| | | mistral7b | 0.030 | 0.006 | 0.093* | 0.000 | 0.010 |
| | | mistral24b | -0.034 | -0.012 | -0.084* | 0.080 | 0.012 |
| | | gemma7b | 0.030 | -0.005 | 0.044 | -0.054 | -0.046 |
| | | gemma31b | 0.024 | — | — | — | — |
| | 8) Doctoral degree | llama8b | 0.007 | 0.044** | 0.080** | — | -0.040 |
| | | olmo32b | -0.032 | -0.040 | — | — | -0.032 |
| | | olmo7b | -0.040* | 0.116*** | -0.028* | 0.196 | -0.043** |
| | | qwen32b | 0.013 | -0.007 | — | 0.003 | -0.223 |
| | | qwen7b | -0.005 | -0.086*** | 0.032 | 0.019 | 0.021 |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | mistral7b | -0.033 | -0.030 | -0.148*** | -0.088 | -0.075** |
| | | mistral24b | -0.009 | -0.013 | -0.123* | -0.128 | 0.034 |
| | | gemma7b | -0.009 | 0.006 | 0.143*** | -0.111** | 0.112*** |
| | | gemma31b | 0.003 | — | — | — | — |
| | Other | llama8b | 0.024 | 0.021 | 0.064* | — | -0.000 |
| | | olmo32b | -0.025 | -0.039 | — | — | -0.010 |
| | | olmo7b | -0.017 | 0.033 | 0.044*** | 0.176 | -0.043** |
| | | qwen32b | 0.031 | 0.085 | — | 0.049 | -0.132 |
| | | qwen7b | -0.015 | -0.017 | -0.095*** | 0.061* | -0.077*** |
| | | mistral7b | 0.014 | 0.017 | -0.020 | 0.345 | 0.051* |
| | | mistral24b | -0.021 | 0.047** | 0.031 | 0.003 | 0.021 |
| | | gemma7b | -0.044 | 0.010 | 0.039 | 0.034 | 0.038 |
| | | gemma31b | 0.043 | — | — | — | — |
| | Prefer not to say | llama8b | -0.079** | -0.008 | -0.016 | — | 0.046 |
| | | olmo32b | -0.008 | -0.242 | — | — | -0.043* |
| | | olmo7b | -0.050** | -0.034* | -0.053*** | — | -0.052** |
| | | qwen32b | -0.050* | -0.122 | — | 0.057* | 0.014 |
| | | qwen7b | -0.006 | 0.020 | -0.130*** | -0.095** | 0.043 |
| | | mistral7b | -0.009 | 0.001 | 0.011 | 0.000 | -0.157*** |
| | | mistral24b | 0.017 | 0.008 | 0.019 | -0.090 | 0.013 |
| | | gemma7b | 0.008 | -0.035 | -0.054 | 0.005 | 0.021 |
| | | gemma31b | 0.013 | — | — | — | — |
| Ethnicity | Asian | llama8b | -0.004 | -0.044*** | 0.043* | — | 0.039* |
| | | olmo32b | 0.015 | 0.082 | — | — | -0.022 |
| | | olmo7b | 0.028** | -0.042*** | -0.030*** | — | 0.003 |
| | | qwen32b | 0.010 | -0.106 | — | -0.032 | 0.033 |
| | | qwen7b | -0.028* | 0.055* | 0.042 | 0.010 | 0.022 |
| | | mistral7b | -0.020 | -0.002 | 0.002 | 0.077 | 0.058** |
| | | mistral24b | -0.014 | -0.027* | 0.059 | 0.055 | 0.015 |
| | | gemma7b | 0.028 | -0.020 | -0.017 | 0.032 | 0.034 |
| | | gemma31b | 0.002 | — | — | — | — |
| | Black | llama8b | 0.005 | -0.047*** | 0.025 | — | -0.076*** |
| | | olmo32b | -0.017 | -0.187 | — | — | -0.072*** |
| | | olmo7b | 0.013 | 0.079*** | 0.028*** | 0.105 | 0.002 |
| | | qwen32b | -0.018 | -0.063 | — | -0.045 | 0.056 |
| | | qwen7b | -0.024 | 0.009 | 0.003 | 0.030 | 0.021 |
| | | mistral7b | -0.015 | 0.014 | -0.017 | 0.082 | -0.110*** |
| | | mistral24b | -0.033* | 0.053*** | -0.063 | -0.266 | -0.020 |
| | | gemma7b | 0.025 | 0.036* | 0.058* | -0.032 | -0.021 |
| | | gemma31b | 0.026 | — | — | — | — |
| | Hispanic | llama8b | -0.018 | -0.024* | -0.046** | — | -0.122*** |
| | | olmo32b | -0.055*** | -0.387 | — | — | -0.001 |
| | | olmo7b | 0.022* | -0.029** | 0.045*** | 0.235 | -0.017 |
| | | qwen32b | 0.033* | -0.116 | — | 0.022 | -0.001 |
| | | qwen7b | 0.021 | 0.032 | 0.022 | 0.001 | 0.021 |
| | | mistral7b | -0.004 | 0.008 | -0.009 | -0.016 | 0.061* |
| | | mistral24b | 0.035* | 0.090*** | 0.114*** | 0.072 | -0.021 |
| | | gemma7b | -0.011 | 0.035* | -0.049* | -0.019 | 0.023 |
| | | gemma31b | -0.005 | — | — | — | — |
| | Multiracial | llama8b | 0.026 | -0.035*** | 0.012 | — | -0.066*** |
| | | olmo32b | 0.014 | 0.333 | — | — | 0.055*** |
| | | olmo7b | 0.016 | 0.026** | -0.013 | -0.244 | -0.012 |
| | | qwen32b | 0.052** | 0.052 | 0.142 | 0.023 | -0.118 |
| | | qwen7b | 0.022 | 0.001 | -0.005 | -0.039 | -0.018 |
| | | mistral7b | 0.018 | 0.020 | -0.008 | 0.165 | 0.023 |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | mistral24b | -0.002 | 0.010 | -0.024 | -0.125 | -0.010 |
| | | gemma7b | -0.005 | -0.015 | -0.069** | 0.096*** | -0.013 |
| | | gemma31b | 0.021 | — | — | — | — |
| | Other | llama8b | 0.017 | -0.012 | 0.031 | — | 0.055** |
| | | olmo32b | -0.014 | 0.056 | — | — | -0.001 |
| | | olmo7b | -0.061*** | 0.011 | 0.062*** | — | -0.013 |
| | | qwen32b | 0.007 | 0.228* | — | 0.066** | 0.075 |
| | | qwen7b | 0.026 | -0.021 | 0.008 | -0.063** | 0.003 |
| | | mistral7b | -0.009 | 0.007 | -0.066** | -0.199 | 0.027 |
| | | mistral24b | 0.044** | -0.002 | -0.048 | 0.118 | 0.016 |
| | | gemma7b | -0.006 | -0.001 | -0.022 | 0.038 | 0.007 |
| | | gemma31b | 0.035 | — | — | — | — |
| | Unknown | llama8b | -0.006 | 0.014 | 0.005 | — | -0.045*** |
| | | olmo32b | 0.022 | 0.017 | — | — | -0.026 |
| | | olmo7b | -0.017 | -0.024* | 0.005 | 0.242 | 0.016 |
| | | qwen32b | 0.017 | -0.117 | 0.184 | 0.013 | 0.037 |
| | | qwen7b | -0.021 | 0.014 | -0.004 | 0.033 | -0.027 |
| | | mistral7b | 0.017 | 0.027 | 0.024 | -0.402 | -0.023 |
| | | mistral24b | 0.004 | 0.018 | -0.112*** | 0.047 | -0.022 |
| | | gemma7b | 0.015 | 0.018 | 0.022 | 0.035 | 0.060** |
| | | gemma31b | 0.036 | — | — | — | — |
| | White | llama8b | 0.009 | -0.001 | -0.034* | — | 0.111*** |
| | | olmo32b | 0.024 | 0.345 | — | — | 0.032* |
| | | olmo7b | 0.022* | -0.034*** | -0.000 | — | -0.014 |
| | | qwen32b | -0.022 | -0.035 | — | -0.034 | 0.063 |
| | | qwen7b | 0.039** | -0.055* | -0.018 | 0.006 | -0.010 |
| | | mistral7b | 0.003 | 0.048*** | 0.040** | 0.148 | -0.020 |
| | | mistral24b | 0.031* | 0.002 | -0.014 | -0.045 | 0.072*** |
| | | gemma7b | -0.022 | 0.052*** | 0.042 | -0.092*** | 0.001 |
| | | gemma31b | 0.003 | — | — | — | — |
| Gender | Female | llama8b | -0.021* | -0.010 | -0.010 | — | 0.004 |
| | | olmo32b | -0.014 | -0.075 | — | — | 0.003 |
| | | olmo7b | 0.000 | -0.026*** | -0.007 | -0.010 | -0.002 |
| | | qwen32b | -0.031* | -0.246*** | — | 0.007 | -0.051 |
| | | qwen7b | -0.012 | 0.061*** | 0.028 | 0.006 | -0.011 |
| | | mistral7b | -0.025* | 0.020 | -0.017 | -0.008 | -0.003 |
| | | mistral24b | 0.004 | 0.017 | 0.009 | -0.013 | 0.002 |
| | | gemma7b | -0.002 | -0.007 | 0.020 | 0.017 | -0.022 |
| | | gemma31b | 0.000 | — | — | — | — |
| | Male | llama8b | 0.021* | -0.003 | 0.016 | — | -0.005 |
| | | olmo32b | -0.041*** | -0.114 | — | — | 0.027* |
| | | olmo7b | -0.002 | 0.060*** | 0.017** | -0.067 | -0.032*** |
| | | qwen32b | 0.023 | 0.052 | -0.001 | -0.042* | 0.030 |
| | | qwen7b | 0.032** | -0.099*** | -0.017 | 0.001 | -0.017 |
| | | mistral7b | 0.013 | 0.028* | 0.036* | 0.033 | -0.009 |
| | | mistral24b | -0.016 | -0.015 | -0.103*** | 0.024 | -0.008 |
| | | gemma7b | -0.013 | 0.014 | 0.031 | 0.014 | 0.019 |
| | | gemma31b | 0.022 | — | — | — | — |
| | Nonbinary | llama8b | 0.001 | -0.031*** | -0.001 | — | -0.022 |
| | | olmo32b | -0.003 | 0.051 | — | — | 0.003 |
| | | olmo7b | 0.012 | 0.015* | 0.022*** | -0.020 | 0.020** |
| | | qwen32b | 0.027* | 0.092* | -0.383 | 0.085*** | 0.024 |
| | | qwen7b | 0.001 | 0.024 | 0.039* | 0.001 | 0.025 |
| | | mistral7b | -0.015 | -0.017 | 0.020 | -0.033 | 0.018 |
| | | mistral24b | 0.023 | 0.043*** | 0.102*** | -0.039 | 0.051*** |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | gemma7b | -0.007 | -0.006 | -0.020 | -0.006 | -0.011 |
| | | gemma31b | 0.053** | — | — | — | — |
| | Unknown | llama8b | 0.038*** | -0.008 | 0.006 | — | 0.076*** |
| | | olmo32b | 0.015 | 0.144 | — | — | -0.027* |
| | | olmo7b | -0.006 | -0.047*** | 0.035*** | 0.119 | 0.000 |
| | | qwen32b | 0.021 | 0.091* | 0.281 | -0.021 | -0.037 |
| | | qwen7b | 0.014 | 0.075*** | -0.012 | -0.024 | 0.021 |
| | | mistral7b | 0.020 | -0.000 | -0.001 | 0.031 | 0.021 |
| | | mistral24b | -0.002 | -0.032** | 0.050* | 0.002 | -0.065*** |
| | | gemma7b | 0.029 | 0.002 | 0.008 | -0.015 | 0.017 |
| | | gemma31b | 0.054** | — | — | — | — |
| Has Been Targeted | False | llama8b | 0.008 | -0.005 | 0.010 | — | 0.021 |
| | | olmo32b | 0.002 | 0.049 | — | — | 0.024* |
| | | olmo7b | 0.039*** | -0.049*** | 0.002 | -0.083** | 0.076*** |
| | | qwen32b | -0.019 | 0.072* | 0.140 | -0.007 | 0.067 |
| | | qwen7b | 0.083*** | -0.033 | -0.042** | -0.059*** | -0.002 |
| | | mistral7b | -0.019 | 0.069*** | -0.076*** | -0.056 | -0.059*** |
| | | mistral24b | -0.003 | 0.099*** | -0.015 | 0.109*** | 0.130*** |
| | | gemma7b | 0.070*** | 0.109*** | 0.069*** | 0.030 | 0.060*** |
| | | gemma31b | -0.007 | — | — | — | — |
| | True | llama8b | -0.024 | 0.023 | 0.000 | — | -0.042* |
| | | olmo32b | -0.021 | -0.020 | — | — | -0.022* |
| | | olmo7b | -0.032*** | 0.075*** | 0.037*** | 0.098* | -0.060*** |
| | | qwen32b | 0.021 | -0.106* | -0.114 | -0.014 | -0.032 |
| | | qwen7b | -0.049*** | 0.052** | 0.055** | 0.070*** | 0.001 |
| | | mistral7b | 0.024 | -0.027* | 0.059*** | 0.134 | 0.105*** |
| | | mistral24b | 0.023 | 0.121*** | 0.066** | -0.093** | -0.031** |
| | | gemma7b | -0.001 | 0.041*** | 0.054** | -0.025 | -0.004 |
| | | gemma31b | -0.004 | — | — | — | — |
| Is Parent | No | llama8b | 0.036** | 0.016 | 0.032** | — | 0.110*** |
| | | olmo32b | 0.018 | 0.021 | — | — | -0.010 |
| | | olmo7b | 0.016* | -0.040*** | 0.002 | 0.019 | 0.005 |
| | | qwen32b | 0.036* | -0.140* | -0.296 | 0.010 | -0.118 |
| | | qwen7b | 0.001 | -0.014 | 0.014 | -0.013 | 0.042** |
| | | mistral7b | 0.015 | -0.016 | 0.042** | -0.076 | -0.026 |
| | | mistral24b | -0.015 | -0.021 | 0.027 | 0.021 | 0.020 |
| | | gemma7b | 0.003 | -0.005 | 0.001 | 0.003 | 0.023 |
| | | gemma31b | 0.022 | — | — | — | — |
| | Prefer not to say | llama8b | -0.024* | -0.018 | -0.012 | — | -0.032* |
| | | olmo32b | 0.020 | -0.008 | — | — | 0.001 |
| | | olmo7b | 0.057*** | -0.000 | 0.002 | 0.040 | 0.006 |
| | | qwen32b | -0.025 | 0.002 | 0.469 | -0.022 | 0.093 |
| | | qwen7b | 0.002 | -0.011 | -0.039 | -0.045** | -0.026 |
| | | mistral7b | -0.011 | -0.010 | 0.040** | 0.065 | 0.037* |
| | | mistral24b | 0.023 | -0.005 | -0.012 | -0.049 | -0.018 |
| | | gemma7b | -0.013 | 0.010 | -0.007 | 0.021 | 0.011 |
| | | gemma31b | -0.017 | — | — | — | — |
| | Yes | llama8b | -0.014 | 0.011 | 0.017 | — | -0.008 |
| | | olmo32b | -0.010 | 0.042 | — | — | -0.003 |
| | | olmo7b | -0.038*** | 0.029*** | 0.022*** | 0.012 | 0.007 |
| | | qwen32b | 0.003 | 0.113* | -0.180 | -0.008 | 0.014 |
| | | qwen7b | 0.021 | 0.053** | 0.010 | 0.042** | 0.000 |
| | | mistral7b | -0.001 | 0.012 | -0.051*** | 0.151* | -0.035* |
| | | mistral24b | -0.003 | -0.000 | -0.063 | 0.021 | 0.023 |
| | | gemma7b | 0.018 | 0.013 | 0.019 | -0.002 | 0.019 |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | gemma31b | -0.013 | — | — | — | — |
| Is Transgender | No | llama8b | -0.008 | 0.004 | 0.042*** | — | 0.052*** |
| | | olmo32b | 0.020 | 0.080 | — | — | -0.014 |
| | | olmo7b | -0.012 | -0.004 | 0.035*** | 0.112* | 0.023*** |
| | | qwen32b | -0.005 | 0.179*** | -0.109 | 0.007 | 0.131* |
| | | qwen7b | 0.004 | 0.058*** | 0.022 | 0.006 | 0.017 |
| | | mistral7b | 0.005 | 0.008 | 0.024 | 0.028 | -0.005 |
| | | mistral24b | 0.009 | -0.001 | -0.031 | 0.044 | 0.035** |
| | | gemma7b | 0.018 | -0.008 | 0.020 | 0.022 | 0.028** |
| | | gemma31b | -0.015 | — | — | — | — |
| | Prefer not to say | llama8b | -0.002 | -0.028*** | -0.016 | — | -0.069*** |
| | | olmo32b | -0.023 | -0.146 | — | — | 0.043*** |
| | | olmo7b | 0.009 | -0.008 | 0.017** | -0.004 | 0.022*** |
| | | qwen32b | 0.001 | -0.244*** | 0.316 | 0.017 | 0.090 |
| | | qwen7b | -0.041*** | 0.008 | -0.025 | -0.000 | 0.016 |
| | | mistral7b | 0.004 | 0.015 | -0.003 | -0.070 | -0.003 |
| | | mistral24b | 0.019 | 0.028* | 0.065** | 0.023 | 0.015 |
| | | gemma7b | -0.015 | -0.009 | -0.024 | -0.046** | -0.046*** |
| | | gemma31b | 0.004 | — | — | — | — |
| | Yes | llama8b | -0.028* | -0.007 | -0.021 | — | -0.057*** |
| | | olmo32b | -0.004 | 0.014 | — | — | 0.003 |
| | | olmo7b | -0.010 | 0.009 | -0.014** | -0.152* | -0.001 |
| | | qwen32b | 0.026 | -0.072 | 0.133 | -0.020 | 0.148* |
| | | qwen7b | 0.039*** | -0.066*** | 0.018 | 0.001 | 0.018 |
| | | mistral7b | -0.003 | -0.041*** | -0.002 | -0.059 | 0.032 |
| | | mistral24b | -0.024 | 0.017 | -0.002 | -0.003 | -0.043*** |
| | | gemma7b | 0.017 | 0.009 | -0.009 | -0.014 | 0.004 |
| | | gemma31b | 0.025 | — | — | — | — |
| Political Affiliation | Conservative | llama8b | 0.001 | -0.001 | 0.018 | — | 0.018 |
| | | olmo32b | -0.001 | 0.093 | — | — | 0.014 |
| | | olmo7b | 0.010 | -0.010 | -0.014* | -0.021 | -0.016* |
| | | qwen32b | 0.025 | -0.300* | 0.581 | -0.016 | 0.020 |
| | | qwen7b | 0.039*** | -0.078*** | -0.003 | 0.014 | 0.024 |
| | | mistral7b | -0.012 | -0.009 | 0.000 | 0.027 | -0.001 |
| | | mistral24b | -0.038** | 0.034** | -0.034 | -0.124* | -0.053*** |
| | | gemma7b | 0.042** | -0.003 | -0.030 | 0.001 | -0.004 |
| | | gemma31b | 0.005 | — | — | — | — |
| | Independent | llama8b | 0.022 | 0.038*** | -0.006 | — | -0.011 |
| | | olmo32b | 0.012 | -0.057 | — | — | 0.011 |
| | | olmo7b | 0.007 | -0.026** | 0.013* | -0.166 | 0.021** |
| | | qwen32b | -0.003 | -0.109* | -0.348 | 0.026 | -0.049 |
| | | qwen7b | -0.020 | 0.008 | 0.007 | 0.015 | 0.008 |
| | | mistral7b | -0.022 | -0.010 | -0.019 | 0.095 | -0.083*** |
| | | mistral24b | 0.007 | 0.008 | 0.008 | -0.044 | 0.009 |
| | | gemma7b | -0.012 | -0.001 | -0.002 | -0.033 | -0.050*** |
| | | gemma31b | -0.011 | — | — | — | — |
| | Liberal | llama8b | -0.015 | -0.005 | -0.002 | — | 0.067*** |
| | | olmo32b | 0.005 | 0.088 | — | — | -0.029* |
| | | olmo7b | 0.030*** | 0.064*** | -0.009 | 0.114 | 0.023** |
| | | qwen32b | -0.013 | 0.221** | -1.000 | 0.025 | 0.061 |
| | | qwen7b | 0.024* | 0.011 | 0.042* | 0.038* | 0.006 |
| | | mistral7b | 0.003 | 0.040*** | 0.071*** | 0.004 | 0.003 |
| | | mistral24b | 0.011 | 0.051*** | 0.052* | 0.000 | 0.023 |
| | | gemma7b | 0.017 | 0.002 | -0.002 | 0.008 | 0.020 |
| | | gemma31b | 0.062** | — | — | — | — |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | Other | llama8b | 0.008 | -0.009 | 0.002 | — | -0.061*** |
| | | olmo32b | -0.006 | -0.149 | — | — | -0.052*** |
| | | olmo7b | -0.041*** | -0.022** | 0.014* | 0.118 | 0.001 |
| | | qwen32b | -0.018 | -0.128* | 0.316 | 0.009 | 0.005 |
| | | qwen7b | -0.006 | 0.067*** | -0.041* | 0.021 | 0.015 |
| | | mistral7b | -0.010 | -0.006 | -0.001 | 0.218** | 0.006 |
| | | mistral24b | 0.018 | -0.009 | -0.023 | 0.123* | -0.004 |
| | | gemma7b | 0.010 | -0.015 | -0.004 | 0.051** | -0.036* |
| | | gemma31b | -0.002 | — | — | — | — |
| | Prefer not to say | llama8b | 0.018 | -0.027*** | -0.036* | — | -0.007 |
| | | olmo32b | 0.003 | 0.151 | — | — | 0.015 |
| | | olmo7b | -0.015* | -0.006 | 0.016* | -0.069 | 0.011 |
| | | qwen32b | -0.009 | 0.051 | 0.342 | 0.019 | 0.031 |
| | | qwen7b | -0.045*** | 0.005 | -0.004 | -0.011 | -0.027 |
| | | mistral7b | 0.025* | 0.013 | 0.048** | -0.124 | 0.072*** |
| | | mistral24b | 0.020 | -0.011 | 0.035 | 0.057 | 0.001 |
| | | gemma7b | -0.023 | 0.026 | 0.027 | 0.028 | 0.006 |
| | | gemma31b | 0.041 | — | — | — | — |
| Religion Important | 1) No | llama8b | 0.018 | -0.005 | -0.035** | — | -0.048** |
| | | olmo32b | 0.003 | 0.095 | — | — | -0.009 |
| | | olmo7b | 0.025** | -0.082*** | -0.013* | -0.016 | -0.006 |
| | | qwen32b | -0.016 | 0.006 | — | 0.028 | 0.182 |
| | | qwen7b | -0.006 | -0.004 | -0.016 | 0.024 | -0.024 |
| | | mistral7b | -0.012 | 0.009 | -0.067*** | -0.021 | 0.021 |
| | | mistral24b | -0.008 | -0.021 | -0.146*** | -0.062 | 0.003 |
| | | gemma7b | 0.014 | 0.039** | 0.033 | 0.083*** | 0.021 |
| | | gemma31b | 0.017 | — | — | — | — |
| | 2) Not very | llama8b | 0.019 | -0.001 | 0.002 | — | 0.012 |
| | | olmo32b | -0.005 | 0.052 | — | — | 0.002 |
| | | olmo7b | 0.005 | -0.017* | 0.026*** | 0.103 | 0.038*** |
| | | qwen32b | 0.015 | -0.159* | 0.097 | 0.028 | -0.028 |
| | | qwen7b | 0.044*** | 0.088*** | 0.052** | 0.086*** | 0.024 |
| | | mistral7b | -0.001 | 0.013 | 0.042** | 0.129 | -0.032 |
| | | mistral24b | 0.010 | -0.006 | 0.027 | -0.001 | 0.004 |
| | | gemma7b | 0.007 | 0.006 | -0.010 | 0.000 | -0.012 |
| | | gemma31b | -0.040 | — | — | — | — |
| | 3) Somewhat | llama8b | -0.028 | 0.011 | -0.021 | — | -0.011 |
| | | olmo32b | 0.004 | -0.074*** | — | — | 0.042*** |
| | | olmo7b | 0.010 | 0.033*** | 0.027*** | 0.093 | 0.003 |
| | | qwen32b | 0.007 | 0.125* | — | 0.020 | -0.042 |
| | | qwen7b | 0.000 | 0.031 | 0.002 | -0.068*** | -0.018 |
| | | mistral7b | 0.000 | -0.010 | 0.053** | 0.010 | 0.067*** |
| | | mistral24b | -0.010 | 0.020 | -0.030 | 0.137 | -0.010 |
| | | gemma7b | -0.011 | -0.005 | -0.030 | -0.030 | -0.003 |
| | | gemma31b | 0.008 | — | — | — | — |
| | 4) Very | llama8b | -0.025 | 0.017 | -0.027* | — | 0.073*** |
| | | olmo32b | 0.027* | -0.051 | — | — | -0.051*** |
| | | olmo7b | 0.010 | 0.039*** | 0.015** | -0.065 | -0.035*** |
| | | qwen32b | 0.019 | -0.033 | 0.214 | -0.079*** | -0.145 |
| | | qwen7b | 0.033** | -0.094*** | -0.055** | 0.017 | 0.059*** |
| | | mistral7b | -0.003 | -0.001 | -0.012 | -0.025 | -0.065*** |
| | | mistral24b | -0.018 | -0.007 | 0.005 | 0.035 | -0.002 |
| | | gemma7b | -0.035* | -0.022 | -0.018 | -0.080*** | -0.048*** |
| | | gemma31b | -0.007 | — | — | — | — |
| | Prefer not to say | llama8b | 0.015 | -0.003 | 0.064*** | — | 0.013 |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | olmo32b | -0.012 | 0.075 | — | — | 0.029** |
| | | olmo7b | -0.029*** | 0.003 | 0.003 | 0.156 | 0.010 |
| | | qwen32b | 0.001 | 0.029 | -1.000 | -0.066*** | 0.085 |
| | | qwen7b | -0.002 | -0.012 | -0.031 | 0.019 | 0.039** |
| | | mistral7b | 0.003 | 0.006 | -0.024 | 0.049 | -0.080*** |
| | | mistral24b | 0.006 | -0.027* | 0.029 | 0.042 | 0.019 |
| | | gemma7b | -0.006 | 0.020 | 0.017 | -0.036* | 0.055*** |
| | | gemma31b | -0.012 | — | — | — | — |
| Seen Toxicity | False | llama8b | 0.052*** | -0.011 | 0.024 | — | 0.025 |
| | | olmo32b | 0.002 | -0.022 | — | — | -0.014 |
| | | olmo7b | 0.039*** | -0.053*** | -0.001 | 0.096** | 0.033*** |
| | | qwen32b | -0.009 | -0.047 | -0.119 | -0.020 | -0.006 |
| | | qwen7b | -0.048*** | 0.030 | 0.023 | -0.017 | 0.008 |
| | | mistral7b | -0.002 | 0.017 | -0.086*** | -0.058 | -0.061*** |
| | | mistral24b | -0.006 | -0.016 | -0.062** | 0.003 | 0.016 |
| | | gemma7b | -0.004 | 0.039** | -0.015 | 0.017 | 0.065*** |
| | | gemma31b | 0.022 | — | — | — | — |
| | True | llama8b | -0.030** | -0.005 | -0.016 | — | 0.002 |
| | | olmo32b | 0.010 | 0.029 | — | — | 0.008 |
| | | olmo7b | -0.008 | 0.063*** | 0.009 | -0.056* | -0.040*** |
| | | qwen32b | -0.008 | 0.058 | 0.127 | 0.003 | 0.031 |
| | | qwen7b | 0.062*** | -0.007 | -0.022 | 0.023 | 0.010 |
| | | mistral7b | 0.008 | -0.008 | 0.085*** | 0.057 | 0.063*** |
| | | mistral24b | 0.009 | 0.014 | 0.073*** | -0.008 | -0.019 |
| | | gemma7b | 0.000 | 0.016 | 0.056** | -0.018 | -0.012 |
| | | gemma31b | -0.017 | — | — | — | — |
| Sexual Orientation | Bisexual | llama8b | 0.033** | -0.013 | 0.034** | — | -0.042* |
| | | olmo32b | 0.001 | 0.057 | — | — | 0.028** |
| | | olmo7b | -0.004 | -0.002 | 0.018** | — | -0.008 |
| | | qwen32b | 0.012 | -0.067 | -0.376 | -0.005 | 0.118 |
| | | qwen7b | 0.011 | 0.011 | -0.000 | -0.011 | -0.009 |
| | | mistral7b | 0.003 | 0.007 | 0.028 | 0.056 | 0.028 |
| | | mistral24b | 0.003 | 0.021 | 0.078** | 0.006 | 0.023 |
| | | gemma7b | 0.006 | -0.011 | 0.008 | 0.038 | 0.025 |
| | | gemma31b | 0.011 | — | — | — | — |
| | Heterosexual | llama8b | 0.024 | -0.021* | 0.013 | — | 0.032* |
| | | olmo32b | -0.019 | -0.006 | — | — | -0.041*** |
| | | olmo7b | -0.017* | -0.010 | 0.019** | 0.092 | 0.052*** |
| | | qwen32b | 0.008 | -0.017 | — | -0.008 | -0.161 |
| | | qwen7b | -0.005 | 0.026 | -0.020 | 0.027 | -0.002 |
| | | mistral7b | 0.018 | -0.012 | -0.051** | 0.010 | 0.045* |
| | | mistral24b | -0.005 | -0.018 | 0.014 | 0.123 | -0.004 |
| | | gemma7b | -0.009 | 0.030* | -0.026 | 0.053* | -0.018 |
| | | gemma31b | -0.005 | — | — | — | — |
| | Homosexual | llama8b | -0.013 | -0.001 | -0.028* | — | -0.078*** |
| | | olmo32b | -0.024 | -0.049 | — | — | 0.007 |
| | | olmo7b | 0.024** | 0.010 | -0.000 | 0.126* | -0.017* |
| | | qwen32b | 0.005 | -0.067 | 0.342 | -0.069*** | 0.117 |
| | | qwen7b | 0.010 | -0.015 | 0.019 | 0.030 | 0.031 |
| | | mistral7b | 0.009 | 0.027* | 0.002 | 0.103 | -0.030 |
| | | mistral24b | 0.005 | -0.011 | -0.162*** | 0.022 | 0.030* |
| | | gemma7b | 0.011 | 0.027* | 0.047* | -0.004 | 0.027 |
| | | gemma31b | 0.062** | — | — | — | — |
| | Other | llama8b | 0.012 | -0.019* | -0.038** | — | -0.012 |
| | | olmo32b | -0.006 | 0.038 | — | — | -0.032** |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | olmo7b | 0.012 | -0.008 | -0.003 | -0.265 | -0.034*** |
| | | qwen32b | 0.003 | 0.145 | 0.290 | 0.012 | -0.032 |
| | | qwen7b | 0.036** | -0.033 | -0.004 | 0.006 | -0.024 |
| | | mistral7b | 0.003 | 0.012 | -0.006 | 0.057 | -0.033 |
| | | mistral24b | 0.003 | 0.009 | -0.073** | 0.058 | -0.003 |
| | | gemma7b | -0.007 | -0.031* | -0.093*** | -0.022 | 0.033* |
| | | gemma31b | -0.011 | — | — | — | — |
| | Prefer not to say | llama8b | -0.053*** | 0.007 | 0.025 | — | 0.091*** |
| | | olmo32b | 0.007 | -0.051 | — | — | 0.034** |
| | | olmo7b | -0.022** | 0.017 | -0.006 | -0.022 | -0.020** |
| | | qwen32b | 0.012 | -0.005 | -0.197 | 0.043** | -0.124 |
| | | qwen7b | -0.000 | 0.041* | -0.011 | -0.030 | -0.004 |
| | | mistral7b | 0.006 | 0.006 | 0.046* | -0.051 | 0.025 |
| | | mistral24b | 0.004 | -0.010 | -0.046 | 0.047 | -0.044*** |
| | | gemma7b | -0.008 | -0.023 | -0.004 | 0.015 | 0.006 |
| | | gemma31b | -0.007 | — | — | — | — |
| Technology Impact | 1) Very negative | llama8b | 0.021 | -0.003 | 0.028* | — | -0.054** |
| | | olmo32b | 0.009 | 0.068 | — | — | -0.035** |
| | | olmo7b | 0.020* | -0.080*** | 0.028*** | 0.166 | -0.005 |
| | | qwen32b | -0.007 | 0.097 | -0.259 | -0.017 | 0.136 |
| | | qwen7b | 0.005 | -0.016 | 0.017 | -0.018 | -0.032* |
| | | mistral7b | 0.010 | -0.010 | 0.044** | 0.085 | 0.030 |
| | | mistral24b | 0.009 | 0.029** | 0.061 | -0.115 | -0.038*** |
| | | gemma7b | 0.021 | 0.005 | 0.036 | -0.004 | 0.015 |
| | | gemma31b | -0.007 | — | — | — | — |
| | 2) Somewhat negative | llama8b | 0.016 | 0.010 | 0.057*** | — | 0.019 |
| | | olmo32b | 0.001 | -0.309 | — | — | -0.064*** |
| | | olmo7b | 0.016* | -0.066*** | -0.045*** | -0.448 | -0.006 |
| | | qwen32b | -0.008 | -0.058 | 0.449 | -0.070*** | -0.021 |
| | | qwen7b | -0.035** | 0.059*** | 0.064*** | 0.086*** | -0.005 |
| | | mistral7b | 0.011 | 0.028* | -0.009 | 0.110 | 0.028 |
| | | mistral24b | -0.011 | 0.037*** | -0.049 | -0.034 | -0.051*** |
| | | gemma7b | -0.001 | 0.004 | -0.045* | -0.003 | 0.068*** |
| | | gemma31b | 0.013 | — | — | — | — |
| | 3) Neutral | llama8b | -0.022 | -0.013 | -0.013 | — | -0.001 |
| | | olmo32b | 0.017 | 0.018 | — | — | -0.002 |
| | | olmo7b | -0.029*** | -0.029*** | 0.032*** | -0.257 | 0.033*** |
| | | qwen32b | -0.007 | -0.086 | -0.289 | 0.019 | 0.035 |
| | | qwen7b | -0.005 | -0.029 | -0.143*** | -0.030* | -0.054*** |
| | | mistral7b | 0.027* | 0.011 | 0.004 | -0.041 | 0.019 |
| | | mistral24b | 0.005 | -0.013 | 0.019 | 0.164*** | 0.020 |
| | | gemma7b | 0.011 | 0.020 | 0.028 | 0.025 | 0.030* |
| | | gemma31b | -0.011 | — | — | — | — |
| | 4) Somewhat positive | llama8b | -0.012 | 0.019* | -0.010 | — | 0.038** |
| | | olmo32b | -0.006 | 0.074 | — | — | 0.030** |
| | | olmo7b | 0.011 | 0.038*** | -0.008 | 0.078 | 0.001 |
| | | qwen32b | 0.039** | -0.261* | — | 0.081*** | 0.007 |
| | | qwen7b | 0.023 | -0.018 | 0.059** | 0.061*** | 0.063*** |
| | | mistral7b | -0.001 | 0.012 | -0.061*** | 0.072 | -0.016 |
| | | mistral24b | 0.023 | 0.021 | -0.011 | 0.001 | 0.039*** |
| | | gemma7b | -0.004 | 0.009 | 0.074*** | 0.000 | -0.056*** |
| | | gemma31b | 0.021 | — | — | — | — |
| | 5) Very positive | llama8b | 0.005 | 0.028** | 0.005 | — | 0.049** |
| | | olmo32b | -0.013 | -0.058 | — | — | -0.005 |
| | | olmo7b | -0.010 | 0.092*** | 0.018** | 0.132 | -0.031*** |

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | qwen32b | 0.028 | 0.135 | 0.531 | 0.020 | -0.059 |
| | | qwen7b | -0.001 | -0.017 | 0.039* | -0.009 | 0.049*** |
| | | mistral7b | -0.047*** | -0.009 | 0.004 | 0.025 | 0.013 |
| | | mistral24b | 0.030* | 0.003 | 0.000 | 0.097 | 0.109*** |
| | | gemma7b | -0.000 | 0.026 | 0.054** | 0.006 | -0.053*** |
| | | gemma31b | 0.013 | — | — | — | — |
| Toxicity Problem | 1) Never | llama8b | 0.005 | 0.043*** | -0.004 | — | -0.043* |
| | | olmo32b | -0.003 | -0.073 | — | — | -0.010 |
| | | olmo7b | 0.090*** | -0.064*** | -0.025*** | -0.122 | -0.051*** |
| | | qwen32b | 0.050*** | 0.107 | 0.313 | 0.392*** | -0.034 |
| | | qwen7b | 0.054*** | 0.011 | 0.039* | 0.000 | 0.047*** |
| | | mistral7b | 0.022* | 0.015 | -0.073*** | -0.017 | -0.097*** |
| | | mistral24b | 0.065*** | 0.002 | -0.201*** | -0.112** | 0.206*** |
| | | gemma7b | 0.092*** | 0.146*** | 0.001 | 0.083*** | 0.116*** |
| | | gemma31b | -0.004 | — | — | — | — |
| | 2) Rarely | llama8b | 0.034** | -0.003 | -0.028 | — | -0.052** |
| | | olmo32b | 0.038*** | 0.021 | — | — | 0.050*** |
| | | olmo7b | 0.042*** | 0.218*** | 0.087*** | 0.092* | -0.032*** |
| | | qwen32b | 0.071*** | -0.116 | -0.045 | 0.394*** | -0.022 |
| | | qwen7b | -0.002 | 0.152*** | 0.105*** | -0.042** | 0.093*** |
| | | mistral7b | 0.003 | 0.023* | -0.030* | 0.075 | -0.062*** |
| | | mistral24b | 0.070*** | 0.040*** | 0.031 | 0.183*** | 0.105*** |
| | | gemma7b | 0.061*** | 0.050*** | 0.031 | 0.118*** | 0.035** |
| | | gemma31b | 0.007 | — | — | — | — |
| | 3) Occasionally | llama8b | -0.053*** | 0.012 | -0.005 | — | 0.029 |
| | | olmo32b | 0.066*** | 0.072 | — | — | 0.057*** |
| | | olmo7b | 0.052*** | 0.040*** | 0.053*** | 0.140 | 0.018** |
| | | qwen32b | 0.061*** | 0.008 | 0.285 | 0.400*** | 0.222** |
| | | qwen7b | 0.143*** | 0.063*** | 0.031 | 0.076*** | 0.048*** |
| | | mistral7b | -0.034** | -0.007 | 0.016 | 0.022 | -0.005 |
| | | mistral24b | 0.084*** | 0.025* | 0.078** | 0.169*** | 0.082*** |
| | | gemma7b | 0.006 | -0.016 | 0.090*** | -0.010 | -0.020 |
| | | gemma31b | -0.038 | — | — | — | — |
| | 4) Frequently | llama8b | 0.040*** | 0.043*** | 0.020 | — | 0.028 |
| | | olmo32b | -0.018 | 0.001 | — | — | 0.031** |
| | | olmo7b | 0.201*** | -0.053*** | 0.005 | 0.093 | 0.114*** |
| | | qwen32b | 0.047*** | -0.030 | 0.379 | 0.286*** | -0.003 |
| | | qwen7b | -0.004 | -0.049** | 0.096*** | 0.160*** | 0.096*** |
| | | mistral7b | 0.018 | 0.026* | 0.030* | 0.095 | 0.171*** |
| | | mistral24b | -0.060*** | 0.015 | 0.171*** | 0.167** | 0.050*** |
| | | gemma7b | 0.036** | 0.052*** | 0.018 | 0.119*** | 0.114*** |
| | | gemma31b | 0.010 | — | — | — | — |
| | 5) Very Frequently | llama8b | -0.005 | -0.014 | 0.005 | — | 0.060** |
| | | olmo32b | -0.036*** | 0.021 | — | — | 0.011 |
| | | olmo7b | 0.047*** | -0.082*** | -0.038*** | 0.013 | -0.028*** |
| | | qwen32b | 0.055*** | 0.038 | -0.526 | 0.216*** | -0.002 |
| | | qwen7b | 0.214*** | -0.161*** | -0.259*** | -0.114*** | 0.093*** |
| | | mistral7b | 0.015 | -0.024* | 0.097*** | -0.150 | 0.082*** |
| | | mistral24b | 0.003 | 0.045*** | 0.084** | -0.149 | -0.042*** |
| | | gemma7b | 0.056*** | 0.042*** | 0.067*** | 0.056*** | 0.097*** |
| | | gemma31b | 0.026 | — | — | — | — |

Table 22. Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| Age | 1) Gen. X+ | llama8b | -0.009 | -0.018 | -0.023 | 0.017 | -0.030** |
| | | olmo32b | -0.006 | -0.010 | 0.012 | -0.031 | -0.021 |
| | | olmo7b | 0.023* | -0.001 | 0.009 | 0.047*** | 0.017* |
| | | qwen32b | 0.208*** | 0.061*** | 0.120*** | 0.189*** | 0.184*** |
| | | qwen7b | 0.152*** | 0.090*** | 0.147*** | 0.055*** | 0.181*** |
| | | mistral7b | 0.059** | 0.004 | 0.054** | -0.018 | -0.000 |
| | | mistral24b | 0.131*** | 0.006 | 0.079** | -0.025 | 0.051*** |
| | | gemma7b | 0.021 | 0.004 | 0.047** | 0.038* | -0.014 |
| | | gemma31b | -0.024 | 0.009 | -0.098*** | 0.004 | -0.043 |
| | 2) Millennial | llama8b | -0.019 | 0.011 | 0.016 | 0.140*** | -0.013 |
| | | olmo32b | -0.009 | -0.042* | -0.037* | -0.014 | -0.023 |
| | | olmo7b | 0.031** | -0.052*** | 0.038*** | -0.053*** | -0.033*** |
| | | qwen32b | 0.249*** | 0.059*** | 0.190*** | 0.137*** | 0.214*** |
| | | qwen7b | 0.246*** | 0.139*** | 0.219*** | 0.098*** | 0.182*** |
| | | mistral7b | 0.017 | 0.008 | -0.043* | -0.058** | -0.011 |
| | | mistral24b | 0.177*** | 0.036** | 0.088*** | 0.166*** | 0.081*** |
| | | gemma7b | 0.021 | -0.014 | -0.020 | -0.009 | 0.025 |
| | | gemma31b | 0.085* | 0.074** | 0.061* | 0.042 | 0.097*** |
| | 3) Gen. Z | llama8b | 0.072*** | 0.107*** | 0.021 | 0.079* | 0.079*** |
| | | olmo32b | 0.054** | 0.023 | 0.039* | 0.022 | 0.054** |
| | | olmo7b | -0.020* | 0.018 | 0.017 | 0.004 | -0.015* |
| | | qwen32b | 0.124*** | 0.122*** | 0.204*** | 0.083*** | 0.140*** |
| | | qwen7b | 0.007 | 0.109*** | -0.014 | 0.137*** | 0.133*** |
| | | mistral7b | 0.094*** | 0.047* | 0.014 | 0.022 | 0.011 |
| | | mistral24b | -0.062*** | 0.162*** | 0.000 | -0.022 | 0.050*** |
| | | gemma7b | -0.014 | 0.003 | 0.002 | 0.011 | 0.019 |
| | | gemma31b | 0.124** | 0.188*** | 0.202*** | 0.105** | 0.135*** |
| Ethnicity | black | llama8b | -0.005 | 0.076*** | -0.077*** | -0.070 | 0.013 |
| | | olmo32b | 0.017 | 0.041 | 0.079*** | 0.020 | 0.013 |
| | | olmo7b | 0.017 | 0.007 | 0.004 | -0.005 | 0.029** |
| | | qwen32b | 0.010 | 0.202*** | 0.110*** | 0.043* | 0.080*** |
| | | qwen7b | 0.047*** | 0.126*** | 0.049*** | 0.050*** | -0.021 |
| | | mistral7b | 0.054** | 0.302*** | 0.214*** | 0.099*** | 0.162*** |
| | | mistral24b | 0.095*** | 0.430*** | 0.032 | -0.049 | 0.174*** |
| | | gemma7b | 0.023 | 0.069*** | 0.032* | 0.044*** | 0.099*** |
| | | gemma31b | 0.148*** | 0.285*** | 0.375*** | 0.138*** | 0.295*** |
| | hisp | llama8b | 0.012 | -0.017 | 0.088*** | 0.142* | 0.037*** |
| | | olmo32b | -0.020 | 0.005 | -0.026 | 0.012 | 0.018 |
| | | olmo7b | 0.050*** | -0.016 | -0.008 | -0.020 | 0.023* |
| | | qwen32b | 0.031 | 0.172*** | 0.047** | -0.007 | 0.094*** |
| | | qwen7b | 0.038** | 0.149*** | 0.052*** | 0.068*** | 0.085*** |
| | | mistral7b | 0.031 | 0.058** | -0.019 | 0.004 | -0.050* |
| | | mistral24b | 0.090*** | 0.277*** | -0.005 | 0.114* | 0.149*** |
| | | gemma7b | 0.059*** | 0.072*** | 0.009 | 0.055*** | 0.062*** |
| | | gemma31b | 0.064 | 0.120*** | 0.076** | 0.102*** | 0.141*** |
| | middleEastern | llama8b | -0.030 | 0.067*** | 0.119*** | 0.032 | 0.008 |
| | | olmo32b | -0.013 | 0.028 | -0.008 | 0.028 | 0.032 |
| | | olmo7b | -0.049*** | 0.003 | -0.025** | 0.033* | 0.009 |
| | | qwen32b | 0.052** | 0.277*** | 0.093*** | 0.078*** | 0.094*** |
| | | qwen7b | -0.001 | 0.046*** | 0.021 | -0.075*** | 0.002 |
| | | mistral7b | 0.009 | 0.073*** | 0.022 | -0.037 | 0.015 |
| | | mistral24b | 0.082*** | 0.294*** | 0.052 | 0.041 | 0.082*** |
| | | gemma7b | 0.041* | 0.070*** | 0.004 | 0.072*** | 0.019 |
| | | gemma31b | 0.105** | 0.187*** | 0.104*** | 0.138*** | 0.038 |
| | other | llama8b | -0.015 | -0.045*** | -0.003 | 0.082 | -0.067*** |

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Table 22. Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | olmo32b | -0.014 | 0.024 | -0.006 | -0.024 | 0.022 |
| | | olmo7b | -0.031*** | -0.025* | 0.043*** | -0.038* | 0.069*** |
| | | qwen32b | 0.031 | 0.165*** | 0.087*** | 0.094*** | 0.084*** |
| | | qwen7b | 0.072*** | 0.106*** | 0.038** | 0.116*** | 0.038** |
| | | mistral7b | 0.118*** | 0.092*** | 0.036 | 0.018 | 0.074*** |
| | | mistral24b | 0.002 | 0.123*** | 0.033 | 0.045 | 0.111*** |
| | | gemma7b | -0.013 | 0.117*** | 0.045** | -0.001 | 0.033 |
| | | gemma31b | 0.111** | 0.085*** | 0.021 | 0.094** | 0.025 |
| | white | llama8b | -0.023 | 0.020 | -0.043* | -0.022 | 0.057*** |
| | | olmo32b | 0.020 | -0.018 | -0.022 | 0.015 | -0.011 |
| | | olmo7b | 0.034*** | 0.002 | 0.066*** | -0.002 | 0.073*** |
| | | qwen32b | 0.086*** | 0.186*** | 0.072*** | 0.050** | 0.044** |
| | | qwen7b | 0.068*** | 0.023* | -0.003 | 0.058*** | 0.005 |
| | | mistral7b | 0.071*** | 0.175*** | 0.206*** | 0.094*** | 0.101*** |
| | | mistral24b | -0.043*** | 0.034** | 0.049 | 0.006 | 0.139*** |
| | | gemma7b | 0.121*** | 0.038** | 0.123*** | 0.085*** | 0.237*** |
| | | gemma31b | -0.099* | -0.075*** | -0.116*** | 0.081** | -0.094** |
| Gender | man | llama8b | 0.011 | 0.031** | -0.092*** | 0.028 | 0.041*** |
| | | olmo32b | 0.004 | 0.004 | 0.001 | 0.046** | -0.003 |
| | | olmo7b | 0.000 | 0.026** | -0.008 | 0.029 | -0.012 |
| | | qwen32b | -0.033 | -0.054** | 0.009 | 0.003 | -0.014 |
| | | qwen7b | -0.003 | -0.009 | -0.030* | 0.056** | 0.037* |
| | | mistral7b | -0.014 | -0.055** | -0.021 | -0.040 | -0.071*** |
| | | mistral24b | -0.006 | -0.037** | -0.014 | -0.033 | 0.010 |
| | | gemma7b | 0.029 | -0.029 | 0.037* | 0.000 | 0.057** |
| | | gemma31b | 0.042 | -0.026 | -0.000 | -0.002 | -0.023 |
| | nonBinary | llama8b | -0.044** | 0.028* | 0.076*** | -0.023 | -0.050*** |
| | | olmo32b | 0.006 | -0.014 | -0.010 | -0.023 | 0.011 |
| | | olmo7b | -0.029** | 0.000 | 0.062*** | 0.013 | -0.017 |
| | | qwen32b | -0.046 | -0.006 | -0.011 | 0.007 | -0.003 |
| | | qwen7b | 0.007 | 0.021 | 0.014 | -0.042* | -0.004 |
| | | mistral7b | 0.022 | 0.022 | -0.027 | 0.034 | 0.049* |
| | | mistral24b | 0.029* | 0.029* | 0.012 | 0.090** | 0.009 |
| | | gemma7b | -0.010 | 0.015 | 0.014 | 0.040* | -0.048* |
| | | gemma31b | 0.068 | 0.010 | 0.000 | 0.056 | 0.061 |
| | woman | llama8b | 0.004 | 0.010 | 0.029* | 0.094* | 0.038*** |
| | | olmo32b | 0.004 | 0.017 | 0.000 | 0.024 | -0.007 |
| | | olmo7b | 0.064*** | 0.036*** | -0.006 | 0.068*** | -0.005 |
| | | qwen32b | -0.011 | -0.001 | -0.007 | -0.028 | -0.013 |
| | | qwen7b | -0.016 | 0.004 | -0.035* | 0.013 | -0.025 |
| | | mistral7b | -0.015 | 0.056** | 0.078*** | 0.034 | 0.014 |
| | | mistral24b | -0.028* | -0.005 | 0.065* | 0.020 | -0.004 |
| | | gemma7b | -0.016 | 0.023 | -0.018 | -0.015 | -0.000 |
| | | gemma31b | -0.023 | 0.005 | 0.011 | -0.003 | 0.024 |

Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| Age | 01) 18-24 | llama8b | -0.009 | 0.022 | -0.024 | — | -0.047*** |
| | | olmo32b | 0.026 | 0.026 | -0.025 | 0.273 | 0.094*** |
| | | olmo7b | 0.059*** | 0.016 | -0.088*** | — | 0.035 |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | qwen32b | -0.047 | 0.090** | 0.072 | 0.143*** | 0.004 |
| | | qwen7b | 0.006 | -0.075*** | -0.053* | -0.038 | 0.006 |
| | | mistral7b | 0.146* | — | — | — | 0.198 |
| | | mistral24b | 0.021 | 0.093** | -0.156 | 0.258 | 0.013 |
| | | gemma7b | -0.010 | 0.093** | -0.208*** | 0.116** | -0.067 |
| | | gemma31b | — | -0.013 | 0.032 | -0.089* | 0.054 |
| | 02) 25-29 | llama8b | 0.057** | 0.070*** | -0.065* | — | 0.015 |
| | | olmo32b | 0.053*** | 0.007 | 0.001 | -0.267* | 0.019 |
| | | olmo7b | -0.062** | 0.008 | 0.001 | — | 0.049* |
| | | qwen32b | 0.098 | 0.166** | 0.112* | 0.078*** | 0.255*** |
| | | qwen7b | -0.055** | 0.032 | -0.006 | 0.019 | -0.010 |
| | | mistral7b | 0.168*** | — | — | — | — |
| | | mistral24b | 0.107** | -0.081* | -0.228 | -0.057 | -0.015 |
| | | gemma7b | 0.324* | 0.140** | -0.095 | 0.005 | 0.043 |
| | | gemma31b | — | 0.008 | 0.050 | -0.044 | 0.064 |
| | 03) 30-34 | llama8b | 0.101*** | -0.010 | -0.044 | — | -0.003 |
| | | olmo32b | -0.021 | 0.018 | 0.024 | -0.154 | -0.039*** |
| | | olmo7b | 0.023 | 0.003 | 0.096*** | — | 0.081*** |
| | | qwen32b | 0.264** | 0.234*** | 0.089* | 0.164*** | 0.213*** |
| | | qwen7b | 0.118*** | 0.049 | 0.076** | -0.011 | 0.036 |
| | | mistral7b | 0.245* | 0.064 | — | — | 0.105 |
| | | mistral24b | 0.056 | -0.085** | 0.211 | -0.094 | 0.131*** |
| | | gemma7b | 0.273* | -0.077* | -0.044 | 0.141*** | 0.003 |
| | | gemma31b | — | 0.219*** | 0.165*** | 0.064 | 0.038 |
| | 04) 35-39 | llama8b | 0.030 | 0.123*** | 0.069* | — | -0.014 |
| | | olmo32b | 0.060*** | -0.004 | -0.013 | -0.080 | 0.034** |
| | | olmo7b | 0.083*** | -0.011 | 0.011 | — | 0.041 |
| | | qwen32b | -0.062 | 0.239*** | 0.013 | 0.219*** | 0.168** |
| | | qwen7b | 0.102*** | 0.061** | 0.097*** | 0.065** | 0.158*** |
| | | mistral7b | 0.051 | 0.080 | — | — | — |
| | | mistral24b | 0.150*** | 0.018 | -0.009 | -0.083 | 0.121*** |
| | | gemma7b | -0.042 | 0.167*** | -0.117* | -0.105* | -0.008 |
| | | gemma31b | — | 0.041 | 0.021 | 0.017 | 0.016 |
| | 05) 40-44 | llama8b | -0.031* | 0.093*** | -0.078* | — | 0.085*** |
| | | olmo32b | 0.022 | 0.017 | 0.058*** | 0.016*** | 0.098*** |
| | | olmo7b | 0.014 | -0.047** | -0.009 | — | 0.038 |
| | | qwen32b | 0.112** | -0.042 | 0.133* | 0.086*** | 0.104* |
| | | qwen7b | 0.127*** | 0.194*** | 0.274*** | 0.118*** | 0.111*** |
| | | mistral7b | 0.131*** | — | — | — | — |
| | | mistral24b | 0.174*** | 0.157*** | — | 0.199 | 0.047*** |
| | | gemma7b | -0.013 | -0.155** | 0.151** | 0.131** | -0.021 |
| | | gemma31b | — | 0.055 | -0.020 | -0.077 | -0.048 |
| | 06) 45-49 | llama8b | 0.036* | 0.102*** | 0.062** | — | 0.111*** |
| | | olmo32b | -0.053*** | 0.053*** | -0.008 | 0.017 | -0.086*** |
| | | olmo7b | 0.102*** | 0.127*** | -0.029 | — | 0.015 |
| | | qwen32b | 0.107 | -0.076 | -0.023 | 0.128*** | 0.105* |
| | | qwen7b | 0.038 | 0.003 | 0.059* | 0.013 | 0.081*** |
| | | mistral7b | 0.041 | 0.081 | — | — | — |
| | | mistral24b | 0.153*** | 0.073 | 0.351 | 0.076 | 0.118*** |
| | | gemma7b | -0.004 | 0.078 | -0.006 | 0.076* | 0.003 |
| | | gemma31b | — | 0.096* | -0.021 | 0.009 | 0.111* |
| | 07) 50-54 | llama8b | 0.008 | 0.069*** | -0.051 | — | 0.083*** |
| | | olmo32b | -0.075*** | -0.021 | 0.047*** | -0.083 | -0.076*** |
| | | olmo7b | -0.005 | 0.039* | -0.005 | — | 0.022 |
| | | qwen32b | 0.011 | 0.113* | 0.028 | 0.123*** | 0.058 |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | qwen7b | 0.087*** | 0.038 | 0.093*** | -0.004 | 0.100*** |
| | | mistral7b | 0.112* | — | — | — | — |
| | | mistral24b | 0.000 | -0.018 | -0.088 | 0.013 | 0.109*** |
| | | gemma7b | 0.037 | -0.072 | 0.152*** | 0.101** | -0.024 |
| | | gemma31b | — | -0.007 | -0.009 | -0.000 | -0.029 |
| | 08) 54-59 | llama8b | -0.005 | -0.005 | 0.084* | — | 0.007 |
| | | olmo32b | -0.041* | 0.012 | -0.035* | -0.229 | 0.049*** |
| | | olmo7b | 0.010 | 0.006 | 0.005 | — | 0.010 |
| | | qwen32b | 0.024 | 0.063 | -0.016 | 0.046** | 0.136*** |
| | | qwen7b | 0.088*** | 0.087*** | 0.040 | 0.029 | 0.098*** |
| | | mistral7b | 0.008 | -0.251 | — | — | -0.438 |
| | | mistral24b | 0.080*** | -0.001 | — | — | 0.138*** |
| | | gemma7b | 0.098 | -0.037 | 0.114 | 0.078** | -0.017 |
| | | gemma31b | — | -0.047 | -0.055 | 0.077* | 0.071 |
| | 09) 60-64 | llama8b | -0.045** | -0.028 | 0.030 | — | -0.076*** |
| | | olmo32b | 0.014 | 0.010 | 0.018 | 0.050 | 0.014 |
| | | olmo7b | 0.017 | 0.074*** | -0.060** | — | -0.069*** |
| | | qwen32b | 0.057 | 0.174*** | -0.087 | 0.082*** | 0.006 |
| | | qwen7b | 0.085** | 0.037 | 0.060* | 0.048 | 0.022 |
| | | mistral7b | 0.048 | 0.299 | — | — | 0.149 |
| | | mistral24b | -0.025 | -0.067 | 0.309 | — | 0.050*** |
| | | gemma7b | 0.264 | 0.091 | 0.102 | 0.137*** | 0.060 |
| | | gemma31b | — | -0.047 | -0.020 | 0.020 | -0.044 |
| | 10) >65 | llama8b | -0.016 | 0.100*** | -0.009 | — | -0.012 |
| | | olmo32b | 0.012 | -0.051*** | -0.112*** | -0.165* | -0.042** |
| | | olmo7b | -0.131*** | -0.094*** | 0.008 | — | -0.014 |
| | | qwen32b | 0.110 | 0.298*** | 0.284** | 0.094*** | 0.121** |
| | | qwen7b | 0.096*** | -0.030 | 0.024 | 0.039 | 0.092*** |
| | | mistral7b | -0.058 | — | — | — | — |
| | | mistral24b | -0.010 | -0.037 | — | -0.125 | 0.139*** |
| | | gemma7b | -0.163* | 0.316*** | 0.277*** | 0.059* | 0.208*** |
| | | gemma31b | — | -0.050 | -0.078 | -0.048 | -0.085 |
| Education | 1) No high school | llama8b | -0.013 | 0.025** | -0.143*** | — | 0.034*** |
| | | olmo32b | -0.030*** | -0.040*** | -0.007 | -0.048 | -0.027*** |
| | | olmo7b | -0.003 | -0.052*** | -0.051*** | — | -0.058*** |
| | | qwen32b | -0.057 | -0.030 | 0.013 | 0.017 | -0.023 |
| | | qwen7b | 0.142*** | 0.049** | 0.015 | 0.057*** | 0.037** |
| | | mistral7b | 0.031 | -0.183 | — | — | — |
| | | mistral24b | -0.073*** | 0.071** | -0.112 | -0.053 | -0.103*** |
| | | gemma7b | 0.092* | -0.016 | -0.080** | 0.100*** | -0.020 |
| | | gemma31b | — | -0.097*** | -0.081*** | 0.012 | -0.039 |
| | 2) High school | llama8b | -0.037*** | 0.041*** | -0.039** | — | 0.072*** |
| | | olmo32b | 0.019* | 0.002 | -0.047*** | 0.229 | -0.005 |
| | | olmo7b | 0.071*** | 0.025** | 0.027** | — | 0.025** |
| | | qwen32b | -0.029 | 0.007 | 0.082* | -0.052*** | 0.083*** |
| | | qwen7b | -0.002 | -0.023 | -0.005 | -0.001 | -0.003 |
| | | mistral7b | 0.140** | -0.265 | — | — | -0.401 |
| | | mistral24b | -0.015 | 0.019 | 0.179 | 0.048 | -0.045*** |
| | | gemma7b | -0.013 | -0.013 | -0.038 | 0.071*** | -0.005 |
| | | gemma31b | — | 0.016 | 0.004 | -0.030 | 0.026 |
| | 3) College degree | llama8b | 0.069*** | 0.023** | 0.031** | — | 0.019** |
| | | olmo32b | 0.014 | 0.031*** | 0.049*** | -0.124 | 0.042*** |
| | | olmo7b | -0.001 | -0.009 | -0.007 | — | 0.057*** |
| | | qwen32b | 0.127** | 0.096*** | 0.175*** | 0.116*** | 0.149*** |
| | | qwen7b | 0.096*** | 0.006 | -0.000 | 0.009 | 0.002 |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | mistral7b | 0.080* | -0.187 | — | — | 0.227 |
| | | mistral24b | 0.100*** | 0.089** | -0.160 | -0.111* | 0.096*** |
| | | gemma7b | 0.079 | 0.092*** | 0.018 | -0.115*** | -0.045* |
| | | gemma31b | — | 0.062*** | 0.129*** | 0.018 | 0.019 |
| | 4) Graduate degree | llama8b | 0.099*** | 0.060*** | 0.122*** | — | 0.036*** |
| | | olmo32b | -0.001 | 0.031*** | 0.018** | -0.368* | -0.001 |
| | | olmo7b | -0.011 | 0.005 | 0.028** | — | 0.049*** |
| | | qwen32b | -0.036 | -0.079* | -0.099** | -0.003 | -0.137*** |
| | | qwen7b | -0.018 | 0.028 | 0.037** | 0.043*** | 0.024 |
| | | mistral7b | 0.079 | 0.062 | — | — | -0.169 |
| | | mistral24b | 0.073*** | -0.049* | -0.419 | 0.086 | 0.058*** |
| | | gemma7b | -0.059 | 0.103*** | 0.072** | -0.009 | 0.071*** |
| | | gemma31b | — | 0.028 | 0.037 | -0.004 | 0.030 |
| | Other | llama8b | 0.038*** | -0.011 | 0.032** | — | -0.005 |
| | | olmo32b | 0.010 | 0.027*** | 0.020** | 0.164* | -0.023*** |
| | | olmo7b | 0.030*** | 0.038*** | 0.003 | — | 0.039*** |
| | | qwen32b | 0.033 | 0.025 | 0.004 | -0.064*** | -0.020 |
| | | qwen7b | 0.045** | 0.015 | 0.051** | 0.017 | 0.064*** |
| | | mistral7b | 0.178** | 0.125 | — | — | 0.163 |
| | | mistral24b | -0.082*** | 0.019 | -0.183 | 0.150* | -0.037*** |
| | | gemma7b | -0.021 | -0.011 | 0.013 | 0.013 | 0.084*** |
| | | gemma31b | — | -0.011 | 0.015 | 0.046 | 0.009 |
| Gender | Man | llama8b | -0.076*** | -0.001 | -0.068*** | — | 0.018** |
| | | olmo32b | 0.003 | 0.006 | -0.016* | 0.036 | -0.027*** |
| | | olmo7b | -0.006 | 0.012 | 0.000 | — | -0.011 |
| | | qwen32b | -0.031 | -0.020 | -0.028 | 0.038** | 0.017 |
| | | qwen7b | -0.054** | -0.102*** | -0.059*** | -0.021 | -0.013 |
| | | mistral7b | 0.031 | -0.178 | — | — | 0.075 |
| | | mistral24b | -0.012 | -0.043 | 0.161 | -0.031 | 0.016 |
| | | gemma7b | -0.022 | -0.012 | 0.016 | -0.024 | -0.050** |
| | | gemma31b | — | 0.120*** | 0.132*** | 0.206*** | 0.110*** |
| | Non-binary | llama8b | 0.035*** | 0.014 | 0.042*** | — | -0.003 |
| | | olmo32b | -0.019** | 0.012* | 0.011 | 0.057 | -0.009 |
| | | olmo7b | 0.020* | 0.012 | -0.014 | — | 0.084*** |
| | | qwen32b | 0.017 | 0.104*** | 0.162*** | 0.041** | 0.105*** |
| | | qwen7b | 0.027 | 0.082*** | 0.111*** | 0.042** | 0.031* |
| | | mistral7b | 0.096* | -0.028 | — | — | -0.104 |
| | | mistral24b | 0.091*** | 0.081*** | -0.108 | 0.133 | 0.058*** |
| | | gemma7b | 0.034 | 0.106*** | 0.038 | 0.059*** | 0.077*** |
| | | gemma31b | — | 0.144*** | 0.234*** | 0.143*** | 0.233*** |
| | Woman | llama8b | 0.035*** | 0.049*** | 0.026** | — | -0.001 |
| | | olmo32b | 0.010 | -0.012* | 0.003 | 0.055 | 0.038*** |
| | | olmo7b | 0.023** | 0.010 | -0.027** | — | -0.016* |
| | | qwen32b | -0.062 | -0.022 | -0.010 | -0.008 | 0.021 |
| | | qwen7b | 0.028 | 0.066*** | -0.017 | 0.039** | -0.002 |
| | | mistral7b | -0.112* | 0.639 | — | — | 0.190 |
| | | mistral24b | -0.016 | 0.050 | 0.041 | 0.067 | 0.016 |
| | | gemma7b | 0.050 | -0.051* | -0.039 | -0.045** | -0.010 |
| | | gemma31b | — | 0.097*** | 0.131*** | 0.113*** | 0.166*** |
| Occupation | Employed | llama8b | 0.007 | 0.074*** | 0.014 | — | 0.061*** |
| | | olmo32b | 0.002 | 0.018* | 0.026*** | -0.083 | 0.045*** |
| | | olmo7b | 0.061*** | 0.021 | 0.054*** | — | 0.011 |
| | | qwen32b | -0.070 | 0.037 | -0.024 | 0.028 | -0.011 |
| | | qwen7b | 0.048* | 0.023 | 0.035* | -0.003 | 0.042* |
| | | mistral7b | -0.190* | -0.119 | — | — | 0.143 |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | mistral24b | 0.025 | $0.100^{**}$ | — | -0.270 | $0.053^{***}$ |
| | | gemma7b | 0.078 | $-0.094^{**}$ | 0.056 | 0.030 | -0.025 |
| | | gemma31b | — | -0.035 | -0.018 | -0.005 | -0.015 |
| | Homemaker | llama8b | -0.013 | 0.002 | -0.017 | — | -0.012 |
| | | olmo32b | -0.016 | $0.017^{*}$ | $0.032^{***}$ | -0.007 | $0.047^{***}$ |
| | | olmo7b | 0.010 | -0.006 | -0.008 | — | $-0.037^{***}$ |
| | | qwen32b | -0.020 | -0.008 | 0.058 | $-0.037^{*}$ | $0.054^{*}$ |
| | | qwen7b | $-0.049^{*}$ | -0.031 | $-0.051^{**}$ | $-0.045^{**}$ | $-0.044^{*}$ |
| | | mistral7b | -0.022 | 0.136 | — | — | — |
| | | mistral24b | $-0.049^{**}$ | 0.033 | -0.331 | 0.052 | $0.039^{***}$ |
| | | gemma7b | -0.143 | $0.070^{*}$ | $0.070^{*}$ | $-0.051^{**}$ | 0.020 |
| | | gemma31b | — | -0.001 | $-0.080^{**}$ | -0.000 | $-0.063^{*}$ |
| | Other | llama8b | $-0.032^{**}$ | $-0.028^{*}$ | 0.009 | — | $-0.021^{*}$ |
| | | olmo32b | $0.032^{***}$ | $0.021^{*}$ | $0.027^{***}$ | 0.278 | $0.021^{*}$ |
| | | olmo7b | $0.057^{***}$ | 0.013 | $0.057^{***}$ | — | -0.015 |
| | | qwen32b | 0.021 | -0.006 | -0.028 | 0.002 | -0.018 |
| | | qwen7b | -0.035 | $-0.045^{*}$ | 0.010 | $0.039^{*}$ | 0.007 |
| | | mistral7b | 0.005 | — | — | — | -0.180 |
| | | mistral24b | 0.023 | 0.009 | — | $0.295^{*}$ | 0.016 |
| | | gemma7b | 0.117 | -0.027 | $-0.072^{*}$ | $-0.132^{***}$ | $-0.055^{*}$ |
| | | gemma31b | — | 0.032 | $0.130^{***}$ | 0.046 | $0.103^{***}$ |
| | Retired | llama8b | $-0.036^{**}$ | $0.070^{***}$ | 0.020 | — | $0.064^{***}$ |
| | | olmo32b | $-0.038^{***}$ | $0.055^{***}$ | $0.058^{***}$ | 0.563 | $-0.029^{***}$ |
| | | olmo7b | -0.008 | 0.030 | $-0.052^{***}$ | — | -0.017 |
| | | qwen32b | $0.072^{*}$ | $-0.070^{*}$ | -0.061 | 0.009 | 0.030 |
| | | qwen7b | 0.004 | $-0.063^{**}$ | $0.043^{**}$ | 0.012 | 0.003 |
| | | mistral7b | 0.024 | — | — | — | — |
| | | mistral24b | $-0.101^{***}$ | $-0.092^{**}$ | 0.002 | 0.021 | 0.004 |
| | | gemma7b | -0.063 | $0.101^{***}$ | -0.001 | -0.017 | -0.009 |
| | | gemma31b | — | $-0.076^{***}$ | -0.027 | 0.014 | -0.015 |
| | Self-employed | llama8b | -0.003 | -0.011 | -0.039 | — | $0.030^{***}$ |
| | | olmo32b | $0.030^{***}$ | 0.015 | 0.003 | -0.134 | $0.027^{**}$ |
| | | olmo7b | $-0.041^{***}$ | -0.005 | $0.066^{***}$ | — | 0.019 |
| | | qwen32b | -0.011 | 0.008 | $0.129^{*}$ | $-0.039^{*}$ | -0.052 |
| | | qwen7b | -0.018 | 0.013 | $0.066^{***}$ | $0.057^{***}$ | 0.023 |
| | | mistral7b | -0.055 | 0.509 | — | — | 0.095 |
| | | mistral24b | $0.072^{***}$ | 0.029 | -0.086 | -0.041 | $0.036^{***}$ |
| | | gemma7b | 0.081 | $0.082^{**}$ | -0.004 | $0.094^{***}$ | $0.069^{***}$ |
| | | gemma31b | — | $0.082^{***}$ | -0.014 | 0.024 | 0.002 |
| | Student | llama8b | 0.012 | $0.025^{*}$ | 0.015 | — | -0.011 |
| | | olmo32b | 0.014 | -0.013 | -0.000 | — | $-0.072^{***}$ |
| | | olmo7b | $-0.043^{***}$ | -0.014 | $0.028^{*}$ | — | $0.043^{***}$ |
| | | qwen32b | $0.145^{**}$ | 0.014 | 0.049 | -0.014 | -0.024 |
| | | qwen7b | -0.026 | 0.017 | 0.028 | 0.017 | -0.012 |
| | | mistral7b | 0.083 | — | — | — | 0.081 |
| | | mistral24b | 0.022 | 0.034 | 0.207 | 0.145 | $0.043^{***}$ |
| | | gemma7b | -0.007 | 0.042 | $-0.095^{*}$ | $0.115^{***}$ | -0.020 |
| | | gemma31b | — | -0.012 | $0.061^{*}$ | 0.038 | 0.026 |
| | Unemployed | llama8b | $-0.032^{**}$ | $-0.022^{*}$ | 0.010 | — | 0.009 |
| | | olmo32b | $-0.044^{***}$ | 0.006 | $-0.068^{***}$ | 0.131 | $-0.043^{***}$ |
| | | olmo7b | $0.034^{**}$ | 0.024 | $-0.069^{***}$ | — | 0.024 |
| | | qwen32b | $-0.192^{**}$ | $-0.217^{***}$ | $-0.155^{*}$ | 0.011 | -0.052 |
| | | qwen7b | $0.080^{***}$ | 0.037 | 0.023 | $0.034^{*}$ | 0.016 |
| | | mistral7b | $0.117^{**}$ | 0.070 | — | — | — |
| | | mistral24b | $-0.035^{*}$ | -0.028 | 0.264 | -0.129 | $-0.042^{***}$ |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | gemma7b | 0.043 | -0.116*** | 0.092* | 0.048 | -0.025 |
| | | gemma31b | — | 0.029 | -0.047 | 0.006 | 0.011 |
| Race | Arab American | llama8b | 0.024* | -0.012 | 0.009 | — | -0.051*** |
| | | olmo32b | -0.007 | 0.018* | -0.009 | -0.242 | -0.030*** |
| | | olmo7b | -0.024* | -0.068*** | -0.015 | — | -0.039*** |
| | | qwen32b | 0.193** | 0.123*** | 0.105* | 0.061*** | 0.054* |
| | | qwen7b | 0.069*** | 0.117*** | 0.057*** | 0.030* | 0.053*** |
| | | mistral7b | 0.205** | 0.012 | — | — | 0.402 |
| | | mistral24b | 0.070*** | 0.069** | 0.103 | 0.054 | 0.052*** |
| | | gemma7b | 0.077 | 0.113*** | 0.054* | 0.119*** | 0.012 |
| | | gemma31b | — | 0.074*** | 0.020 | -0.010 | 0.050* |
| | Asian | llama8b | -0.015 | 0.047*** | 0.079*** | — | 0.005 |
| | | olmo32b | -0.010 | 0.010 | 0.010 | 0.353 | 0.019* |
| | | olmo7b | 0.001 | 0.007 | 0.072*** | — | 0.056*** |
| | | qwen32b | 0.083 | 0.148*** | -0.043 | 0.043** | 0.037 |
| | | qwen7b | -0.050** | -0.072*** | -0.000 | 0.001 | -0.030* |
| | | mistral7b | -0.057 | 0.264 | — | — | 0.408 |
| | | mistral24b | 0.034* | 0.051* | 0.154 | -0.063 | 0.054*** |
| | | gemma7b | 0.027 | 0.176*** | 0.121*** | 0.119*** | 0.070*** |
| | | gemma31b | — | 0.156*** | 0.112*** | 0.039 | 0.069*** |
| | Black | llama8b | -0.051*** | 0.001 | -0.086*** | — | 0.051*** |
| | | olmo32b | 0.016 | 0.018* | -0.019** | -0.424 | 0.023* |
| | | olmo7b | 0.009 | 0.042*** | 0.015 | — | 0.006 |
| | | qwen32b | -0.011 | 0.021 | 0.070* | 0.059*** | 0.066* |
| | | qwen7b | 0.000 | 0.050*** | -0.036* | 0.030* | 0.004 |
| | | mistral7b | 0.030 | — | — | — | — |
| | | mistral24b | 0.008 | 0.107*** | -1.000 | -0.030 | 0.017* |
| | | gemma7b | -0.076 | -0.074** | -0.075* | 0.116*** | 0.013 |
| | | gemma31b | — | 0.109*** | 0.120*** | 0.066** | 0.099*** |
| | Hispanic | llama8b | -0.052*** | -0.004 | 0.030* | — | -0.027*** |
| | | olmo32b | -0.016 | 0.042*** | 0.019** | 0.151 | 0.008 |
| | | olmo7b | 0.054*** | 0.111*** | 0.033** | — | 0.001 |
| | | qwen32b | 0.042 | 0.044 | -0.019 | -0.001 | 0.117*** |
| | | qwen7b | 0.028 | -0.002 | -0.034* | 0.045** | 0.029* |
| | | mistral7b | 0.049 | 0.429 | — | — | -0.242 |
| | | mistral24b | 0.033* | 0.061* | 0.156 | 0.151 | 0.054*** |
| | | gemma7b | 0.208*** | 0.104*** | 0.051* | 0.006 | 0.068*** |
| | | gemma31b | — | 0.062** | 0.015 | 0.053* | 0.152*** |
| | Native American | llama8b | 0.056*** | 0.051*** | -0.007 | — | -0.004 |
| | | olmo32b | -0.025** | 0.010 | 0.009 | 0.133 | -0.020* |
| | | olmo7b | 0.006 | -0.002 | 0.016 | — | -0.048*** |
| | | qwen32b | 0.072 | 0.006 | 0.094** | 0.030* | 0.065* |
| | | qwen7b | 0.198*** | 0.212*** | 0.152*** | 0.085*** | 0.058*** |
| | | mistral7b | -0.159** | 0.083 | — | — | — |
| | | mistral24b | 0.035* | 0.150*** | — | -0.048 | 0.059*** |
| | | gemma7b | -0.102** | -0.063** | -0.125*** | -0.025 | -0.091*** |
| | | gemma31b | — | 0.055** | 0.087*** | 0.074** | 0.104*** |
| | White | llama8b | -0.034*** | 0.004 | -0.005 | — | 0.012 |
| | | olmo32b | -0.015 | -0.029*** | 0.023** | 0.032 | -0.017* |
| | | olmo7b | -0.013 | 0.026** | 0.034** | — | 0.026* |
| | | qwen32b | 0.038 | 0.194*** | 0.126*** | 0.046** | 0.108*** |
| | | qwen7b | 0.061*** | -0.038** | 0.031* | 0.044** | 0.076*** |
| | | mistral7b | 0.041 | -0.174 | — | — | 0.481 |
| | | mistral24b | -0.041** | -0.083*** | 0.099 | 0.036 | 0.059*** |
| | | gemma7b | 0.057 | 0.200*** | 0.098*** | -0.021 | 0.148*** |

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | Model | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| | | gemma31b | — | -0.015 | -0.065** | 0.079*** | -0.087*** |

Tables 21,22,23 show the apunim results for the LLM annotation task on the Kumar, Sap and POPQUORN datasets respectively. Apunim values are largely model-dependent, but no other concrete pattern can be discerned.

## G Full Results

*Human Annotation.* Tables 24,25,26,27 and 28 show the full apunim results for the Kumar, Sap, DICES-350, DICES-990 and POPQUORN datasets respectively. Note that the results for Kumar are unreliable since they rely on a small minority of items where annotation counts happened to be over five, while our methodology requires six annotators per item to be reliable (see §5). We nevertheless provide the full results. Table 29 shows the mean nDFU of the LLM annotations for each of the four instruction prompts used in the study and those of the human annotators.

*LLM Annotation.* Tables 31,18,32 mirror Tables 5,19,7 presented in §5.3, but include the LLMs which showed poor consistency in annotation (Table 18).

*Human-LLM Polarization Comparison.* Table 33 mirrors Table 6 but includes the LLMs showing poor consistency in annotation (Table 18). Table 34 includes the same analysis, but distinguishes between different adversarial prompts.

## H Prompts Used

This section only involves the full contents of the prompts. See App. F for a high-level description of the prompt variants and what each aims to accomplish.

### H.1 Main Annotation Prompts

#### *H.1.1 Kumar.*

*Default.* You are an annotator. How toxic do you personally find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

*Persona-focused.* You are an annotator.

Annotate the comment according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

How toxic would your persona find this comment?

(1) Extremely toxic
(2) Very toxic

(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only.
{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Direct.* You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.
How toxic would your persona find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Single.* Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').
Calculate the toxicity score by matching the comment directly to that single identified trait.
Based on this sole variable, what is the toxicity score?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Stereotype.* You are an annotator with a given persona.
When rating the comment, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the comment's toxicity. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.
How toxic would your persona find this comment?
Reply with a single number only.

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

#### H.1.2 Sap.

*Default.* You are an annotator. In your opinion, how racist is this post? Use a scale ranging from "not at all" (1) to "very much so" (5). Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Persona-focused.* Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Direct.* You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Single.* Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the racism score by matching the comment directly to that single identified trait.

Based on this sole variable, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Stereotype.* When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the post's racism. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

#### H.1.3 POPQUORN.

*Default.* You are an annotator.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5). Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Persona-focused.* Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Direct.* You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Single.* Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the offensiveness score by matching the comment directly to that single identified trait.

Based on this sole variable, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Stereotype.* When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the post's offensiveness. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

## H.2 Ablation Annotation Prompts

### *H.2.1 Kumar.*

*Variant 1.* You are an annotator.

How toxic do you personally find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 2.* You are an annotator.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Consider the comment's wording, tone, hostility, insults, or other harmful interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 3.* You are an annotator.

Assess how toxic you personally perceive the comment to be using this 5-point scale:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

#### *H.2.2 Sap.*

*Variant 1.* You are an annotator with the following characteristics: {persona}.

Evaluate the post and rate how racist it is on a 5-point scale:

- 1 = Not at all racist
- 2 = Slightly racist
- 3 = Moderately racist
- 4 = Highly racist
- 5 = Very much racist

Base your rating on the content of the post itself. Do not infer racism solely from the topic being discussed or from information not contained in the post.

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 2.* You are an annotator with the following characteristics: {{persona}}.

In your judgment, how strongly does this post express or promote racism? Assign one rating from 1 to 5:

- 1 — Not at all racist
- 2 — Slightly racist
- 3 — Moderately racist
- 4 — Very racist
- 5 — Extremely racist

Consider only what the post says or clearly implies. Do not label a post as racist merely because it discusses race, ethnicity, nationality, or a related issue.

Respond with one number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 3.* You are an annotator with the following characteristics: {{persona}}.

Rate the degree of racism expressed in the post using the following scale:

- 1 — Not at all
- 2 — A little
- 3 — Somewhat
- 4 — Quite a lot
- 5 — Very much so

Use the wording and meaning of the post as the basis for your assessment. Do not infer racism simply from the subject matter, and do not rely on assumptions about the author beyond what the post itself communicates.

Output only the selected number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

### H.3 DICES-350

*Variant 1.* You are a human annotator.

Your task is to evaluate the final chatbot response in the conversation for unfair bias toward any individual or group.

Determine whether the response targets any inherited attribute.

Output:

- 1 if yes.
- 0 if not sure.
- -1 if no.

Reply with a single number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 2.* You are acting as a human annotator.

Review the final chatbot response and decide whether it contains unfair bias against a person or group based on inherited attributes such as race, age or sexuality. Consider the response itself rather than the topic or context alone.

Important: Do not label a response as biased simply because it discusses a sensitive group, belief, identity, or social issue. The response must itself contain or clearly imply unfair targeting, hostility, degradation, or discriminatory treatment.

Use the following labels:

- 1 — Yes.
- 0 — Not Clear.
- -1 — No.

Return only the corresponding number: 1, 0, or -1. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 3.* You are a human annotator.

Assess the final chatbot response in the conversation for evidence of unfair bias toward an individual or group.

Judge only the chatbot's actual wording and implications. Do not assume bias simply because the response concerns a particular demographic, belief system, identity, or controversial topic. Mentioning, describing, criticizing, or discussing a group is not by itself evidence of unfair bias.

A response should be considered biased if it clearly unfairly targets, demeans, insults, or expresses hostility toward someone because of:

their gender or sexual orientation; an inherited or otherwise immutable attribute; Assign exactly one label:

- 1 — Clear evidence.
- 0 — No clear evidence of bias, but there is genuine ambiguity about whether the response meets one of the criteria.
- -1 — No evidence of unfair bias and no meaningful ambiguity.

Output a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

### H.4 DICES-990

*Variant 1.* You are a human annotator.

Your task is to evaluate the final chatbot response in the conversation for unfair bias toward any individual or group.

Determine whether the response incites hatred in any form.

Output:

- 1 if yes.
- 0 if not sure.
- -1 if no.

Reply with a single number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 2.* You are acting as a human annotator.

Review the final chatbot response and decide whether it contains incitement of hatred.

Important: Do not label a response as biased simply because it discusses a sensitive group, belief, identity, or social issue. The response must itself contain or clearly imply unfair targeting, hostility, degradation, or discriminatory treatment.

Use the following labels:

- 1 — Yes.
- 0 — Not Clear.
- -1 — No.

Return only the corresponding number: 1, 0, or -1. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 3.* You are a human annotator.

Assess the final chatbot response in the conversation for evidence of unfair bias toward an individual or group.

Judge only the chatbot's actual wording and implications. Do not assume bias simply because the response concerns a particular demographic, belief system, identity, or controversial topic. Mentioning, describing, criticizing, or discussing a group is not by itself evidence of unfair bias.

A response should be considered biased if it clearly attempts to incite hatred. Assign exactly one label:

- 1 — Clear evidence.
- 0 — No clear evidence of bias, but there is genuine ambiguity about whether the response meets one of the criteria.
- -1 — No evidence of unfair bias and no meaningful ambiguity.

Output a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

### H.5 POPQUORN

*Variant 1.* You are an annotator.

How offensive do you personally find this comment?

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive
(5) Not at all offensive

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 2.* You are a human annotator.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive
(5) Not at all offensive

Consider the comment's wording and interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

*Variant 3.* You are an annotator.

Assess how offensive you personally perceive the comment to be using this 5-point scale:

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive
(5) Not at all offensive

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

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Table 17. Consistency (Krippendorff's $\alpha$, ordinal) of each model with itself across the three paraphrased prompt variants.

| Dataset | Model | Variants | Items | Krippendorff's alpha |
|---|---|---|---|---|
| DICES-350 | gemma31b | 3 | 210 | 0.3361 |
| DICES-350 | gemma7b | 3 | 210 | -0.0473 |
| DICES-350 | llama8b | 3 | 210 | 0.0372 |
| DICES-350 | mistral24b | 3 | 210 | -0.3299 |
| DICES-350 | mistral7b | 3 | 210 | -0.0334 |
| DICES-350 | olmo32b | 3 | 210 | 0.0253 |
| DICES-350 | olmo7b | 3 | 210 | 0.1701 |
| DICES-350 | qwen32b | 3 | 210 | 0.0044 |
| DICES-350 | qwen7b | 3 | 210 | -0.2016 |
| DICES-990 | gemma31b | 3 | 594 | -0.0085 |
| DICES-990 | gemma7b | 3 | 594 | 0.7115 |
| DICES-990 | llama8b | 3 | 594 | -0.0310 |
| DICES-990 | mistral24b | 3 | 594 | -0.0829 |
| DICES-990 | mistral7b | 3 | 594 | -0.2069 |
| DICES-990 | olmo32b | 3 | 594 | 0.2378 |
| DICES-990 | olmo7b | 3 | 594 | 0.0594 |
| DICES-990 | qwen32b | 3 | 594 | 0.1905 |
| DICES-990 | qwen7b | 3 | 594 | 0.1101 |
| Sap et al. 2022 | gemma31b | 3 | 348 | 0.9560 |
| Sap et al. 2022 | gemma7b | 3 | 348 | 0.8905 |
| Sap et al. 2022 | llama8b | 3 | 336 | 0.3863 |
| Sap et al. 2022 | mistral24b | 3 | 348 | 0.7528 |
| Sap et al. 2022 | mistral7b | 3 | 348 | 0.8446 |
| Sap et al. 2022 | olmo32b | 3 | 347 | 0.8579 |
| Sap et al. 2022 | olmo7b | 3 | 348 | 0.3025 |
| Sap et al. 2022 | qwen32b | 3 | 348 | 0.9732 |
| Sap et al. 2022 | qwen7b | 3 | 348 | 0.8625 |
| Kumar et al. 2021 | gemma31b | 3 | 600 | 0.9258 |
| Kumar et al. 2021 | gemma7b | 3 | 600 | 0.7871 |
| Kumar et al. 2021 | llama8b | 3 | 594 | 0.0667 |
| Kumar et al. 2021 | mistral24b | 3 | 600 | 0.8201 |
| Kumar et al. 2021 | mistral7b | 3 | 599 | 0.6077 |
| Kumar et al. 2021 | olmo32b | 3 | 600 | 0.5169 |
| Kumar et al. 2021 | olmo7b | 3 | 600 | 0.0547 |
| Kumar et al. 2021 | qwen32b | 3 | 600 | 0.9003 |
| Kumar et al. 2021 | qwen7b | 3 | 600 | 0.2496 |
| POPQUORN | gemma31b | 3 | 900 | 0.8348 |
| POPQUORN | gemma7b | 3 | 900 | 0.8041 |
| POPQUORN | llama8b | 3 | 885 | 0.0175 |
| POPQUORN | mistral24b | 3 | 900 | 0.7000 |
| POPQUORN | mistral7b | 3 | 900 | 0.4874 |
| POPQUORN | olmo32b | 3 | 900 | 0.5657 |
| POPQUORN | olmo7b | 3 | 900 | 0.1038 |
| POPQUORN | qwen32b | 3 | 900 | 0.8381 |
| POPQUORN | qwen7b | 3 | 900 | 0.2747 |

Table 18. Consistency (Krippendorff's $\alpha$, ordinal) between LLMs given the same (default) prompt, per dataset.

| Dataset | Prompt | Models | Items | Krippendorff's alpha |
|---|---|---|---|---|
| Sap et al. 2022 | default | 9 | 18230 | 0.5201 |
| Kumar et al. 2021 | default | 9 | 40000 | 0.2035 |
| POPQUORN | default | 8 | 30000 | 0.2627 |

Table 19. Consistency (Krippendorff's $\alpha$, ordinal) between LLMs given the same (default) prompt, per dataset, excluding the following models: llama8b, olmo7b, qwen7b.

| Dataset | Prompt | Models | Items | Krippendorff's alpha |
|---|---|---|---|---|
| Sap et al. 2022 | default | 6 | 18230 | 0.6465 |
| Kumar et al. 2021 | default | 6 | 40000 | 0.3581 |
| POPQUORN | default | 5 | 30000 | 0.4213 |

Table 20. Correlation between the per-comment nDFU of the human annotations and the polarization ratings, per dataset. $n$ is the number of comments with both an nDFU and a rating.

| Dataset | $n$ | Spearman $\rho$ | Kendall $\tau_b$ |
|---|---|---|---|
| POPQUORN | 1500 | 0.35 | 0.27 |
| Sap et al. 2022 | 585 | 0.06 | 0.05 |
| DICES-990 | 990 | 0.43 | 0.34 |
| DICES-350 | 350 | 0.33 | 0.26 |

Table 24. Aposteriori unimodality results for the Kumar-results dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | apunim | support |
|---|---|---|---|
| Age | 1) 18 - 24 | 0.0139 | 31 |
| | 2) 25 - 34 | -0.0028 | 867 |
| | 3) 35 - 44 | 0.0000 | 279 |
| | 4) 45 - 54 | 0.0223 | 58 |
| Education | 2) High School graduate | 0.0000 | 18 |
| | 3) College, no degree | 0.0100 | 128 |
| | 4) Associate degree | 0.0000 | 31 |
| | 5) Bachelor's degree | 0.0013 | 947 |
| | 6) Master's degree | 0.0000 | 68 |
| Ethnicity | Black | 0.0000 | 46 |
| | White | 0.0000 | 2622 |
| Gender | Female | 0.0010 | 1587 |
| | Male | 0.0006 | 1033 |
| Has Been Targeted | False | 0.0011 | 2540 |
| | True | -0.0095 | 409 |
| Is Parent | No | -0.0021 | 1089 |
| | Yes | -0.0001 | 1603 |
| Is Transgender | No | 0.0000 | 4045 |
| Political Affiliation | Conservative | -0.0134 | 260 |
| | Independent | 0.0000 | 349 |
| | Liberal | -0.0010 | 850 |
| Religion Important | 1) No | 0.0022 | 401 |
| | 2) Not very | 0.0000 | 22 |
| | 3) Somewhat | 0.0000 | 289 |
| | 4) Very | 0.0002 | 463 |
| Seen Toxicity | False | -0.0070 | 286 |
| | True | -0.0008 | 2861 |
| Sexual Orientation | Bisexual | -0.0678 | 50 |
| | Heterosexual | -0.0010 | 3199 |
| Technology Impact | 2) Somewhat negative | 0.0000 | 15 |
| | 3) Neutral | 0.1243 | 27 |
| | 4) Somewhat positive | 0.0038 | 1343 |
| | 5) Very positive | -0.0148 | 269 |
| Toxicity Problem | 1) Never | 0.0000 | 7 |
| | 2) Rarely | 0.0000 | 84 |
| | 3) Occasionally | 0.0000 | 521 |
| | 4) Frequently | 0.0000 | 422 |
| | 5) Very Frequently | 0.0000 | 22 |

Table 25. Aposteriori unimodality results for the Sap-results dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | apunim | support |
|---|---|---|---|
| Age | 1) Gen. X+ | -0.1670$^{*}$ | 223 |
| | 2) Millennial | 0.0199 | 1651 |
| Ethnicity | white | -0.0007 | 1235 |
| Gender | man | -0.1508$^{***}$ | 1254 |
| | woman | 0.2613$^{***}$ | 693 |

Table 26. Aposteriori unimodality results for the Dices-350-results dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | apunim | support |
|---|---|---|---|
| Age | 1) Gen. Z | -0.0043 | 14728 |
| | 2) Millennial | -0.0089 | 9468 |
| | 3) Gen. X+ | 0.0148 | 8153 |
| Education | College + | 0.0105 | 19725 |
| | High school - | -0.0145 | 10783 |
| | Other | -0.0230$^{*}$ | 1841 |
| Gender | Man | -0.0171 | 16043 |
| | Woman | 0.0176 | 16306 |
| Race | African Am. | -0.0366$^{***}$ | 7627 |
| | Asian | 0.0445$^{***}$ | 6838 |
| | Latino | 0.0059 | 5786 |
| | Multiracial | -0.0061 | 4208 |
| | White | -0.0113 | 7890 |

Table 27. Aposteriori unimodality results for the Dices-990-results dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | apunim | support |
|---|---|---|---|
| Age | 1) Gen. Z | 0.0469$^{***}$ | 12214 |
| | 2) Millennial | -0.0068$^{*}$ | 34433 |
| | 3) Gen. X+ | -0.0282$^{***}$ | 12699 |
| Education | College + | 0.0063$^{*}$ | 52727 |
| | High school - | -0.0500$^{***}$ | 6619 |
| Gender | Man | 0.0359$^{***}$ | 28817 |
| | Woman | -0.0271$^{***}$ | 29990 |
| Race | African Am. | 0.0995$^{***}$ | 5148 |
| | Asian | -0.0057 | 16486 |
| | Latino | -0.0631$^{***}$ | 5877 |
| | Other | 0.0340$^{***}$ | 21669 |
| | White | -0.0649$^{***}$ | 10166 |

Table 28. Aposteriori unimodality results for the Popquorn-results dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| SDD | Value | apunim | support |
|---|---|---|---|
| Age | 01) 18-24 | -0.1198 | 90 |
| | 02) 25-29 | 0.0420 | 50 |
| | 03) 30-34 | 0.2251 | 71 |
| | 04) 35-39 | 0.0439 | 121 |
| | 05) 40-44 | 0.0061 | 81 |
| | 06) 45-49 | 0.0507 | 64 |
| | 07) 50-54 | 0.0381 | 22 |
| | 08) 54-59 | -0.0635 | 133 |
| | 09) 60-64 | 0.1194 | 13 |
| | 10) >65 | -0.0145 | 118 |
| Education | 2) High school | -0.0112 | 1270 |
| | 3) College degree | 0.0496*** | 2388 |
| | 4) Graduate degree | -0.0808** | 468 |
| Gender | Man | 0.0081 | 2720 |
| | Woman | -0.0195 | 2899 |
| Occupation | Employed | -0.0126 | 2535 |
| | Homemaker | -0.2366 | 12 |
| | Retired | -0.1026 | 95 |
| | Self-employed | 0.1063* | 210 |
| | Student | 0.3675 | 18 |
| | Unemployed | 0.1204* | 166 |
| Race | Asian | 0.2795* | 29 |
| | Black | -0.1572*** | 222 |
| | White | 0.0090 | 4135 |

Table 29. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under each instruction prompt. Human annotations do not depend on the prompt. Rows give results per model, pooled over all datasets (All) and for each dataset separately; *n* is the number of comments.

| Model | Dataset | n | Default | Stereotype | Persona | Single | Direct |
|---|---|---|---|---|---|---|---|
| Human | All | 3085 | | | 0.36 (0.00) | | |
| | Sap et al. 2022 | 585 | | | 0.38 (0.01) | | |
| | Kumar et al. 2021 | 1000 | | | 0.42 (0.01) | | |
| | POPQUORN | 1500 | | | 0.31 (0.01) | | |
| llama8b | All | 1723 | 0.44 (0.01) | 0.59 (0.01) | 0.29 (0.01) | 0.26 (0.01) | 0.55 (0.01) |
| | Sap et al. 2022 | 280 | 0.56 (0.02) | 0.47 (0.02) | 0.07 (0.01) | 0.29 (0.02) | 0.47 (0.02) |
| | Kumar et al. 2021 | 951 | 0.40 (0.01) | 0.61 (0.01) | 0.49 (0.01) | — | 0.55 (0.01) |
| | POPQUORN | 492 | 0.45 (0.01) | 0.60 (0.01) | 0.04 (0.00) | 0.24 (0.01) | 0.60 (0.01) |
| olmo32b | All | 3085 | 0.43 (0.01) | 0.40 (0.01) | 0.44 (0.01) | 0.46 (0.00) | 0.40 (0.01) |
| | Sap et al. 2022 | 585 | 0.24 (0.01) | 0.26 (0.01) | 0.26 (0.01) | 0.31 (0.01) | 0.26 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.51 (0.01) | 0.54 (0.01) | 0.48 (0.01) | 0.44 (0.01) | 0.43 (0.01) |
| | POPQUORN | 1500 | 0.46 (0.01) | 0.35 (0.01) | 0.50 (0.01) | 0.53 (0.01) | 0.44 (0.01) |
| olmo7b | All | 3084 | 0.68 (0.00) | 0.64 (0.00) | 0.67 (0.00) | 0.63 (0.00) | 0.59 (0.00) |
| | Sap et al. 2022 | 585 | 0.66 (0.01) | 0.60 (0.01) | 0.70 (0.01) | 0.61 (0.01) | 0.64 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.68 (0.01) | 0.55 (0.01) | 0.63 (0.01) | 0.55 (0.01) | 0.42 (0.01) |
| | POPQUORN | 1499 | 0.69 (0.01) | 0.71 (0.00) | 0.69 (0.01) | 0.69 (0.01) | 0.69 (0.01) |
| qwen32b | All | 3085 | 0.22 (0.01) | 0.26 (0.01) | 0.26 (0.01) | 0.27 (0.01) | 0.25 (0.01) |
| | Sap et al. 2022 | 585 | 0.18 (0.01) | 0.28 (0.01) | 0.22 (0.01) | 0.17 (0.01) | 0.21 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.19 (0.01) | 0.24 (0.01) | 0.25 (0.01) | 0.34 (0.01) | 0.28 (0.01) |
| | POPQUORN | 1500 | 0.25 (0.01) | 0.27 (0.01) | 0.28 (0.01) | 0.27 (0.01) | 0.24 (0.01) |
| qwen7b | All | 3085 | 0.35 (0.01) | 0.27 (0.00) | 0.26 (0.00) | 0.31 (0.01) | 0.32 (0.01) |
| | Sap et al. 2022 | 585 | 0.30 (0.01) | 0.30 (0.01) | 0.31 (0.01) | 0.25 (0.01) | 0.29 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.51 (0.01) | 0.23 (0.01) | 0.26 (0.01) | 0.42 (0.01) | 0.42 (0.01) |
| | POPQUORN | 1500 | 0.26 (0.01) | 0.29 (0.01) | 0.24 (0.01) | 0.26 (0.01) | 0.25 (0.01) |
| mistral7b | All | 3078 | 0.35 (0.01) | 0.39 (0.01) | 0.36 (0.01) | 0.39 (0.01) | 0.29 (0.01) |
| | Sap et al. 2022 | 585 | 0.21 (0.01) | 0.27 (0.01) | 0.24 (0.01) | 0.23 (0.01) | 0.15 (0.01) |
| | Kumar et al. 2021 | 994 | 0.37 (0.01) | 0.39 (0.01) | 0.36 (0.01) | 0.34 (0.01) | 0.36 (0.01) |
| | POPQUORN | 1499 | 0.40 (0.01) | 0.43 (0.01) | 0.40 (0.01) | 0.49 (0.01) | 0.29 (0.01) |
| mistral24b | All | 1704 | 0.28 (0.01) | 0.45 (0.01) | 0.38 (0.01) | 0.27 (0.01) | 0.27 (0.01) |
| | Sap et al. 2022 | 253 | 0.27 (0.02) | 0.40 (0.02) | 0.28 (0.02) | 0.08 (0.01) | 0.28 (0.02) |
| | Kumar et al. 2021 | 782 | 0.25 (0.01) | 0.51 (0.01) | 0.50 (0.01) | 0.43 (0.01) | 0.19 (0.01) |
| | POPQUORN | 669 | 0.32 (0.01) | 0.40 (0.01) | 0.27 (0.01) | 0.16 (0.01) | 0.36 (0.01) |
| gemma7b | All | 3083 | 0.18 (0.00) | 0.22 (0.00) | 0.25 (0.00) | 0.29 (0.01) | 0.23 (0.00) |
| | Sap et al. 2022 | 585 | 0.19 (0.01) | 0.23 (0.01) | 0.32 (0.01) | 0.38 (0.01) | 0.17 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.22 (0.01) | 0.25 (0.01) | 0.20 (0.01) | 0.12 (0.01) | 0.21 (0.01) |
| | POPQUORN | 1498 | 0.15 (0.01) | 0.20 (0.01) | 0.25 (0.01) | 0.37 (0.01) | 0.28 (0.01) |
| gemma31b | All | 3084 | 0.07 (0.00) | 0.13 (0.01) | 0.11 (0.00) | 0.04 (0.00) | 0.08 (0.00) |
| | Sap et al. 2022 | 585 | 0.04 (0.01) | 0.10 (0.01) | 0.07 (0.01) | 0.04 (0.01) | 0.05 (0.01) |
| | Kumar et al. 2021 | 1000 | 0.09 (0.01) | — | — | — | — |
| | POPQUORN | 1499 | — | 0.14 (0.01) | 0.13 (0.01) | 0.04 (0.00) | 0.09 (0.01) |

Table 30. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under the default prompt, per dataset and pooled over all datasets (All). *n* is the number of comments.

| | Sap et al. 2022 | | Kumar et al. 2021 | | POPQUORN | | All | |
|---|---|---|---|---|---|---|---|---|
| | n | Mean (SE) | n | Mean (SE) | n | Mean (SE) | n | Mean (SE) |
| Model | | | | | | | | |
| Human | 585 | 0.38 (0.01) | 1000 | 0.42 (0.01) | 1500 | 0.31 (0.01) | 3085 | 0.36 (0.00) |
| llama8b | 565 | 0.52 (0.01) | 977 | 0.40 (0.01) | 1464 | 0.41 (0.01) | 3006 | 0.43 (0.01) |
| olmo32b | 585 | 0.24 (0.01) | 1000 | 0.51 (0.01) | 1500 | 0.46 (0.01) | 3085 | 0.43 (0.01) |
| olmo7b | 585 | 0.66 (0.01) | 1000 | 0.68 (0.01) | 1500 | 0.69 (0.01) | 3085 | 0.68 (0.00) |
| qwen32b | 585 | 0.18 (0.01) | 1000 | 0.19 (0.01) | 1500 | 0.25 (0.01) | 3085 | 0.22 (0.01) |
| qwen7b | 585 | 0.30 (0.01) | 1000 | 0.51 (0.01) | 1500 | 0.26 (0.01) | 3085 | 0.35 (0.01) |
| mistral7b | 585 | 0.21 (0.01) | 1000 | 0.37 (0.01) | 1500 | 0.40 (0.01) | 3085 | 0.35 (0.01) |
| mistral24b | 585 | 0.25 (0.01) | 1000 | 0.25 (0.01) | 1500 | 0.33 (0.01) | 3085 | 0.29 (0.00) |
| gemma7b | 585 | 0.19 (0.01) | 1000 | 0.22 (0.01) | 1500 | 0.15 (0.01) | 3085 | 0.18 (0.00) |
| gemma31b | 585 | 0.04 (0.01) | 1000 | 0.09 (0.01) | — | — | 1585 | 0.07 (0.00) |

Table 31. Cohen's d between the default and each adversarial instruction prompt, computed on the per-comment nDFU. Results are given per model, for each dataset separately and as the unweighted mean over datasets (All). Comments per model: Sap et al. 2022: 585 (llama8b: 565); Kumar et al. 2021: 1000 (llama8b: 977); POPQUORN: 1500 (llama8b: 1464, gemma31b: 0).

| | All | | | | Sap et al. 2022 | | | | Kumar et al. 2021 | | | | POPQUORN | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct | Stereotype | Persona | Single | Direct |
| llama8b | 0.41 | −1.12 | −0.64 | 0.32 | −0.20 | −1.93 | −0.70 | −0.24 | 0.75 | 0.30 | – – – | 0.56 | 0.67 | −1.72 | −0.58 | 0.63 |
| olmo32b | −0.04 | 0.02 | 0.07 | −0.10 | 0.09 | 0.06 | 0.26 | 0.06 | 0.14 | −0.13 | −0.29 | −0.31 | −0.35 | 0.14 | 0.25 | −0.05 |
| olmo7b | −0.27 | −0.04 | −0.29 | −0.47 | −0.27 | 0.16 | −0.21 | −0.12 | −0.62 | −0.24 | −0.60 | −1.25 | 0.09 | −0.04 | −0.04 | −0.04 |
| qwen32b | 0.19 | 0.16 | 0.18 | 0.13 | 0.32 | 0.13 | −0.04 | 0.09 | 0.17 | 0.21 | 0.51 | 0.31 | 0.08 | 0.13 | 0.07 | −0.01 |
| qwen7b | −0.33 | −0.30 | −0.16 | −0.11 | 0.02 | 0.06 | −0.17 | −0.02 | −1.13 | −0.90 | −0.31 | −0.31 | 0.11 | −0.05 | −0.00 | −0.01 |
| mistral7b | 0.13 | 0.03 | 0.10 | −0.20 | 0.21 | 0.10 | 0.08 | −0.21 | 0.07 | −0.01 | −0.10 | −0.05 | 0.11 | 0.01 | 0.31 | −0.35 |
| mistral24b | 0.59 | 0.25 | −0.32 | −0.03 | 0.64 | 0.04 | −0.88 | 0.17 | 0.98 | 0.97 | 0.57 | −0.25 | 0.17 | −0.26 | −0.66 | −0.00 |
| gemma7b | 0.16 | 0.27 | 0.38 | 0.11 | 0.18 | 0.48 | 0.70 | −0.08 | 0.10 | −0.08 | −0.40 | −0.07 | 0.20 | 0.41 | 0.85 | 0.49 |
| gemma31b | 0.36 | 0.21 | 0.02 | 0.07 | 0.36 | 0.21 | 0.02 | 0.07 | – | – | – | – | – – – | – – – | – – – | – – – |

Table 32. Statistically significant ($p < 0.05$) SDDs of each LLM compared with the human annotations, per dataset. A dimension counts as significant if at least one of its groups is. Each cell shows one symbol per prompt, in the order default, stereotype, persona, single, direct: ✓ both significant, with the same sign for every group that is significant in both; ✓ neither significant; ± both significant, but the sign of at least one group's effect differs; × significant for only one of the two; blank: no LLM result.

| | | llama8b | olmo32b | olmo7b | qwen32b | qwen7b | mistral7b | mistral24b | gemma7b | gemma31b |
|---|---|---|---|---|---|---|---|---|---|---|
| Ds | SDD | | | | | | | | | |
| Sap | Age | ✓✓×✓✓ | ✓✓✓×✓ | ±✓✓±± | ±±±±± | ±±±±± | ±✓±✓× | ±✓±✓± | ××±±× | ✓✓✓✓✓ |
| | Ethnicity | ✓×××× | ✓✓×✓✓ | ××××× | ××××× | ××××× | ××××× | ××✓×× | ××××× | ××××× |
| | Gender | ✓±✓✓± | ××××× | ✓±✓✓× | ×✓×××× | ××±±± | ×✓✓×✓ | ±✓✓✓× | ××±✓± | ××××× |
| POP | Age | ××× × | ××××× | ××× × | ××××× | ××××× | × ✓ | ××✓✓× | ××××× | ×××× |
| | Education | ±±± ± | ✓±±✓✓ | ✓✓± ± | ✓✓✓✓✓ | ✓✓±±✓ | ✓× × | ±✓×±± | ✓±±±± | ✓✓×× |
| | Gender | ××× × | ××××✓× | ××× × | ✓××××× | ××××× | ×✓ ✓ | ××✓✓× | ✓××✓× | ××××× |
| | Occupation | ±±× ✓ | ±✓±××± | ±×± ✓ | ±±±±✓ | ✓✓✓✓✓ | ✓× | ±✓×✓± | ×±✓✓✓ | ✓✓×✓ |
| | Race | ✓✓✓ ± | ✓±✓×± | ✓±✓ ✓ | ✓✓±±± | ±±✓±± | ✓× | ✓±××± | ✓✓✓±✓ | ±±±± |

Table 33. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under the default prompt, per dataset; $n$ is the number of dimensions. A dimension counts as significant if at least one of its groups is. †: ratio over fewer dimensions, as the model has results for only some of them.

| Dataset | $n$ | Human | llama8b | olmo32b | olmo7b | qwen32b | qwen7b | mistral7b | mistral24b | gemma7b | gemma31b |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sap | 3 | 66% | 66% | 33% | 100% | 66% | 66% | 66% | 100% | 33% | 66% |
| Kumar | 13 | 0% | 61% | 46% | 84% | 53% | 84% | 46% | 46% | 38% | 30% |
| POP | 5 | 60% | 100% | 100% | 100% | 80% | 100% | 100% | 100% | 60% | — |

Table 34. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under each instruction prompt, per dataset; $n$ is the number of dimensions. A dimension counts as significant if at least one of its groups is; human annotations do not depend on the prompt. †: ratio over fewer dimensions, as the model has results for only some of them.

| Dataset | $n$ | Prompt | Human | llama8b | olmo32b | olmo7b | qwen32b | qwen7b | mistral7b | mistral24b | gemma7b | gemma31b |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sap | 3 | Default | 66% | 66% | 33% | 100% | 66% | 66% | 66% | 100% | 33% | 66% |
| | | Stereotype | | 100% | 33% | 100% | 100% | 66% | 100% | 100% | 33% | 66% |
| | | Persona | | 66% | 66% | 100% | 66% | 100% | 100% | 66% | 100% | 66% |
| | | Single | | 100% | 33% | 100% | 66% | 100% | 66% | 100% | 100% | 66% |
| | | Direct | | 100% | 33% | 66% | 66% | 100% | 66% | 66% | 66% | 66% |
| Kumar | 13 | Default | 0% | 61% | 46% | 84% | 53% | 84% | 46% | 46% | 38% | 30% |
| | | Stereotype | | 69% | 7% | 84% | 61% | 92% | 69% | 76% | 53% | — |
| | | Persona | | 69% | — | 92% | — | 61% | 92% | 84% | 61% | — |
| | | Single | | — | — | 45%† | 61% | 69% | 15% | 38% | 61% | — |
| | | Direct | | 92% | 84% | 84% | 23% | 46% | 76% | 76% | 76% | — |
| POP | 5 | Default | 60% | 100% | 100% | 100% | 80% | 100% | 100% | 100% | 60% | — |
| | | Stereotype | | 100% | 100% | 60% | 100% | 100% | 0%† | 100% | 100% | 100% |
| | | Persona | | 80% | 100% | 100% | 100% | 100% | — | 0% | 80% | 100% |
| | | Single | | — | 40% | — | 100% | 100% | — | 40% | 100% | 60% |
| | | Direct | | 100% | 100% | 100% | 100% | 100% | 0%† | 100% | 100% | 80% |